Paths in Polynomial Lemniscates: A Degree-Seven Counterexample and Radial Connections

Proofs and examples for Erdős Problem #1041, with a Lean proof for the polynomial constructed by `ani`

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Abstract

We give the rational estimates and geometric proof for the degree-seven counterexample to Erdős Problem 1041 constructed by `ani`. A double-cover argument gives a Hausdorff-measure lower bound for every connected set joining the two roots in its unique two-root component. For trinomials, collinear roots and two sparse families we describe contained paths. We also retain ordinary analytic arguments for critical-value criteria, with their unformalised inputs identified beside the statements. Critical-value moments, counterexamples to prescribed paths and questions about flow-strip geometry complete the record.

The short paper develops the counterexample and the trinomial argument. Here Section 2 supplies the exact tests for the polynomial, while Sections 6 and 9 give further explicit paths. The area argument in Section 4 and the isolated-value construction in Section 7 are ordinary analytic proofs, still requiring formalisation. Their conclusions appear as remarks. The local notes identify the assumptions of the existing Lean deductions; Section C collects the formal and computational references.

1 The question and the degree-seven counterexample

For a monic polynomial f, let $\Omega_f = \{|f| < 1\}$; the notation E_f used in the segment arguments denotes the same open set. We write $K_t(f) = \{|f| \le t\}$ for a closed sublevel. Erdős, Herzog and Piranian [4, Problem 5, p. 139] asked:

Problem 1.1 (Erdős #1041)**.** Let $f(z) = \prod_{i=1}^{n}(z - z_i)$ be monic of degree $n \ge 2$, with $z_i \in \mathbb{D}$, the open unit disc. Must two of the roots be joined by a curve of length less than 2 lying in the open lemniscate $\{z \in \mathbb{C} : |f(z)| < 1\}$?

Boundary roots already give an elementary failure of strict containment. For $f(z) = z^2 - 1$, every continuous path from -1 to 1 meets the imaginary axis, where $|f(iy)| =$

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$1+y^2 \geq 1$. The two endpoints are therefore unjoinable in $\{|f| < 1\}$, although the diameter lies in $\{|f| \leq 1\}$. This illustrates the distinction between the open and closed lemniscates.

A repeated zero gives a constant path between two occurrences. Thus the question concerns distinct zeros. The catalogue of Bloom [3] records the problem and its discussion. Pendyala [11, Theorem 1 and Lemma 1] proved the degree-four case by chords or radial segments through the centre of a smallest enclosing disc. The degree-seven construction of `ani` has a different geometry.

Theorem 1.2 (`ani`'s counterexample)**.** *The monic polynomial in* (2) *has seven distinct roots in the open unit disc. Every connected subset of its strict unit lemniscate containing two roots has one-dimensional Hausdorff measure greater than* 2*.*

Five components contain one zero each; the remaining component contains two, and the polynomial has degree two on it. A quadratic estimate near its critical point counts two inverse images, with multiplicity, for each value on an outward slit. Since the component has degree two, these are all its inverse images of that value. Hence the entire preimage of the slit in the component lies in the small disc where the local estimate applies.

On each of the two inverse sheets, radial projection gives a lower bound for the Hausdorff measure of a connected set joining the zeros. The sheets are disjoint, so the two bounds can be added. Their sum exceeds 2. Section 2 supplies the estimates and proof. Applied to a continuous path image, this also excludes a rectifiable path of total variation less than 2.

The theorem concerns a fixed polynomial. The small-parameter family reported by `ani` is not proved here, and independent human review of correspondence with the 1958 wording is not recorded. The supplied formal statements and their scope are listed in Section C.

For trinomials the root identity keeps every root-to-origin segment inside the lemniscate, giving a short broken line between any pair of zeros. The later analytic remarks seek a pair by following sublevel components: one argument uses a small least critical-value modulus, and another uses an isolated simple critical value. Their conclusions have additional formalisation requirements. A bound on critical-value means alone makes no choice of a contained path.

2 The degree-seven construction

The construction has two separate quantitative requirements. The two roots in the selected component must have combined distance from its critical point greater than 2, and the preimage of an outward value slit must be small enough to preserve that surplus. A narrow neck alone would not suffice: for $p(z) = z^2 - r^2$, $0 < r < 1$, the slit $(-1, -r^2]$ has preimage $\{iy : |y| < \sqrt{1-r^2}\}$, while the root segment $[-r, r]$ still has length $2r < 2$ and lies in $\{|p| < 1\}$.

We first prove the estimate that separates these requirements. We then verify root locations, critical values and the membership of the chosen pair in one component. The final fixed-parameter estimates compare the root-distance surplus with the neck loss, and distinguish both from the radial contraction $1 - \rho = \varepsilon^8$.

2.1 A two-sheeted component

The quantity $\delta = 1 - |p(c)|$ measures the remaining length of the slit in the value plane. The local quadratic coefficient converts it into the spatial scale $\sqrt{\delta/M}$. The radius h records where that quadratic comparison is valid. Degree two will turn the local preimage estimate into a statement about the whole component.

Lemma 2.1 (a bottleneck estimate)**.** *Let p be a polynomial and U a component of $\{|p| < 1\}$ on which p is a proper map of degree two. Suppose its only critical point c is simple and $v = p(c) \neq 0$. Write*

$$p(c+z) - v = z^2 A(z), \qquad M = |A(0)|, \qquad \delta = 1 - |v|.$$

Suppose $h > 0$, $|A(z)/A(0) - 1| \le 1/4$ for $|z| \le h$, and $\delta < Mh^2/4$. If a, b are the two zeros in U, every connected $K \subset U$ containing a, b satisfies

$$\mathcal{H}^1(K) \ge |a-c| + |b-c| - \frac{8}{3}\sqrt{\delta/M}.$$

Proof. Cut the value disc along $J = \{tv : 1 \le t < 1/|v|\}$. Its complement is simply connected and contains 0. The restriction of p above this complement is an unramified two-sheeted covering, hence is the disjoint union of two biholomorphic sheets U_1, U_2, containing a, b respectively.

Put $r_0 = (4/3)\sqrt{\delta/M} < 2h/3$. On $|z| = r_0$,

$$|p(c+z) - v| \ge \tfrac{3}{4} M r_0^2 = \tfrac{4}{3}\delta > \delta.$$

For each $w \in J$, Rouché's theorem therefore gives two zeros, counted with multiplicity, of $p(c+z) - w$ in $|z| < r_0$. They remain in U: as w moves out from v along J, the zeros continue from c and cannot cross $|z| = r_0$. Since the degree on U is two, these are all its preimages of w. Consequently $p^{-1}(J) \cap U \subset B(c, r_0)$.

For $r_0 < t < |a-c|$, the set $U_1 \cap \{|z-c| > t\}$ has relative boundary in U contained in the circle $|z-c| = t$. If that circle missed $K \cap U_1$, then $K \cap U_1 \cap \{|z-c| > t\}$ would be both open and closed in K. It contains a and excludes b, contradicting connectedness. Thus the radial map $z \mapsto |z-c|$ sends $K \cap U_1$ onto a set containing $(r_0, |a-c|)$ whenever that interval is nonempty. This map is 1-Lipschitz, so

$$\mathcal{H}^1(K \cap U_1) \ge (|a-c| - r_0)_+.$$

The same argument applies to b and U_2. Although the two radial intervals can overlap on the real line, the sets $K \cap U_1$ and $K \cap U_2$ lie in disjoint open subsets of the plane. Hausdorff outer measure adds across that separation, even when K is not assumed measurable. Consequently

$$\begin{aligned} \mathcal{H}^1(K) &\ge \mathcal{H}^1(K \cap U_1) + \mathcal{H}^1(K \cap U_2) \\ &\ge (|a-c| - r_0)_+ + (|b-c| - r_0)_+ \\ &\ge |a-c| + |b-c| - 2r_0. \end{aligned}$$

Substituting the value of r_0 proves the bound. Connectedness supplied the intervals; no path or parametrisation of K was required. □

The diagram in the short paper distinguishes the value-plane slit from the two radial projections used above.

2.2 Polynomial and roots

The perturbation of $z^7 - 1$ is small on the unit circle but has a nontrivial shape at $z = \varepsilon w$: the terms $\varepsilon^4 a z^3$, $\varepsilon^5 b z^2$ and $\varepsilon^6 c_0 z$ all become order ε^7 there. The paired conjugate terms give the antisymmetry used to locate the roots. These two features let us control roots near the unit circle and critical points near the origin by different tests.

All constants are rational, so both sets of tests can be made without rounding. Set

$$s = 10^{-6}, \qquad \varepsilon = s^2 = 10^{-12}, \qquad \rho = 1 - s^{16} = 1 - 10^{-96},$$

$$A_0 = -\frac{329507}{1600}, \qquad B_0 = \frac{551827}{800}, \qquad C_0 = -\frac{23013813}{32000},$$

and define

$$a = A_0 - is, \qquad b = iB_0 + \frac{9}{5}s, \qquad c_0 = -C_0 - \frac{162}{25}is.$$

The polynomial is

$$F(z) = z^7 - 1 + \varepsilon^4(az^3 - \bar{a}z^4) + \varepsilon^5(bz^2 - \bar{b}z^5) + \varepsilon^6(c_0 z - \bar{c}_0 z^6), \tag{1}$$

$$f(z) = \rho^7 F(z/\rho). \tag{2}$$

The identity $F(z) = -z^7\overline{F(1/\bar{z})}$ is useful, but does not by itself place the zeros on the unit circle. For that purpose put $\chi(x) = (1 + ix)/(1 - ix)$. Direct expansion gives $(1 - ix)^7 F(\chi(x)) = 2iH(x)$, where

$$\begin{aligned} H(x) = {} & 7x - 35x^3 + 21x^5 - x^7 - \varepsilon^4(1 + x^2)^3(s + A_0 x) \\ & + \varepsilon^5(1 + x^2)^2[B_0(1 - 3x^2) + (9s/5)(x^3 - 3x)] \\ & + \varepsilon^6(1 + x^2)[C_0(5x - 10x^3 + x^5) - (162s/25)(1 - 10x^2 + 5x^4)]. \end{aligned}$$

The signs at the following rational endpoints are obtained by substitution. Both endpoints in each row are divided by 10^4.

j	Left endpoint	Right endpoint	Left sign	Right sign
0	-1	1	$-$	$+$
1	4815	4816	$+$	$-$
2	12539	12540	$-$	$+$
3	43812	43813	$+$	$-$
4	-43813	-43812	$+$	$-$
5	-12540	-12539	$-$	$+$
6	-4816	-4815	$+$	$-$

The seven intervals give distinct real zeros x_j of H, so $\zeta_j = \chi(x_j)$ are all seven zeros of F. They have modulus one. Hence the zeros $b_j = \rho\zeta_j$ of f are distinct and lie in the open unit disc.

2.3 Critical points and local estimates

The critical points cluster at scale ε. We resolve them by writing $F(\varepsilon w) = -1 + \varepsilon^7 Q(w)$, where

$$Q(w) = c_0 w + b w^2 + a w^3 - \varepsilon \bar{a} w^4 - \varepsilon^3 \bar{b} w^5 - \varepsilon^5 \bar{c}_0 w^6 + w^7.$$

For a centre v_j, let $T_k(v_j)$ be the coefficient of z^k in $Q(v_j + z)$. If $Q(w) = \sum q_k w^k$, these coefficients are given by the finite formula

$$T_k(v_j) = \sum_{\ell=k}^{7} \binom{\ell}{k} q_\ell v_j^{\ell-k}.$$

Use the following centres, whose displayed decimals are exact, and radius $r = 10^{-6}$:

j	$\operatorname{Re} v_j$	$\operatorname{Im} v_j$
0	3.1898275786	−0.4999999979
1	−3.1898275845	−0.5000000021
2	0.0000000039	0.8232474662
3	0.0000002274	1.8078590858
4	−0.0000002265	1.8838591455
5	0.0000000011	−3.5149656976

These six discs are disjoint. For a Gaussian rational $z = u + iv$, put $N(z) = |u| + |v|$ and $m(z) = \max(|u|, |v|)$, so $m(z) \le |z| \le N(z)$. Define

$$E_j = N(T_1(v_j)) + \sum_{k=2}^{6} (k+1) N(T_{k+1}(v_j)) r^k, \qquad L_j = 2m(T_2(v_j))r.$$

The next table proves $E_j < L_j$ in every disc. Rouché's theorem, applied to the linear term of Q', gives exactly one critical point in each.

To determine its critical-value modulus, use

$$\mathcal{H}(w) = \frac{1-\rho^{14}}{2\rho^{14}\varepsilon^7} + \operatorname{Re} Q(w) - \frac{\varepsilon^7}{2}|Q(w)|^2, \qquad |f(\rho\varepsilon w)|^2 = 1 - 2\rho^{14}\varepsilon^7 \mathcal{H}(w).$$

For $|w - v_j| \le r$, set $V_j = \sum_{k=1}^{7} N(T_k(v_j)) r^k$. Substituting $\operatorname{Re} T_0(v_j) \pm V_j$ for $\operatorname{Re} Q(w)$ and $N(T_0(v_j)) + V_j$ for $|Q(w)|$ gives the following rational bounds. An interval in the last column contains $10^9 \mathcal{H}(w)$ throughout the whole disc.

j	$10^9 E_j <$	$10^6 L_j >$	$10^9 \mathcal{H}(w)$
0	423	8340	$[-502, -486]$
1	433	8340	$[-502, -486]$
2	12	378	$[3556, 3558]$
3	3	43	$[-75, -74]$
4	4	48	$[-864, -863]$
5	1298	16812	$[-1566, -1548]$

It follows that f has just one critical point in $\{|f|<1\}$, namely $c_s=\rho\varepsilon w_s$ with $|w_s-v_2|<r$. Its critical value is nonzero, since the polynomial has simple roots, and no critical point lies on $|f|=1$.

Expanding the same Taylor coefficients throughout $|w-v_2|\le r$ gives

$$|T_2(w)|>189.1,\qquad (|T_3(w)|,|T_4(w)|,|T_5(w)|,|T_6(w)|,|T_7(w)|)<(190,20,15,6,2).$$

In the last coordinate the exact value is $|T_7|=1$. In particular, for $h=\rho\varepsilon/10$ and $f(c_s+z)-f(c_s)=z^2A_s(z)$,

$$|A_s(0)|>180\rho^5\varepsilon^5,\qquad \left|\frac{A_s(z)}{A_s(0)}-1\right|<\frac14\quad(|z|\le h). \tag{3}$$

Indeed the numerator of the relative error is at most $190/10+20/100+15/1000+6/10^4+1/10^5<189.1/4$ in scaled coordinates. The positive critical-value bound in the table also gives

$$0<\delta=1-|f(c_s)|\le 1-|f(c_s)|^2<\rho^7\varepsilon^7\frac{36}{5\cdot10^6}.$$

Consequently

$$\sqrt{\delta/|A_s(0)|}<\frac{\rho\varepsilon}{5000},\qquad \delta<|A_s(0)|h^2/4. \tag{4}$$

2.4 The two roots in the component

The local estimates identify a critical point, but we still have to find the two roots in its component. The connections from c_s to b_3,b_6 use a small disc and four segments near the critical point, short segments to the root directions, and radial segments out to the roots. The following ordinary argument identifies the component directly; the Lean source uses barriers. Put

$$\begin{aligned}
\operatorname{Re} w_q &= \frac{392265832599170384843469532006 47}{10^{40}},\\
\operatorname{Im} w_q &= \frac{823247466194484895367291784947280007 8083}{10^{40}},\\
x_3 &= w_q-6+3i,\qquad x_6=w_q+5-5i,\\
m_3 &= \frac{-7207750943219353009 89+347106991294046496381i}{10^{20}},\\
m_6 &= \frac{4987918414869868244 20-625465185974423846967i}{10^{20}}.
\end{aligned}$$

The elementary inequality $1-\rho^{14}\ge 14\rho^{14}\varepsilon^8$ shows that

$$\mathcal{H}_0(w)=7\varepsilon+\operatorname{Re}Q(w)-\frac{\varepsilon^7}{2}|Q(w)|^2\le\mathcal{H}(w).$$

Thus $\mathcal{H}_0>0$ suffices for containment. To check a segment from u to v, expand $\mathcal{H}_0(u+t(v-u))=\sum_{k=0}^{14}a_kt^k$. Its Bernstein coefficients of degree 14 are

$$\beta_j=\sum_{k=0}^{j}a_k\frac{\binom{j}{k}}{\binom{14}{k}},\qquad \mathcal{H}_0(u+t(v-u))=\sum_{j=0}^{14}\beta_j\binom{14}{j}t^j(1-t)^{14-j}.$$

All weights on the right are nonnegative and sum to one for $0 \le t \le 1$. Exact substitution gives the lower bounds

Segment	Lower bound for every β_j
$[w_q, x_3]$	$3556874/10^{12}$
$[x_3, m_3]$	640506
$[w_q, x_6]$	$3556874/10^{12}$
$[x_6, m_6]$	139485

The formula for T_k above, followed by the displayed formula for β_j, specifies these checks using rational arithmetic alone; no sampling between endpoints is involved. The same Taylor test at w_q, with radius 10^{-8}, gives one zero of Q' in that disc and $\mathcal{H}_0 > 3556874/10^{12}$ throughout it. The disc is contained in $|w - v_2| < 10^{-6}$, so this zero is w_s. For each $j = 3, 6$, we can therefore follow the segments from w_s to w_q, then to x_j and m_j, staying where $\mathcal{H}_0 > 0$. It remains to connect m_j to the corresponding root.

For the remaining segments no large polynomial expansion is needed. The Cayley intervals give $|m_j/8 - \zeta_j| < 1/10$ and $|m_j/8| < 101/100$, for $j = 3, 6$. If x lies between $m_j/8$ and ζ_j, then $|x| \le 101/100$ and $|x - \zeta_j| < 1/10$. From $F(\zeta_j) = 0$ and $|\zeta_j| = 1$ we have $|\zeta_j^7 - 1| < 10^{-40}$. Telescoping $x^7 - \zeta_j^7$ therefore gives

$$\operatorname{Re} x^7 \ge 1 - \frac{7}{10}\left(\frac{101}{100}\right)^6 - 10^{-40} > \frac{1}{4}.$$

The coefficient bounds

$$(|q_1|, \dots, |q_7|) \le (720, 690, 206, 10^{-9}, 10^{-30}, 10^{-55}, 1) =: (U_1, \dots, U_7)$$

imply

$$\sum_{k=1}^{6} U_k (8.08)^k < 160000, \qquad \operatorname{Re} Q(8x) > 350000, \qquad |Q(8x)| < 2500000.$$

Hence $\mathcal{H}_0(8x) > 0$. This connects m_j to $8\zeta_j$. Finally, for $8\varepsilon \le t \le 1$, the root equation gives

$$F(t\zeta_j) = t^7 - 1 + \sum_{k=1}^{6} \varepsilon^{7-k} q_k \zeta_j^k (t^k - t^7),$$

so

$$|F(t\zeta_j)| \le 1 - t^7 + t^7 \sum_{k=1}^{6} \frac{U_k}{8^{7-k}} < 1, \qquad \sum_{k=1}^{6} \frac{U_k}{8^{7-k}} < \frac{3}{40}.$$

Scaling by ρ joins $\rho\varepsilon\, 8\zeta_j$ to b_j within $\{|f| < 1\}$. The concatenated segments connect c_s to both roots.

Every component of a polynomial lemniscate at a regular level is simply connected. The component-wise Riemann–Hurwitz formula [8, proof of Proposition 2.1] says that a component containing k zeros has $k - 1$ critical points, counted with multiplicity. The critical point table therefore shows that the component U of c_s contains exactly b_3, b_6, and every other component contains one root.

2.5 The length inequality

It remains to compare the root distances with the loss at the slit. For a unit complex number ζ, $|\zeta - \varepsilon w_s| \geq 1 - \varepsilon \operatorname{Re}(\bar{\zeta} w_s)$. The Cayley intervals and $|w_s - v_2| < 10^{-6}$ give

$$\operatorname{Re}((\bar{\zeta}_3 + \bar{\zeta}_6) w_s) < -\frac{143}{1000}.$$

For completeness, this last bound can be obtained without trigonometric estimates: use the midpoint of each rational interval and the derivative bound $|\chi'(x)| = 2/(1 + x^2)$ to enclose ζ_j, then substitute the disc for w_s. These enclosures in fact give an upper bound below -0.28635. Thus

$$|b_3 - c_s| + |b_6 - c_s| > \rho \left(2 + \frac{143}{1000} \varepsilon \right).$$

Lemma 2.1 and (4) now yield

$$\mathcal{H}^1(K) > \rho \left(2 + \left(\frac{143}{1000} - \frac{8}{15000} \right) \varepsilon \right) > 2$$

for every connected subset of U containing both roots. The last strict inequality follows by substituting $\rho = 1 - 10^{-96}$ and $\varepsilon = 10^{-12}$; its excess over 2 exceeds $1.42 \cdot 10^{-13}$. A connected set containing any two roots must lie in one component, so this proves the assertion for the whole strict lemniscate. In particular it applies to the image of every continuous root-to-root path. Since the Hausdorff measure of a rectifiable path image is at most its total variation, both interpretations of path length are excluded.

The construction above fixes $s = 10^{-6}$. It establishes no interval of parameters for the small-parameter family reported by `ani`. The formal files also prove the assertion for arbitrary preconnected sets; the argument here uses the equivalent connected-subset formulation and ordinary Rouché and covering arguments. No new Lean build or independent human review is asserted by this exposition.

3 Trinomials

At a zero of a trinomial, the root equation eliminates the middle coefficient. Along the segment from the origin to that zero, the remaining expression is a nonnegative linear combination of two values of modulus less than one. The sum of the coefficients is at most one, which gives the required containment.

Theorem 3.1 (trinomial root connections)**.** *Let n, m be integers with $1 \leq m < n$, and let $f(z) = z^n + az^m + b$ have every zero in $\mathbb{D}$. For every zero ζ, the segment $[0, \zeta]$ lies in E_f. Consequently any two zeros ζ_1, ζ_2 are joined in E_f by the broken line $\zeta_1 \to 0 \to \zeta_2$, of length $|\zeta_1| + |\zeta_2| < 2$.*

Proof. Vieta's formula gives $|b| < 1$. At a zero ζ the root equation $\zeta^n + a\zeta^m + b = 0$ eliminates the middle coefficient:

$$f(t\zeta) = b(1 - t^m) + \zeta^n (t^n - t^m). \tag{5}$$

For $0 \le t < 1$ the two scalar weights $1 - t^m$ and $t^m - t^n$ are nonnegative and sum to $1 - t^n$, so

$$|f(t\zeta)| \le |b|(1 - t^m) + |\zeta|^n(t^m - t^n) < 1 - t^n \le 1.$$

At $t = 1$ the value is zero. Concatenating two such segments gives the length assertion. □

No separate bound on a is needed, since the root equation eliminates it before we take absolute values. The conclusion concerns segments to zeros and makes no assertion that E_f is star-shaped.

Summation by parts. For $f(z) = \sum_{k=0}^{n} c_k z^k$ and a zero ζ, put $S_j = \sum_{k=0}^{j} c_k \zeta^k$. Finite summation by parts gives

$$f(t\zeta) = \sum_{j=0}^{n-1} (t^j - t^{j+1}) S_j \qquad (0 \le t \le 1), \tag{6}$$

since the coefficient of $c_k\zeta^k$ on the right is $t^k - t^n$ and $\sum_{k<n} c_k\zeta^k = -c_n\zeta^n$. The weights in (6) are nonnegative and sum to $1 - t^n$, so the whole radial segment lies in the closed unit sublevel set whenever every partial sum lies in the closed unit disc. For a trinomial, $S_j = b$ for $j < m$ and $S_j = -\zeta^n$ for $m \le j < n$; these two values may coincide. This is exactly the estimate above. With two intermediate coefficients, the assumption that all zeros lie in the unit disc need not bound every partial sum by one. The next example shows that a prescribed radial segment can then escape.

Example 3.2 (an escaping root spoke)**.** Let $0 < r < 1$ satisfy $r^6 > 320/327$, and set

$$f_r(z) = z^6 + \tfrac{1}{5}r^2z^4 - \tfrac{1}{5}r^4z^2 - r^6 = (z^2 - r^2)(z^4 + \tfrac{6}{5}r^2z^2 + r^4).$$

Every zero has modulus r: for the quartic factor put $z = rw$, so that the squared roots solve $v^2 + (6/5)v + 1 = 0$, whose two roots are complex conjugates of product one. Nevertheless

$$f_r(r/2) = -\frac{327}{320}\, r^6,$$

so the segment $[0, r]$ leaves E_{f_r}. This rules out a universal assertion about every origin-to-zero segment. It exhibits no counterexample to the existence of some short connection.

Sources and scope. Lean checks the identity (6), the constant-term bound, the radial estimate, their combination with the length bound and the evaluation in Example 3.2. No novelty claim is made for this elementary trinomial argument.

4 A small least critical value

We follow a component from its first merger as the level of $|f|$ increases. Under the assumption that it contains no path shorter than 2 between distinct roots, conformal length bounds force the roots apart. Packing then forces area growth beyond Pólya's upper bound. The reduction to this comparison is an ordinary analytic argument; the

exact computation at its end does not formalise the reduction. We retain the resulting criterion as an unformalised remark. Throughout this section

$$\mu = \min_{f'(c)=0} |f(c)|$$

is the least critical-value modulus.

Remark 4.1 (unformalised small-critical-value criterion). Let f be squarefree and monic of degree $n \geq 2$ with $\mu \leq 13/25$. Then two distinct roots of f are joined inside $\{|f| < 1\}$ by a rectifiable curve of length strictly below 2. No hypothesis is placed on the locations of the roots, on the number of roots in any component, or on the capacity of any component.

No Lean proof in the supplied sources establishes this whole assertion. The analytic reduction and the rational comparison below remain separate unformalised inputs; checking the final exponential inequality alone does not check that reduction.

For $f(z) = z^n - b$ with $0 < |b| < 1$, the only critical point is 0 and $\mu = |b|$. Thus this criterion includes $|b| \leq 13/25$ but excludes $13/25 < |b| < 1$, although the trinomial theorem gives the required path in both ranges. Its value is that it also applies to polynomials with arbitrary coefficient patterns, including roots outside the unit disc. For example, $(z-3)^n - 1/2$, $n \geq 2$, has $\mu = 1/2$ and all its roots satisfy $|z| > 2$. The connection is required to stay in $|f(z)| < 1$; the example places no restriction on its distance from the origin.

Remark 4.2 (unformalised scale-free consequence). Every squarefree monic f of degree $n \geq 2$ has two distinct roots joined inside $\{|f| < (25/13)\mu\}$ by a curve of length below $2((25/13)\mu)^{1/n}$.

Lean proves this rescaling under the input `LowCriticalThirteenTwentyFifths`, namely Remark 4.1. That input is the unformalised path criterion itself. It has no named published theorem as its supplied justification.

Proof. Apply Remark 4.1 to $s^{-n}f(sz)$ with $s = ((25/13)\mu)^{1/n}$, whose least critical-value modulus is 13/25, and scale back. Squarefreeness gives $\mu > 0$, so the scaling factor is positive. □

The proof has two steps. Failure of the length bound first forces the roots to be far apart in the hyperbolic metric of a sublevel component. Packing those roots gives a lower bound for their number. Averaging paths between adjacent boundary points then converts that root count into an area-growth inequality. The rational certificate shows that this growth would exceed Pólya's area bound before the value level reaches 1.

Consequences of having no short path. Assume no two distinct roots are joined inside $\{|f| < 1\}$ by a curve of length below 2. Choose a critical point at level μ and two descending inverse arcs into distinct one-root components of $\{|f| < \mu\}$; below μ each component maps conformally onto the value disc, and distinct local inverse arcs at the first critical point enter distinct components, since two inverse images of a nearby regular value in one component would contradict its degree one. Their union is a compact connected set containing two roots. Fix a level $t \in (\mu, 1)$ at which no critical value has

modulus t, so that the level is regular. Let C_t be the component of $\{|f| < t\}$ containing that set, let $k \geq 2$ be its root count, and put $x = \log(t/\mu)$ and $a = \mathrm{Area}(C_t)/\pi$; Pólya's inequality [2, Theorem 1] gives $a \leq t^{2/n} < 1$. The component is a Jordan domain. List its roots as $\zeta_1, \dots, \zeta_k$, choose a Riemann map $\varphi : \mathbb{D} \to C_t$, and write $b_j = \varphi^{-1}(\zeta_j)$. The component-wise form of [8, (2.2)] is

$$\frac{f(\varphi(z))}{t} = e^{i\theta} \prod_{j=1}^{k} \frac{z - b_j}{1 - \overline{b_j} z}.$$

Indeed, regularity extends φ across the Jordan boundary. The quotient of the two sides without the phase has no zero or pole on the closed disc and has boundary modulus one. The maximum principle, applied to that quotient and its reciprocal, makes it a unimodular constant. The product has degree k, the root count of this component, not necessarily the full degree n. Distances between the b_j below are hyperbolic distances in $\mathbb{D}$, normalised by $d(0, s) = 2\,\mathrm{artanh}\, s$ for $0 \leq s < 1$. Intrinsic distances in C_t are infima of Euclidean lengths of curves in that component.

The length estimate converts area into a bound for the image of a compact real interval $I \subset (-1, 1)$. Here is the coefficient calculation underlying the Bergman-kernel argument. Write $\varphi'(z) = \sum_{\ell \geq 0} \alpha_\ell z^\ell$. Injectivity and the area formula give $\sum_{\ell \geq 0} |\alpha_\ell|^2/(\ell+1) = a$. To express length as a linear integral, choose $\omega(x) = \overline{\varphi'(x)}/|\varphi'(x)|$ on I; the denominator is nonzero because φ is conformal. Cauchy–Schwarz gives

$$\mathrm{length}(\varphi(I))^2 = \left| \sum_{\ell \geq 0} \alpha_\ell \int_I \omega(x) x^\ell \, dx \right|^2 \leq a \sum_{\ell \geq 0} (\ell + 1) \left| \int_I \omega(x) x^\ell \, dx \right|^2.$$

Compactness of I permits termwise integration of the series. Since $\sum_{\ell \geq 0} (\ell + 1)(xy)^\ell = (1 - xy)^{-2} > 0$ for real $x, y \in I$, taking absolute values in the resulting double integral yields

$$\mathrm{length}(\varphi(I))^2 \leq a \int_I \int_I \frac{dx\, dy}{(1 - xy)^2}.$$

This is precisely the positivity of the disc Bergman kernel $K(z, w) = \pi^{-1}(1 - z\overline{w})^{-2}$ on the real diameter. For $I = [u, v] \subset (-1, 1)$, direct integration gives

$$\int_u^v \int_u^v \frac{dx\, dy}{(1 - xy)^2} = \log \frac{(1 - uv)^2}{(1 - u^2)(1 - v^2)}.$$

Thus $I = [0, r]$ gives $\mathrm{length}(\varphi([0, r]))^2 \leq a \log(1/(1 - r^2))$. Moving two preimages to $\pm s$ and taking $I = [-s, s]$ instead gives

$$\mathrm{length}(\varphi([-s, s]))^2 \leq 4a\, \mathrm{artanh}(s^2).$$

A pair of roots at hyperbolic distance d may be placed at $\pm \tanh(d/4)$. Their connecting curve then has length less than 2 whenever $\mathrm{artanh}(\tanh^2(d/4)) < 1/a$. Under our assumption that no such curve exists, we therefore have

$$d(b_i, b_j) \;\geq\; D := 4\,\mathrm{artanh}\,\sqrt{\tanh(1/a)}, \qquad \cosh(D/2) = e^{2/a}. \tag{7}$$

There is also a point h in the chosen compact connected set whose intrinsic distance in C_t from every root is at least 1. Otherwise the open intrinsic balls of radius 1 about the roots would cover that set. Two balls with distinct centres cannot meet: paths from their centres to a common point would concatenate to length below 2. But a connected set containing two distinct roots cannot be covered by these disjoint open sets. This contradiction proves the assertion. Now choose the Riemann map with $\varphi(0) = h$, and continue to write $b_j = \varphi^{-1}(\zeta_j)$. Put $d_j = d(0, b_j)$ and $\lambda(d) = -\log\tanh(d/2)$. The one-root estimate at h gives $\tanh(d_j/2) \geq \sqrt{1 - e^{-1/a}}$. The centre h lies on the chosen inverse arcs, so $0 < |f(h)| \leq \mu$; it is not a root because its intrinsic distance from every root is at least 1. The Blaschke identity at 0 gives $|f(h)|/t = \prod_j |b_j|$, hence $\sum_j \lambda(d_j) = \log(t/|f(h)|) \geq \log(t/\mu) = x$. Together these estimates give

$$\lambda(d_j) \leq \frac{\delta(a)}{2}, \quad \delta(a) = -\log(1 - e^{-1/a}), \qquad \sum_{j=1}^{k} \lambda(d_j) \;\geq\; x. \tag{8}$$

A lower bound for the number of roots. Equation (8) already gives $k \geq 2x/\delta(a)$: the required sum is at least x, and each root contributes at most $\delta(a)/2$. Improving this estimate requires the pairwise separation, not just the distance of each root from the origin. Testing only consecutive pairs in angular order loses essential information. For $k = 2m$, put m points at hyperbolic radius d_{low} and m at radius $d_{\text{low}} + D$, alternating in cyclic order. The reverse triangle inequality separates every consecutive pair by at least D, regardless of the angular gaps, whereas

$$\sum_j \lambda(d_j) = m(\lambda(d_{\text{low}}) + \lambda(d_{\text{low}} + D)) \longrightarrow \infty \qquad (m \to \infty).$$

The even number of points is needed for cyclic alternation. These point configurations test only the consecutive-pair constraints; they need not satisfy separation for nonconsecutive pairs or arise from a polynomial component. To use the missing pairwise information, we impose the packing bound on every circle centred at the origin.

Lemma 4.3 (circle-slice packing)**.** *Under* (7)*, for every* $r > 0$*,*

$$\sum_{j=1}^{k} w(d_j, r) \leq \pi, \qquad w(d, r) = \arccos\Big(\operatorname{clamp} \frac{\cosh d \cosh r - \cosh(D/2)}{\sinh d \sinh r}\Big),$$

where clamp *truncates its argument to* $[-1, 1]$*.*

Proof. The open balls $B_j = B_{\text{hyp}}(b_j, D/2)$ are pairwise disjoint, since a common point would force $d(b_i, b_j) < D$. Fix $r > 0$. By the hyperbolic law of cosines the point of the hyperbolic circle of radius r at angle θ lies in B_j exactly when $\cosh d_j \cosh r - \sinh d_j \sinh r \cos(\theta - \theta_j) < \cosh(D/2)$. The set of angles satisfying this inequality has measure $2w(d_j, r)$: clamping gives measure zero for an empty intersection and 2π for a full circle, with tangent endpoint sets of measure zero. The intersections of the balls with this circle are disjoint, so their angular measures sum to at most 2π. □

Remark 4.4 (unformalised application of a bound for the number of roots). Fix radii $r_1, \dots, r_p > 0$ and weights $\sigma_1, \dots, \sigma_p \geq 0$, put $\Sigma = \sum_i \sigma_i$ and

$$U = \sup_{d \geq d_{\mathrm{low}}(a)} \Big[\lambda(d) - \sum_i \sigma_i\, w(d, r_i)\Big], \qquad \lambda(d_{\mathrm{low}}(a)) = \frac{\delta(a)}{2}.$$

If $U > 0$, then failure forces $k \geq (x - \pi\Sigma)/U$.

Lean proves the geometric inequality from separation (7) and the individual and summed bounds (8). It does not derive those hypotheses from the absence of a short polynomial-root path. The latter application is the unformalised part of this remark.

Proof. By (8), Lemma 4.3 and the radius bound,

$$x \leq \sum_j \lambda(d_j) = \sum_j \Big[\lambda(d_j) - \sum_i \sigma_i w(d_j, r_i)\Big] + \sum_i \sigma_i \sum_j w(d_j, r_i) \leq kU + \pi\Sigma. \qquad \square$$

A nonnegative weighted sum of the circle-slice inequalities bounds every admissible configuration at once. The weights give a usable bound only when Σ and the supremum U have rigorous bounds. In particular, checking the expression on a finite grid does not bound the supremum between grid points. Zero weights recover only the individual bound $k \geq 2x/\delta(a)$; nonzero weights incorporate the circle-packing information and can improve the result. A certificate at area $a' \geq a$ also applies at a. Indeed, $d_{\mathrm{low}}(a) \geq d_{\mathrm{low}}(a')$, so the admissible range of d only shrinks. Also $D(a) \geq D(a')$, which enlarges every slice angle $w(d, r_i)$ at fixed d, r_i. Since the weights are nonnegative, the expression defining U can only decrease. Thus the same certified pair (Σ, U) remains valid. The numerical comparison also uses two bounds that do not involve chosen weights. They are useful where the circle-slice certificate is weaker. We derive them here so that every root-count input to the comparison is visible.

Two further root-count bounds. Order the distances as $d_1 \leq \cdots \leq d_k$. The triangle inequality gives $d_j \geq D - d_1$ for $j \geq 2$. The function λ is decreasing and convex, since $\lambda'(d) = -1/\sinh d$ and $\lambda''(d) = \cosh d/\sinh^2 d > 0$. On $[d_{\mathrm{low}}, D/2]$, the convex function $\lambda(d) + (k-1)\lambda(D-d)$ takes its maximum at an endpoint. If $d_1 \geq D/2$, each summand is at most $\lambda(D/2)$ instead. (The definitions give $d_{\mathrm{low}} < D/2$.) Thus

$$x \leq \sum_j \lambda(d_j) \leq \max\left\{\frac{\delta(a)}{2} + (k-1)\lambda(D - d_{\mathrm{low}}),\ k\lambda(D/2)\right\},$$

and consequently

$$k \geq \min\left\{1 + \frac{x - \delta(a)/2}{\lambda(D - d_{\mathrm{low}})},\ \frac{x}{\lambda(D/2)}\right\}.$$

The minimum is essential: the two expressions arise from alternative positions of the nearest root, not from simultaneous restrictions.

For the second bound, integrate over the disjoint hyperbolic balls of radius $D/2$. Put $E = e^{2/a} - 1$, so each ball has hyperbolic area $2\pi E$. At most one contains the origin; its contribution to $\sum_j \lambda(d_j)$ is at most $\delta(a)/2$. On every other ball, $-\log|z|$ is harmonic. To average it, first move the ball's centre to 0 by a disc automorphism. Hyperbolic area is

invariant under this change of coordinates and has radial density $4/(1-|z|^2)^2$ there. The ordinary mean-value property on each centred Euclidean circle therefore gives hyperbolic area mean $\lambda(d_j)$. If the original ball has 0 on its boundary, use increasing smaller concentric balls and monotone convergence.

There are $m = k - 1$ nonexceptional balls if one contains the origin, and $m = k$ otherwise. Their union has area $2\pi mE$. Each superlevel set of $-\log|z|$ is a centred ball, so its intersection with any set of this area has area at most the smaller of the two areas. The centred ball of area $2\pi mE$ attains that upper bound at every superlevel. Integrating these intersection areas over the superlevel parameter therefore bounds the integral over the union by the integral over that centred ball. Its radius R satisfies $\cosh R = 1 + mE$. With $v = \cosh r - 1$, the identity $-\log\tanh(r/2) = \frac{1}{2}\log(1 + 2/v)$ reduces the radial integral to $(2E)^{-1}\int_0^{mE}\log(1 + 2/v)\,dv$. Evaluating it gives

$$\begin{aligned}\frac{1}{2\pi E}\int_{B_{\mathrm{hyp}}(0,R)} -\log|z|\,dA_{\mathrm{hyp}} &= \frac{\frac{mE}{2}\log(1 + 2/(mE)) + \log(1 + mE/2)}{E}\\ &\le \frac{1 + \log(1 + mE/2)}{E},\end{aligned}$$

where the improper integral is finite because $v\log v \to 0$ as $v \downarrow 0$, and the last inequality uses $\log(1+u) \le u$ for $u > 0$. Adding the possible exceptional contribution and inverting each case yields

$$k \ge \min\left\{1 + \frac{2}{E}(e^{E(x-\delta(a)/2)-1} - 1),\ \frac{2}{E}(e^{Ex-1} - 1)\right\}.$$

Again the minimum accounts for the two possible cases. This estimate uses area rearrangement, whereas the preceding one uses only ordered distances; neither replaces the circle-slice inequalities.

The comparison takes the maximum of 2, $2x/\delta(a)$, these two bounds, and the certified circle-slice bounds. Each may first be rounded up because k is an integer. Any of these estimates derived with an upper bound for a remains valid at the actual smaller area: the root configuration still satisfies the weaker separation and radius conditions at that upper bound.

From the root count to an area contradiction. At a direction avoiding the finitely many critical-value arguments, lift the value radius from 0 to $te^{i\theta}$ from each of the k roots in C_t and split each lift at level μ. Put $A_0 = \mathrm{Area}(C_t \cap \{|f| < \mu\})$. This is the sum of the areas of the k one-root components inside C_t, not of all the root components of f; in particular, $0 \le A_0 \le \pi a$. On each of these components f/μ is conformal, since it is a proper map with no critical point onto a simply connected disc. Take $\psi(z) = \sum_{\ell\ge0}\gamma_\ell z^\ell$ to be its inverse, so $f(\psi(z)) = \mu z$ and $\psi(0)$ is the selected root. The radial curves in this coordinate therefore trace exactly the required value-ray lifts. Cauchy–Schwarz in the radial variable and Parseval give

$$\mathrm{mean}_\theta\Big(\int_0^1 |\psi'(re^{i\theta})|\,dr\Big)^2 \le \sum_{\ell\ge1}\frac{\ell^2|\gamma_\ell|^2}{2\ell-1} \le \sum_{\ell\ge1}\ell|\gamma_\ell|^2 = \frac{\mathrm{Area(component)}}{\pi}.$$

First stop the radial integrals short of the unit circle, then pass to the endpoint by monotone convergence. Thus no smoothness of the boundary at level μ is assumed. Summing the k square-root area bounds gives a mean total low lift length at most $\sqrt{kA_0/\pi}$. At a regular intermediate level u, let $A_C(u)$ be the area of $C_t \cap \{|f| < u\}$ and $P_C(u)$ its total boundary length. The argument principle and the coarea formula give

$$P_C(u)^2 \le 2\pi k\, u\, A_C'(u), \tag{9}$$

because $|dz| = u\, d(\arg f)/|f'|$ on the level curve, its total argument variation is $2\pi k$, and Cauchy–Schwarz applies. Integrating $P_C(u)/(2\pi u)$ from μ to t bounds the mean high lift length by $\sqrt{kx(\pi a - A_0)/(2\pi)}$. Cauchy–Schwarz combines the low and high bounds as

$$\sqrt{\frac{kA_0}{\pi}} + \sqrt{\frac{kx(\pi a - A_0)}{2\pi}} \le \sqrt{ka(1 + x/2)}.$$

Choose a direction whose total lift length is at most $\sqrt{ka(x+2)/2}$. Order its k boundary endpoints cyclically and join each adjacent pair using the two lifts and the intervening boundary arc. Each path lies in $\overline{C_t} \subset \{|f| < 1\}$ and, under failure, has length at least 2. Each lift occurs twice in the sum of these lengths, while the boundary occurs once. Therefore

$$2k \le \sqrt{2ka(x+2)} + P_C(t).$$

Now let $t = \mu e^x$ vary, and write $a(x) = \mathrm{Area}(C_{\mu e^x})/\pi$ for the component containing the chosen first pair; its root count k may increase at later mergers. On a regular interval, $a'(x) = tA_C'(t)/\pi$. Applying (9) at the outer level and rearranging therefore gives

$$a'(x) \;\ge\; \frac{1}{2\pi^2}\Big[\, 2\sqrt{k} - \sqrt{2a(x)(x+2)}\,\Big]_+^2 \tag{10}$$

on each regular interval for the component containing the selected pair, using the largest of the root-count bounds just derived. At merger levels this component only gains area, which strengthens the integrated comparison. Pólya's inequality gives $a \le 1$ until the level reaches 1. Thus an integrated lower bound greater than 1 before $t = 1$ contradicts the assumption.

No positive initial area has to be assumed. While $a \le 1$, for $0 < x \le 3/10^5$ the bound $k \ge 2$ alone gives

$$a'(x) \ge \frac{1}{2\pi^2}\left(2\sqrt{2} - \sqrt{2(2 + 3/10^5)}\right)^2 > \frac{1}{30}$$

at regular levels. For example, the last inequality follows using $\pi < 22/7$, $\sqrt{2} > 140/99$ and $\sqrt{2(2+3/10^5)} < 20001/10000$. Integration, including the nonnegative merger jumps, gives $a(3/10^5) > 10^{-6}$. This is the universal starting bound used by the certificate.

The comparison is integrated cell by cell without assuming that the independently certified lower bounds increase with the table index. On a cell $[x_\ell, x_r]$, suppose $a(x_\ell) \ge a_\iota$ and choose a trial upper bound m for the area. Let g be a certified lower bound for the right side of (10) throughout that cell under the trial assumption $a \le m$. If $m < a_\iota + (x_r - x_\ell)g$, then $a(x_r) > m$. Indeed, the contrary assumption $a(x_r) \le m$ would imply

$a \le m$ throughout the cell by monotonicity. Integrating the differential inequality, with the nonnegative merger jumps, would then give $a(x_r) > m$, a contradiction.

Certificate. The comparison gives an upper bound X on the logarithmic time needed for the forced area to exceed 1. It therefore gives a contradiction whenever $\mu e^X < 1$. The stated constant 13/25 is a convenient rational choice with room to spare, not an optimality claim. A program evaluates every accepted inequality in exact rational arithmetic, with directed rounding on the transcendental evaluations. Its fixed rational brackets for π and $\log 2$ have different elementary checks: Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ for the former, and $\log 2 = 2\sum_{j\ge 0} 3^{-(2j+1)}/(2j+1)$ for the latter. Alternating-series bounds and a geometric tail bound, respectively, suffice to verify them. Its full run uses 18 area-table levels, 14 radii in each weighted bound, 126 attempted choices of weights, maximum step 1/400, and the initial lower area 10^{-6} at $x = 3/10^5$. Only accepted rational certificates enter the comparison; its early steps may be shorter than 1/400. The full run certifies

$$X = \frac{635762889599}{1000000000000} < 0.6357629, \qquad \frac{13}{25}e^X < 1,$$

which gives Remark 4.1. The full replay recorded on 29 September 2026 used the immutable program cited above. It accepted all 126 dual certificates and reproduced this value of X; no new replay is claimed for the present editorial revision. The retained certificate data give the exact radii and nonnegative weights, together with their sums and certified upper bounds. The program's Git blob identifier is `e9b4998884575a31ce0d1f6762a210ae2d617103`. That replay also ran the comparison regression and the observer-containing-circle audit. The comparison check is exact; the geometry audit combines an exact full-circle boundary test with floating identity checks. No new per-degree thresholds were computed.

A coarser run gives $X = 664373027131/1000000000000$ and the weaker threshold 51/100. The floating optimiser proposes weights but supplies no proof. The argument uses only the subsequently certified rational bounds on their sum and on the supremum U. Each accepted pair supplies a weight sum Σ and a certified upper bound for the supremum U in Remark 4.4. For orientation, the pairs of the coarser run at $a = 1$, rounded to six decimal places, are $(0.085674, 0.045865)$, $(0.126361, 0.021829)$, $(0.163157, 0.010513)$ and $(0.208928, 0.004318)$, giving $kU + \pi\Sigma$ approximately 0.4526, 0.5716, 0.6808 and 0.7945 at $k = 4, 8, 16, 32$, respectively. These rounded pairs do not themselves certify an inequality; the comparison uses the unrounded rational values and their directed bounds. To bound the supremum over d, the checker subdivides intervals and bounds each term in the direction needed for an upper bound. The function λ is decreasing. For fixed r, the slice angle $w(d, r)$ has no strict interior minimum. Indeed, before clipping the angle to 0 or π, its cosine is

$$H(d) = \frac{\cosh d \cosh r - \cosh(D/2)}{\sinh d \sinh r}, \qquad H'(d) = \frac{\cosh(D/2)\cosh d - \cosh r}{\sinh^2 d \sinh r}.$$

The numerator of H' is increasing. Thus H has at most one interior minimum, and $w = \arccos H$ has at most one interior maximum; clipping preserves the endpoint-minimum

property. On an interval $[d_-, d_+]$, therefore,

$$\lambda(d) - \sum_i \sigma_i w(d, r_i) \le \lambda(d_-) - \sum_i \sigma_i \min\{w(d_-, r_i), w(d_+, r_i)\}.$$

Beyond $d = \max_i r_i + D/2$, all the slice angles vanish, so only the decreasing function λ remains. These are interval bounds, not values sampled on a grid. The closing inequality is independently checkable: with $X = 635762889599/10^{12}$ and $e^X \le \sum_{j\le 14} X^j/j! + (X^{15}/15!)/(1 - X/16)$, rational arithmetic gives $(13/25)e^X < 0.982000386 < 1$. The same majorant also gives

$$\frac{529}{1000} e^X < 0.998996547 < 1.$$

Thus the full certificate in fact gives the path conclusion for $\mu \le 529/1000$, without changing the geometric argument or its hypotheses. This is an arithmetic consequence of the certified stopping time. We retain $13/25$ in the theorem and its scaling corollaries as the simpler constant.

Sources and scope. The analytic chain above is ordinary mathematics. Its general inputs are the Riemann mapping theorem, the Bergman kernel, the argument principle, the coarea formula, and Pólya's area inequality $\mathrm{Area}\{|f| \le t\} \le \pi t^{2/n}$ [1, printed pp. 280–282], [2, Theorem 1]. The exact rational certificate is checked by a separate exact-arithmetic program, not by the Lean kernel. The two additional root-count bounds above are the ordered-distance estimate in the earlier distance comparison and the area-rearrangement estimate in the hyperbolic packing argument. Their derivations above are part of the analytic proof. No independent review of this theorem is recorded. Prior art for the assembled statement is unassessed. No novelty is claimed for the slice inequality, which follows directly from disjointness of the balls and the hyperbolic law of cosines. A research note carries the full analytic argument, the table of certified weighted bounds and the instructions for running the certificate. The stopping-time test requires $\mu < e^{-X}$; the stated $13/25$ theorem and the $529/1000$ consequence are two rational cutoffs within that range. Other theorems apply to some polynomials outside it, so failure of this numerical test does not identify the class left untreated by all the results. Fixed-degree arguments remain stronger at $n = 4$ and $n = 5$, where the corresponding thresholds are $61/100$ and $139/250$; from $n = 6$ on the all-degree constant $13/25$ is the better statement. These threshold arguments do not settle the unrestricted existence of a short curve joining two roots.

Remark 4.5 (unformalised scaled path bound). Every squarefree monic polynomial f of degree $n \ge 2$ has two distinct zeros joined by a rectifiable curve of length less than $(5/2)\mu^{1/n}$ in $\{|f| < (25/13)\mu\}$, where $\mu = \min_{f'(c)=0} |f(c)|$.

Above degree two, Lean assumes `LowCriticalThirteenTwentyFifths`, the criterion in Remark 4.1. It checks the rescaling and the degree-two case without supplying that missing analytic proof.

Proof. For $n \ge 3$, apply Remark 4.1 to $g(z) = s^{-n} f(sz)$ with $s = ((25/13)\mu)^{1/n}$. Its least critical modulus is $13/25$; rescaling gives length less than $2(25/13)^{1/n}\mu^{1/n} < (5/2)\mu^{1/n}$, since $25/13 < (5/4)^3$. For $n = 2$, write $f(z) = (z - h)^2 - d^2$. The two radial segments

through h have total length $2|d| = 2\mu^{1/2}$ and lie in the closed level μ, which is inside the stated open level. □

The next construction retains the root count and capacity of a chosen component, at the cost of a larger universal constant.

5 A path estimate from area and boundary length

We can also average inverse-ray lengths and boundary arcs directly. This ordinary construction keeps both the number of roots and the capacity of the selected component in the estimate. Its formal source assumes the construction in degrees above two, so the resulting path bounds below are unformalised remarks.

The scaled packing criterion, Remark 4.5, has smaller constants: $5/2 < 71/10$ for length and $25/13 < 2$ for the containment level. The advantage of the direct argument is its explicit dependence on component geometry. Its $\mu \le 1/2$ specialisations are already within the range of Remark 4.1. For a monic degree-n polynomial put

$$K_t = \{z : |f(z)| \le t\}, \qquad \mu = \min_{f'(c)=0} |f(c)|, \qquad \rho = \mu^{1/n}.$$

Remark 5.1 (unformalised path bound at level 2μ). For every monic polynomial f of degree $n \ge 2$, two zero occurrences are joined by a possibly degenerate path of length at most

$$\frac{71}{10}\rho$$

inside $K_{2\mu}$. If f is squarefree, their locations are distinct. If $\mu \le 1/2$, the construction may be chosen inside $\{|f| < 1\}$ with length at most 5.7.

In degrees $n \ge 3$ Lean assumes `CFAPathConstruction`, the two-parameter path bound (CF) below. Its averaging construction is proved only in the ordinary argument. The degenerate case, degree two and numerical inequalities have separate formal proofs.

The constant is the rationally certified specialization of a two-parameter bound. If the selected component contains $k \ge 2$ roots, then for every $r \in (0,1)$ and $\lambda > 1$ the proof constructs a path in $K_{\lambda\mu}$ whose length is at most

$$\sqrt{\frac{2}{k}}\left(\frac{\sqrt{2}\,r}{(1-r)^2} + \lambda^{1/n}\left(\sqrt{\log(\lambda/r)} + \frac{\pi}{\sqrt{\log\lambda}}\right)\right)\rho. \tag{CF}$$

For $n \ge 3$, the choice $\lambda = 2$, $r = 3/20$ makes the bracket less than $71/10$: the largest case is $n = 3$, where exact rational bounds give $66517563/9392500 < 71/10$. Degree two has the exact root-segment bound 2ρ.

Proof. A repeated zero gives the constant path, so assume f is squarefree and $\mu > 0$. Degree two is the segment of length 2ρ in K_μ; assume $n \ge 3$. Put $T = \lambda\mu$, let C_T be the component of K_T containing a critical point at level μ, and write $A(\sigma) = \mathrm{Area}(K_\sigma \cap C_T)$. We will first choose a regular level t, then a direction in the value plane, and finally an adjacent pair of roots. This order ensures that the three estimates below apply to the same path.

At a regular level σ, a component C' containing k' roots has $|dz| = \sigma\, d\varphi/|f'|$ on its boundary and total argument variation $2\pi k'$. Cauchy–Schwarz, followed by the coarea formula, gives

$$\mathcal{H}^1(\partial C')^2 \le 2\pi k'\, \sigma A'(\sigma). \tag{CF1}$$

Here A' sums the area derivatives of all the components in C_T, so it bounds the contribution from C'. Pólya's inequality $A(T) \le \pi T^{2/n}$ [2, Theorem 1] and averaging with respect to $d\sigma/\sigma$ on (μ, T) give a regular level t such that $tA'(t) \le \pi T^{2/n}/\log\lambda$. The component C_t containing the chosen first merger has $k \ge 2$ roots and

$$\mathcal{H}^1(\partial C_t) \le \pi\sqrt{2k/\log\lambda}\, T^{1/n}.$$

Averaging over levels avoids estimating the area derivative at the critical level itself.

Choose a direction avoiding all critical-value arguments. Lift its radial segment from each of the k roots to ∂C_t, splitting each lift at value modulus $r\mu$. Below μ, the inverse branches of f/μ are univalent. Their derivatives at 0 are $\mu/f'(z_i)$, where z_i ranges over the roots. The area formula for their disjoint images gives

$$\sum_i \frac{\mu^2}{|f'(z_i)|^2} \le \rho^2. \tag{CF2}$$

Indeed, the squared derivative at the centre is at most the image area divided by π, and the total image area is at most $\pi\mu^{2/n}$. Koebe distortion and Cauchy–Schwarz therefore bound the sum of the lengths below $r\mu$, uniformly in the direction. Write $\ell_i^{\mathrm{high}}(\theta)$ for the length of the ith piece above $r\mu$ in direction θ. At an intermediate regular level, apply (CF1) to all the lower components contained in C_t; their root counts sum to k. Integrating their total perimeter with measure $d\sigma/(2\pi\sigma)$ gives the mean of $\sum_i \ell_i^{\mathrm{high}}$. Cauchy–Schwarz in σ, using an area increment at most $\mathrm{Area}(C_t) \le \pi T^{2/n}$, gives

$$\frac{1}{2\pi}\int_0^{2\pi} \sum_i \ell_i^{\mathrm{high}}(\theta)\, d\theta \le \sqrt{\frac{k\,\mathrm{Area}(C_t)}{2\pi}\log\frac{t}{r\mu}} \le \sqrt{\frac{k}{2}}\, T^{1/n}\sqrt{\log(\lambda/r)}.$$

Choose a direction avoiding the critical-value arguments at which the sum is at most this bound. The excluded directions form a finite set and hence have measure zero.

Order its k endpoints cyclically on ∂C_t. Each adjacent pair of endpoints gives a path consisting of two lifted segments and the intervening boundary arc. In the sum over these k paths, every lifted segment occurs twice and the boundary arcs partition ∂C_t. Thus some pair has length at most the sum of the following three bounds:

$$\frac{2}{k}\sum_i \ell_i^{\mathrm{low}} \le \frac{2r\rho}{\sqrt{k}(1-r)^2},$$

$$\frac{2}{k}\sum_i \ell_i^{\mathrm{high}} \le \sqrt{\frac{2}{k}}\sqrt{\log(\lambda/r)}\, T^{1/n},$$

$$\frac{\mathcal{H}^1(\partial C_t)}{k} \le \sqrt{\frac{2}{k}}\frac{\pi T^{1/n}}{\sqrt{\log\lambda}}.$$

The first holds in every direction, the second in the direction just chosen, and the third at the previously selected level. Substituting $T^{1/n} = \lambda^{1/n}\rho$ proves (CF). For the final assertion, the repeated-zero case is already a constant path, and in degree two the root segment has length $2\rho = 2\sqrt{\mu} \le \sqrt{2}$. For $n \ge 3$, take $\lambda = 2$ and $r = 3/20$. Since $k \ge 2$ and $\mu \le 1/2$, (CF) gives

$$\begin{aligned}\text{length} &\le \frac{60\sqrt{2}}{289}\mu^{1/n} + (2\mu)^{1/n}\left(\sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}}\right)\\ &\le \frac{60\sqrt{2}}{289} + \sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}}\\ &= 5.676476\ldots < 5.7.\end{aligned}$$

Here $2\mu \le 1$, and the mean-value choice may be taken at a regular level strictly below the top of its positive-measure window. Thus the entire path lies in $\{|f| < 1\}$. □

Before replacing the right side of (CF) by the constant $71/10$, we can retain its dependence on the selected component. Its number k of roots occurs in the factor $\sqrt{2/k}$, and its logarithmic capacity bounds the area used in the estimate. Each gives sufficient conditions for the length bound to be less than 2.

The number of roots in the first merged component can be much smaller than the degree. At a first merger caused by one simple critical point it is 2, so the following thresholds do not apply. For $z^n - b$, however, all n roots merge at the same level, giving $k_0 = n$. For example, $n \ge 17$ and $0 < |b| \le 1/2$ satisfy the first pair of hypotheses. These conditions illustrate what the area argument gains from simultaneous mergers; the preceding unformalised $13/25$ criterion already covers these three ranges without the additional lower bound on the number of roots.

Remark 5.2 (unformalised root-count refinement). Let f be monic with every root in the open unit disc, let c_* be a critical point with $|f(c_*)| = \mu$, and let k_0 be the number of roots, counted with multiplicity, in the component of K_μ containing c_*. Then Erdős #1041 holds for f in each of the three cases

$$\mu \le \tfrac{1}{2} \text{ and } k_0 \ge 17, \qquad \mu \le \tfrac{1}{4} \text{ and } k_0 \ge 12, \qquad \mu \le \tfrac{1}{8} \text{ and } k_0 \ge 10.$$

Lean assumes `CFAArityConstruction`, the construction (CF) with the stated lower bound on the component's root count. The formal rational inequalities do not supply this analytic input.

Proof. If $\mu = 0$, a repeated zero gives the constant path. Assume henceforth $\mu > 0$. Every selected component C_t in the proof of Remark 5.1 contains the first-merge component, so $k \ge k_0$. For the first case take $\lambda = 2$, $r = 13/100$; then $(2\mu)^{1/n} \le 1$ and $\rho \le 1$, and the exact bounds $\sqrt{2} < 283/200$, $\sqrt{\log(200/13)} < 5/3$ and $\pi/\sqrt{\log 2} < (22/7)/(104/125) = 1375/364$ make the bracket in (CF) at most $15668813/2755116$, whose square is $34 - 12570881068535/7590664173456 < 34$. Hence $k_0 \ge 17$ gives squared length below $(2/17)\cdot 34 = 4$. For the second case take $\lambda = 4$, $r = 3/25$: the bracket is below $6075221/1273888$, whose square is $24 - 2038665078215/1622790636544 < 24$, and $2/k_0 \le$

$1/6$ gives squared length below 4. For the third take $\lambda = 8$, $r = 11/100$: the bracket is below $55629121/12475575$, whose square is $20 - 18200328379859/155639971580625 < 20$, and $2/k_0 \le 1/5$ again gives squared length below 4. In each case $\lambda\mu \le 1$, so the freedom in the choice of the regular level t keeps $t < 1$ and the containment strict. □

The next criterion uses logarithmic capacity, a measure of the size of a compact set that scales linearly under dilation. It compares the size of one component with that of the entire polynomial sublevel set. If that sublevel set is connected, the ratio κ below is 1; a cutoff such as $\kappa \le 1/3$ therefore requires the component to have smaller capacity than the full sublevel set. A small number of roots in the component does not imply this capacity bound. The ratio requires $\mu > 0$. When $\mu = 0$, f has a repeated zero and the constant path already gives the conclusion; there is no capacity ratio to evaluate.

Remark 5.3 (unformalised capacity refinement). Keep the hypotheses of Remark 5.2 with $0 < \mu \le 1/2$, let C be the component of $\{|f| < 2\mu\}$ containing c_*, and put $\kappa = \operatorname{cap}(\overline{C})/(2\mu)^{1/n}$. If $\kappa \le \tau_{k_0}$, where

$$\tau_k = \frac{\sqrt{2k} - A}{B}, \qquad A = \frac{283}{3610}, \qquad B = \frac{52029}{9100},$$

then Erdős #1041 holds for f. In particular $\kappa \le 1/3$ suffices for every root count $k_0 \ge 2$, and the rational cutoffs $2/5$, $12/25$, $1/2$, $7/12$, $16/25$, $2/3$, $7/10$ suffice at $k_0 = 3, \dots, 9$, rising to $39/40$ at $k_0 = 16$.

Lean assumes `CFACapacityConstruction`, the averaging construction with the component area bounded through its capacity. It checks the numerical cutoffs. The construction from the polynomial hypotheses remains an additional input.

Proof. Repeat the averaging proof of Remark 5.1 with $A(\sigma) = \operatorname{Area}(K_\sigma \cap C)$ for $\sigma < 2\mu$. This keeps the chosen paths in the open component C, even if distinct components have touching boundaries at level 2μ. Replace the global area input by the component form $\operatorname{Area}(C) \le \pi \operatorname{cap}(\overline{C})^2 = \pi\kappa^2(2\mu)^{2/n}$ of the area–capacity inequality [2, Theorem 6]. Only the high-lift and boundary terms acquire the factor κ; the low-lift term (CF2) is unchanged. Taking $\lambda = 2$, $r = 1/20$, and the exact bounds above together with $\log 40 < 12641/3402 < (97/50)^2$, gives length $< \sqrt{2/k_0}\,(A + B\kappa)$. The definition of τ_{k_0} makes the right side at most 2, and for each displayed rational q_k integer arithmetic gives $(A + Bq_k)^2 < 2k$. Discarding $\sqrt{2/k_0} \le 1$ altogether gives the uniform cutoff $\kappa \le 1/3$. □

By the component-capacity formula, $\kappa = e^{-\Sigma/n}$ with Σ the sum of exterior Green function values at the roots excluded from C. Indeed, let Ω be the unbounded component of $\widehat{\mathbb{C}} \setminus \overline{C}$, let G be its Green function with pole at infinity, extended by zero off Ω, and let $g(\cdot, a)$ be its Green function with pole at $a \in \Omega$. The function $\frac{1}{n}\log\frac{|f|}{2\mu} - G + \frac{1}{n}\sum_a g(\cdot, a)$, summed over the roots $a \in \Omega$ with multiplicity, extends harmonically to Ω, is bounded, and vanishes on $\partial\Omega \subseteq \partial C$, so it vanishes identically; its value at infinity is $\log\kappa + \frac{1}{n}\sum_a G(a)$ by the symmetry $g(\infty, a) = G(a)$. Failure of these component-sensitive criteria alone, in the range $\mu \le 1/2$, forces $k_0 \le 16$ and exterior Green sum $\Sigma < n\log(40/39)$ when $k_0 = 16$. This is a limitation of the area-based criteria. The low-critical theorem already excludes an actual counterexample throughout that range.

The proof of Remark 5.1 averages over levels in $[\mu, \lambda\mu]$. To keep the path at the first critical level instead, one would need a perimeter estimate of the following kind. The regular-level argument (CF1) gives no finite pointwise bound at the critical level: even at a simple critical point, the area derivative diverges logarithmically.

The addendum [26, Proposition 2] reports a Runge-approximation construction of monic level-one components with one simple zero and arbitrarily long analytic boundaries. It imposes neither a unit root disc nor a global lower bound on critical-value moduli. Its proof is not available, and the report is not an input to the path estimates here. The question below requires a stronger hypothesis than the existence of one unramified component: the level must lie below every critical-value modulus of the polynomial. A construction without that global condition cannot by itself answer the question.

Problem 5.4 (perimeter below the smallest critical-value modulus)**.** Let f be squarefree and monic of degree $n \geq 2$, and put $\mu = \min_{f'(c)=0} |f(c)|$. Is there a constant β, independent of n, f and $0 < \sigma < \mu$, such that every component C of $\{|f| \leq \sigma\}$ satisfies

$$\mathcal{H}^1(\partial C) \leq \beta \sigma^{1/n}?$$

Equivalently, for each univalent inverse branch defined on $\mathbb{D}$ by $f(\phi_j(w)) = \sigma w$, is

$$\int_0^{2\pi} |\phi_j'(e^{it})|\, dt \leq \beta \sigma^{1/n} \qquad (1 \leq j \leq n)?$$

There is no root-location hypothesis. If such constants exist, write β_* for their infimum.

All these components contain one zero. The condition $\sigma < \mu$ controls all inverse branches at once; it is stronger than requiring only the selected component to contain one zero. The examples below compare particular constants, not the necessity of this global hypothesis for every possible perimeter bound. For $1 < R < \mu/\sigma$, the area estimate gives

$$\frac{1}{2\pi} \int_0^{2\pi} |\phi_j'(e^{it})|\, dt \leq \frac{(\sigma R)^{1/n}}{\sqrt{R^2 - 1}}.$$

Indeed, circular means of $|\phi_j'|^2$ increase, the image of the annulus $1 < |w| < R$ has area at least $\pi(R^2 - 1)$ times the mean at 1, and Pólya's preimage-area bound [2, Theorem 1] bounds that area by $\pi(\sigma R)^{2/n}$. The loss as $R \downarrow 1$ belongs to this argument and does not prove an endpoint divergence.

For $d > 0$ and $f(z) = z^2 - d^2$, one has $\mu = d^2$ and $\rho = d$. For $\sigma < d^2$ the component of $\{|f| \leq \sigma\}$ around d contains one root. Its parametrisations $d\sqrt{1 + (\sigma/d^2)e^{i\theta}}$ converge uniformly, as $\sigma \uparrow d^2$, to one loop of Bernoulli's lemniscate $|z^2 - d^2| = d^2$, of length $d\,\Gamma(1/4)^2/(2\sqrt{\pi})$ by the substitution $\varphi(w) = d\sqrt{1 + w}$ and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. At $\sigma = d^2$ the two loops meet at the critical point, and the closed sublevel set is a single component containing both roots. Lower semicontinuity of length under the stated uniform convergence gives $\beta_* \geq \Gamma(1/4)^2/(2\sqrt{\pi}) = 3.70814935\ldots$; convergence of the lengths themselves is not required. For $z^n - r^n$, polar parametrisation of one limiting loop gives its length divided by r as

$$\frac{2^{1/n}}{n} \int_{-\pi/2}^{\pi/2} (\cos\theta)^{1/n - 1}\, d\theta = \frac{2^{1/n}}{n} \sqrt{\pi}\, \frac{\Gamma(1/(2n))}{\Gamma(1/(2n) + 1/2)}.$$

The evaluation is Euler's beta integral (5.12.1), in its trigonometric form (5.12.2). To verify the claimed decrease with n, put $p = 1/n$. The expression is $2^{p+1}\sqrt{\pi}\,\Gamma(1+p/2)/\Gamma((1+p)/2)$; its logarithmic derivative is

$$\log 2 + \tfrac{1}{2}\left(\frac{\Gamma'(1+p/2)}{\Gamma(1+p/2)} - \frac{\Gamma'((1+p)/2)}{\Gamma((1+p)/2)}\right) > 0.$$

Here Γ'/Γ is increasing on the positive real axis, by the positive trigamma series (5.15.1). The expression therefore decreases with n and tends to 2.

The comparison with the next example also has an elementary bound, independent of rounded gamma values. One loop of $z^2 - d^2$ has parametrisation

$$z(s) = \sqrt{2}\,d\,\frac{\cos s(1 + i\sin s)}{1+\sin^2 s}, \qquad -\pi/2 \le s \le \pi/2,$$

whose speed is $\sqrt{2}\,d/\sqrt{1+\sin^2 s}$. Convexity puts $(1+x)^{-1/2}$ below its secant on $[0,1]$. Integrating that secant at $x = \sin^2 s$ bounds the loop length divided by d by $(\pi/2)(1+\sqrt{2})$, as used below.

Proposition 5.5 (a counterexample to the proposed one-root perimeter bound)**.** *Let $p(z) = z^8 - (3/2)z$ and let C be the component of $\{|p| \le 1\}$ containing the origin. Then C contains exactly one zero and a neighbourhood of the closed disc of radius $5/8$, so $\mathcal{H}^1(\partial C) > 5\pi/4$. The constant $\Gamma(1/4)^2/(2\sqrt{\pi})$ is at most $(\pi/2)(1+\sqrt{2}) < 5\pi/4$.*

The quadratic loop is exceeded in degree eight, even with the global subcritical hypothesis. For $p(z) = z^8 - (3/2)z$, every critical point c satisfies $c^7 = 3/16$ and $p(c) = -(21/16)c$. Hence

$$\mu = \frac{21}{16}\left(\frac{3}{16}\right)^{1/7} > 1, \qquad \mu^7 = \frac{3\cdot 21^7}{16^8} > 1.$$

Thus $\sigma = 1$ is below every critical-value modulus, not merely a regular level for the selected component. The component C of $\{|p| \le 1\}$ containing the origin lies in $\{|z| < 4/5\}$, because $|p(z)| \ge 6/5 - (4/5)^8 > 1$ on $|z| = 4/5$, and it contains a neighbourhood of the closed disc of radius $5/8$, because $|p(z)| \le (5/8)^8 + 15/16 < 1$ on that disc. The other roots satisfy $|z|^7 = 3/2$, so C contains exactly one root, and $\mathcal{H}^1(\partial C) > 5\pi/4 > (\pi/2)(1+\sqrt{2}) \ge \Gamma(1/4)^2/(2\sqrt{\pi})$. Moreover, for fixed $a > 1$ and all sufficiently large N, the one-root component of $\{|z^N - az| \le 1\}$ contains every disc of radius $r < 1/a$, and a circle of radius between $1/a$ and 1 isolates it by Rouché's theorem; letting $a \downarrow 1$ shows that every admissible β is at least 2π. For the latter family all critical values have modulus $(1-1/N)a(a/N)^{1/(N-1)} \to a > 1$, so it also gives the necessary bound $\beta_* \ge 2\pi$ under the corrected global subcritical hypothesis. Kuznetsova–Tkachev [24, Theorems 1–2] establish Laplace-transform and log-convexity properties for regular level-length functions. Those results do not provide a uniform bound at the first critical level. Total lemniscate length and one-branch endpoint length are different extremal quantities.

The preceding examples do not establish a universal perimeter bound. The implication below applies to an individual polynomial; a constant β independent of the polynomial and degree would give a uniform path bound. The construction uses two components meeting at a first critical point and joins each root to that point at a cost of at most half its component's perimeter. It does not give the constant 2 of the historical question.

Remark 5.6 (unformalised consequence of a perimeter bound). Let f be squarefree and monic of degree $n \geq 2$, and put $\mu = \min_{f'(c)=0} |f(c)| > 0$. Suppose $\beta > 0$ satisfies

$$\mathcal{H}^1(\partial C) \leq \beta\sigma^{1/n}$$

for every $0 < \sigma < \mu$ and every component C of $\{|f| \leq \sigma\}$. Then two distinct roots of f are joined inside $K_\mu = \{|f| \leq \mu\}$ by a rectifiable path of length at most $\beta\mu^{1/n}$.

Lean additionally assumes `SubcriticalSplitExists`: two one-root components whose closures meet at the first critical point, with the required boundary lengths and half-perimeter joins. The ordinary argument below constructs this datum; it is not formalised by the perimeter hypothesis alone.

Proof. Put $\rho = \mu^{1/n}$. On each component of $\{|f| < \mu\}$, f is a proper unramified covering of the disc $\{|w| < \mu\}$, hence univalent. Equivalently, exhaust by regular levels below μ and apply the component count [8, proof of Proposition 2.1]. At a critical point c_* with $|f(c_*)| = \mu$, the local degree $d \geq 2$ gives d nearby preimages of an inward radial value. Univalence places them in different one-root components. Choose two such components U_a, U_b, containing roots a, b and with c_* in both closures; simplicity of c_* is unnecessary. For the boundary passage, take $f(\phi(w)) = \mu w$ on $\mathbb{D}$. Fix $|w_0| = 1$. The finitely many preimages of μw_0 have disjoint neighbourhoods. Properness of f and connectedness of a sufficiently small disc cap force ϕ to stay in one of them as $w \to w_0$. At its preimage c, write

$$f(z) - f(c) = (z-c)^d h(z), \qquad h(c) \neq 0.$$

A local analytic dth root of h makes $(z-c)h(z)^{1/d}$ a conformal coordinate. In each inverse sector, ϕ is therefore an analytic function of $(w - w_0)^{1/d}$ and extends continuously to c. The finitely many critical boundary values are the only exceptional points; at every other boundary value the ordinary inverse function theorem applies. Moreover, near $e^{i\theta_0} = w_0$ the boundary speed is $O(|\theta - \theta_0|^{1/d-1})$, which is integrable even when $d > 2$. Thus the extension is continuous on the closed disc and its boundary curve is rectifiable. The identity $f(\phi(e^{i\theta})) = \mu e^{i\theta}$ makes the boundary values injective, so the curve is Jordan.

As $r \uparrow 1$, the curves $\phi(re^{i\theta})$ converge uniformly to that boundary and have length at most $\beta(\mu r)^{1/n}$ by the hypothesis. Lower semicontinuity of length gives boundary length at most $\beta\rho$ for each component.

A bounded rectifiable Jordan domain U has the following elementary property: any $a \in U$ can be joined to any $c \in \partial U$ in $\overline{U}$ with length at most $\mathcal{H}^1(\partial U)/2$. Choose a line through a but not c, and let $[u, v]$ be the closure of the connected interval in its intersection with U containing a. Of the two boundary arcs from u to v, write A for the one containing c and A' for the other. The paths from a through u or v and then along A to c have total length

$$|a - u| + |a - v| + \text{length}(A) = |u - v| + \text{length}(A) \leq \mathcal{H}^1(\partial U),$$

because $|u - v| \le \mathrm{length}(A')$. One path is at most half this sum. The connected interval ensures containment of the straight parts even when U is not convex.

Apply this property in U_a, U_b with common boundary point c_*. Concatenating the two paths gives length at most $\frac{1}{2}(\mathcal{H}^1(\partial U_a)+\mathcal{H}^1(\partial U_b)) \le \beta\rho$, as required. Only these two component perimeters are used; the subcritical hypothesis bounds them by the limiting argument above. Boundary arcs are allowed, so containment is in K_μ, not the open sublevel. □

Sources and scope. The inputs of the construction in the proof of Remark 5.1 are the capacity identity for polynomial preimages [2, §2], Pólya's area inequality [2, Theorem 1], Cauchy–Schwarz, the Koebe distortion theorem for univalent maps, and averaging over adjacent boundary endpoints. The public source of record is the full area and boundary-length argument, which carries every rational verification quoted above. Its Section 10 gives a weaker conditional bound involving an auxiliary parameter and a logarithm; Remark 5.6 states the stronger $\beta\rho$ consequence proved here. That section's proposed numerical perimeter constant is refuted by the degree-eight example above and is not assumed in the theorem. The unconditional construction retains level 2μ and constant $71/10$. The scaled low-critical corollary has both a smaller constant and a smaller level. The proof retained here is ordinary, with component-sensitive consequences on their stated regions; its rational cutoffs are exact integer inequalities. The earlier perimeter samples concerned only a selected component. They do not test the stronger condition that the level lie below every critical value. The existence of a root-to-root curve of length less than 2 is not established for the unrestricted class by the arguments in this record. The adjacent classical literature on lemniscate length concerns the arclength of the level curve $\{|p| = 1\}$, which is Erdős #114. Fryntov and Nazarov recall its history [15, Introduction]: Dolzhenko's bound $4\pi n$, Pommerenke's bound $74n^2$ of 1961, and Borwein's bound $8\pi en$ [13]. Eremenko and Hayman proved $9.173n$ [14, Theorem 1], Fryntov and Nazarov the asymptotically sharp $2n + o(n)$ [15], and Tao resolved the problem for large n [16]. These results bound the length of a level curve, not the least internal root-pair path. This distinction identifies the quantity being estimated; no claim of novelty follows from the scope or outcome of a literature search.

6 Degree three

For a cubic we can estimate the inverse roots directly after normalising at a critical point of least value modulus. The root-disc hypothesis makes that modulus less than one, and two radial segments in the normalised coordinates give a path shorter than 2.

Theorem 6.1 (the cubic case)**.** *Let $f(z) = \prod_{j=1}^{3}(z - z_j)$ with $|z_j| < 1$, the roots listed with multiplicity. Then two listed root occurrences are joined inside $\{|f| < 1\}$ by a polygonal path of length strictly below 2. If f is squarefree the two are distinct.*

Proof. Multiple roots give a constant path, so assume f squarefree. The Erdős–Herzog–Piranian component lemma [4, remark before Problem 5, p. 139] gives a component of $\{|f| < 1\}$ containing two roots. Join them by a compact path in that open component, and choose a regular value modulus $t < 1$ larger than $\max|f|$ on the path. The component

of $\{|f| < t\}$ containing it has $k \geq 2$ roots and hence $k - 1 \geq 1$ critical points, by the component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1]. One of these critical points has value of modulus below $t < 1$. This choice of t avoids assuming that the level $|f| = 1$ is regular.

Suppose first that f' has two distinct zeros. Choose a critical point c minimising $|f(c)|$, write the other one as $c + \delta$, and put $v = f(c)$, so $0 < |v| < 1$. Monicity gives the exact expansion $f(c + d) = d^3 - \frac{3}{2}\delta d^2 + v$. Choose α with $\alpha^3 = v$ and set $b = \delta/\alpha$; dividing by v gives $f(c + \alpha w)/v = P_b(w)$ with

$$P_b(w) = w^3 - \tfrac{3}{2}bw^2 + 1.$$

At the other critical point $f(c + \delta) = v(1 - b^3/2)$, so minimality of $|v|$ is exactly the hypothesis $|1 - b^3/2| \geq 1$.

Under that hypothesis P_b has at least two zeros, with multiplicity, in the closed unit disc. In the strict region a unit-circle zero w would give $b = \frac{2}{3}(w + w^{-2})$. Write $w^3 = e^{2iu}$; substitution gives

$$|1 - b^3/2|^2 = 1 + \frac{64}{729}\cos^4 u\,(16\cos^2 u - 27) \leq 1.$$

Since $16\cos^2 u - 27 < 0$, equality requires $\cos u = 0$, which gives $b = 0$. Thus the strict region has no unit-circle zero. The radial deformation $b \mapsto sb$, $s \geq 1$, stays in the strict region, since $z = b^3/2$ and $|1 - z|^2 > 1$ give $|1 - s^3 z|^2 - 1 = s^3(s^3|z|^2 - 2\Re z) > 0$. For large s, Rouché's theorem on $|w| = 1$ compares P_{sb} with $-\frac{3}{2}sbw^2$, whose modulus $\frac{3}{2}s|b|$ exceeds $2 \geq |w^3 + 1|$, so P_{sb} has exactly two zeros in the open unit disc; no zero crosses the circle along the deformation. For the equality case with $b \neq 0$, the same radial deformation lies in the strict region for every $s > 1$, and continuity of the root multiset allows $s \downarrow 1$. If $b = 0$, then $P_b(w) = w^3 + 1$ and all three roots lie on the unit circle.

If $P_b(w) = 0$ and $0 \leq t \leq 1$, then $P_b(tw) = 1 - t^2 - t^2(1 - t)w^3$, so $|w| \leq 1$ gives $|P_b(tw)| \leq (1 - t^2) + t^2(1 - t)|w|^3 \leq 1 - t^3 \leq 1$. The whole segment from 0 to w therefore lies in the closed unit sublevel set. Selecting two normalized roots w_1, w_2 with $|w_i| \leq 1$, the two segments from c to $c + \alpha w_i$ lie in $\{|f| \leq |v|\} \subset \{|f| < 1\}$ and have combined length at most $2|\alpha| = 2|v|^{1/3} < 2$.

If instead f' has a double zero c, then $f(c + d) = d^3 + v$. With $\alpha^3 = -v$ and $\omega = e^{2\pi i/3}$, its roots are $c + \alpha$, $c + \alpha\omega$, $c + \alpha\omega^2$. Averaging their squared moduli gives $|c|^2 + |\alpha|^2 < 1$, so $|\alpha| < 1$. Any two segments from c to the roots have total length $2|\alpha| < 2$, and along either segment the value has modulus $|v|(1 - t^3) < 1$ away from the root endpoint. □

The unit level need not be regular, even when all roots lie in the open unit disc. For example, $Q(z) = (z - 1)(z + 1)(z + 9/10)$ has $Q(1/3) = -148/135 < -1$. Its minimum v on $[-9/10, 1]$ occurs at an interior critical point and satisfies $v < -1$. Set $r = |v|^{-1/3} < 1$. The monic cubic $r^3 Q(z/r)$ has the three distinct roots $-r, -9r/10, r$ in the open unit disc and a critical value equal to -1. Its unit level is therefore singular, although the cubic path theorem applies.

Sources and scope. Pendyala [11, Theorem 1, pp. 1–3] proves the separate quartic case; no historical priority for the cubic theorem is asserted here.

7 An isolated simple critical value

A disc containing 0 and one simple critical value selects a component of degree two. Taking a square root removes its local branching and gives a coordinate in which the proposed connecting curve is a real segment. The ordinary argument below then bounds its length by area and capacity. Lean assumes this combined analytic construction as `DiscSepBergmanArea`; it does not derive it from the polynomial hypotheses. We therefore record the path criterion and its numerical consequences as unformalised remarks.

The simplicity assumption means that the chosen critical point is a simple zero of the derivative, so two inverse branches meet there. A multiple chosen critical point is excluded; the other critical points need not be simple. Also, separation of roots in the z-plane is not enough: different critical points can have nearly equal values. An exact quartic example below makes this distinction explicit. For a positive example, take $f(z) = z^3 - 3a^2z$ with $0 < a < 1/\sqrt{3}$. Its roots $0, \pm\sqrt{3}a$ lie in the unit disc. At $c = a$ the critical value is $v = -2a^3$, and the other normalised critical value is -1. Thus the disc centred at 1 with radius $4/3$ contains 0 and 1 but excludes the other critical value. Radius 2 is allowed as well: the other critical value then lies on the boundary. The proof must therefore allow a nonregular outer level.

Remark 7.1 (unformalised isolated-value criterion). Let P be a polynomial of degree $n \geq 3$ whose leading coefficient has modulus one, with

$$P(0) = 1, \qquad P'(0) = 0, \qquad P''(0) \neq 0.$$

Fix $w_0 \in [0,1]$ and $S > \max(w_0, 1 - w_0)$. Suppose every other critical point $d \neq 0$ satisfies

$$|P(d) - w_0| \geq S. \tag{4}$$

Put $p = w_0(1-w_0)$. The two local solutions of $P(Z(\xi)) = 1-\xi^2$, $Z(0) = 0$, continue along the real segment to one injective curve Γ joining two roots. Its endpoints are distinct roots, $\Gamma \subseteq \{|P| \leq 1\}$, and

$$\operatorname{length}(\Gamma)^2 \leq 2\Big(\frac{S}{n-1}\Big)^{2/n} \log\frac{S^2+S+p}{S^2-S+p}. \tag{5}$$

Consequently this curve has length less than 2 whenever

$$\Big(\frac{S}{n-1}\Big)^{2/n} \log\frac{S^2+S+p}{S^2-S+p} < 2. \tag{6}$$

Lean assumes `DiscSepBergmanArea`, which includes the square-root connector, its length estimate and the component-area bound. This input includes the combined analytic construction below; a published Bergman inequality alone does not supply it. Its derivation from the stated polynomial hypotheses remains unformalised.

Proof. Let $Q = D(w_0, S)$ and let U be the component of $P^{-1}(Q)$ containing 0. Both 0 and 1 lie in Q, and (4) says that 0 is the only critical point in U. A component of a polynomial sublevel set is simply connected: on any Jordan curve in the component, the maximum principle bounds the same polynomial throughout its interior, which must therefore

remain in the component. Condition (4) permits critical values on ∂Q, so we must not assume that ∂U is regular. For $1-w_0 < S' < S$, let U' be the component of $\{|P-w_0| < S'\}$ containing 0. This level is regular, and 0 contributes its only unit of ramification. The component count in the proof of [8, Proposition 2.1] therefore gives $\deg(P|_{U'}) = 2$.

These nested components exhaust U. Indeed, a point of U can be joined to 0 by a compact path in U; the maximum of $|P-w_0|$ on that path is strictly below S, so the path eventually lies in U'. For any fixed regular value in Q, all its finitely many preimages in U therefore lie in one such U' once S' is sufficiently large. The proper map $P : U \to Q$ consequently also has degree two. This establishes the count without a smoothness assumption at the outer level.

Set $a = 1 - w_0$. On U, the function $1 - P$ has only its double zero at 0, so it has a single-valued square root

$$\xi(z)^2 = 1 - P(z).$$

Taking this square root removes the double branching at 0. The map ξ is proper into

$$\widetilde{Q} = \{\xi : |\xi^2 - a| < S\}.$$

Away from 0, the identity $2\xi\xi' = -P'$ gives $\xi' \neq 0$; at 0, $(\xi'(0))^2 = -P''(0)/2 \neq 0$. The target is star-shaped about 0: because $S > a$, the convex disc $D(a, S)$ contains 0, so $t^2\xi^2 \in D(a, S)$ whenever $\xi \in \widetilde{Q}$ and $0 \leq t \leq 1$. It is therefore connected and simply connected. A proper local biholomorphism has open and closed image, hence is a covering of this target. Since U is connected, it is a conformal bijection. Its inverse Z continues both local inverse branches through the critical point. Since $[-1, 1] \subset \widetilde{Q}$, the curve $Z([-1, 1])$ joins the two distinct points over $P = 0$, and $P(Z(\xi)) = 1 - \xi^2 \in [0, 1]$ gives its containment.

We next map the parameter domain to the unit disc so that the curve becomes a straight interval. This allows its length to be estimated by the area of U, without replacing the domain by a larger disc. Squaring sends $\widetilde{Q}$ onto $D(a, S)$. To map this disc to $\mathbb{D}$ while fixing 0, compose $w \mapsto (w - a)/S$ with the disc automorphism taking $-a/S$ to 0. The result is the Möbius map

$$w \longmapsto \frac{Sw}{S^2 + aw - a^2}.$$

Taking its square root after substituting $w = \xi^2$ gives the conformal bijection

$$\zeta(\xi) = \xi\sqrt{\frac{S}{S^2 + a\xi^2 - a^2}} : \widetilde{Q} \longrightarrow \mathbb{D},$$

where the square-root factor is chosen positive at 0. It sends $[-1, 1]$ to $[-q, q]$, with

$$q^2 = \frac{S}{S^2 + p}, \qquad S^2 - S + p = (S - w_0)(S - (1 - w_0)) > 0.$$

The last inequality is exactly where $S > \max(w_0, 1 - w_0)$ ensures $0 < q < 1$, as required by the length estimate below. For $\Phi = Z \circ \zeta^{-1} : \mathbb{D} \to U$, the Bergman segment inequality,

proved by the kernel estimate of Section 4 with the double integral taken over $[-q,q]^2$, gives

$$\begin{aligned}\operatorname{length}(\Gamma)^2 &\le \frac{2}{\pi}\log\frac{1+q^2}{1-q^2}\operatorname{Area}(U)\\ &= \frac{2}{\pi}\log\frac{S^2+S+p}{S^2-S+p}\operatorname{Area}(U).\end{aligned}\tag{7}$$

It remains to bound the area of this component. The factor $n-1$ comes from the roots outside it, rather than from Pólya's inequality alone. The auxiliary capacity estimate is as follows: for a monic degree-n polynomial F, a regular value $t>0$ of $|F|$, and a component V of $\{|F|<t\}$ containing $k<n$ zeros counted with multiplicity,

$$\frac{\operatorname{cap}(\overline{V})^n}{t} < \frac{k}{2n-k}.$$

Multiplying F by a unimodular constant leaves the estimate unchanged. This is a separate component estimate, not part of the Riemann–Hurwitz statement cited above. We give the harmonic-measure argument behind it, following Theorem 2 and Corollary 3 of the component-capacity proof.

Let $m=n-k$ and let ψ map $|\zeta|>1$ conformally onto the exterior of $\overline{V}$, with positive leading coefficient $\operatorname{cap}(\overline{V})$. The m excluded roots have preimages ξ_j with $|\xi_j|>1$. Put $b_j=1/\overline{\xi_j}$ and form

$$B(z)=\prod_{j=1}^{m}\frac{z-b_j}{1-\overline{b_j}z}.$$

Reflection in the unit circle shows that $F(\psi(\zeta))/t$ equals $\zeta^n\overline{B(1/\overline{\zeta})}$ up to a constant of modulus one: their zeros and pole at infinity agree, and both have modulus one on the boundary. Comparing leading coefficients gives

$$|B(0)|=\frac{\operatorname{cap}(\overline{V})^n}{t}.$$

Regularity of the level permits analytic continuation of ψ across the boundary. On the unit circle,

$$\frac{d}{d\theta}\arg F(\psi(e^{i\theta}))=n-|B'(e^{i\theta})|>0.$$

Positivity follows because a regular polynomial level is mapped locally in the positive boundary direction. Thus $|B'|<n$ there.

The boundary-fibre identity used next is Lemma 1 of the same component-capacity proof. Here is its harmonic-measure argument. For a continuous real function h on the unit circle, let H be its harmonic extension to the disc. Since $H\circ B$ is harmonic, the mean value property and the Poisson formula give

$$\frac{1}{2\pi}\int_0^{2\pi}h(B(e^{i\theta}))\,d\theta=H(B(0))=\frac{1}{2\pi}\int_0^{2\pi}h(e^{i\phi})\frac{1-|B(0)|^2}{|e^{i\phi}-B(0)|^2}\,d\phi.$$

Differentiating the factors gives the positive angular derivative

$$\frac{d}{d\theta}\arg B(e^{i\theta}) = |B'(e^{i\theta})| = \sum_{j=1}^{m}\frac{1-|b_j|^2}{|e^{i\theta}-b_j|^2} > 0.$$

Changing variables through the m boundary inverse branches in the first integral and comparing the continuous densities therefore yields

$$\sum_{B(\zeta)=w}\frac{1}{|B'(\zeta)|} = \frac{1-|B(0)|^2}{|w-B(0)|^2} \qquad (|w|=1).$$

In particular each boundary fibre consists of m distinct points. The Poisson kernel on the right is smallest at the boundary point opposite $B(0)$. Choosing $w = -B(0)/|B(0)|$ therefore yields

$$\frac{m}{n} < \frac{1-|B(0)|}{1+|B(0)|}, \qquad |B(0)| < \frac{n-m}{n+m} = \frac{k}{2n-k}.$$

Here $B(0) \neq 0$ because every excluded-root preimage is finite. This proves the component estimate and explains how the excluded roots enter it.

For the regular inner components U' above, the degree-two count gives exactly two zeros of $P - w_0$, counted with multiplicity. Since $2 < n$, the preceding estimate applies with $k = 2$ and gives

$$\frac{\operatorname{cap}(\overline{U'})^n}{S'} < \frac{k}{2n-k} = \frac{1}{n-1}.$$

Pólya's area–capacity inequality $\operatorname{Area}(K) \leq \pi \operatorname{cap}(K)^2$ [2, Theorem 6] then yields $\operatorname{Area}(U') < \pi(S'/(n-1))^{2/n}$. The exhaustion proved at the start and continuity of area from below give, without assuming regularity of the limiting boundary,

$$\operatorname{Area}(U) \leq \pi\Big(\frac{S}{n-1}\Big)^{2/n}.$$

Substitution into (7) proves (5), and (6) makes its right side strictly less than 4. □

Remark 7.2 (unformalised numerical consequences). Inequality (6) holds for every $n \geq 3$, every $w_0 \in [0,1]$, and every $4/3 \leq S \leq 2$. Thus, if f is monic with roots in the open unit disc, c is a simple critical point with $v = f(c) \neq 0$ and $|v| < 1$, and

$$\left|\frac{f(d)}{v} - w_0\right| \geq \frac{4}{3}$$

for every other critical point d, then two roots of f are joined inside $\{|f| < 1\}$ by a curve of length strictly below 2.

Lean checks the displayed coefficient estimates and numerical steps. Their path conclusions still require `DiscSepBergmanArea`, the unformalised construction used in Remark 7.1.

Proof. For $S \geq 4/3$ and $p \geq 0$,

$$\frac{S^2+S+p}{S^2-S+p} \leq \frac{S+1}{S-1} \leq 7.$$

Also $S/(n-1) \le 1$ when $S \le 2$ and $n \ge 3$, while $\log 7 < 2$. This proves (6). For the unnormalised polynomial, set $z = c+|v|^{1/n}w$. Then $P(w) = f(z)/v$ has leading coefficient $|v|/v$, of modulus one. Scaling the connector multiplies its length by $|v|^{1/n} < 1$ and takes $\{|P| \le 1\}$ to $\{|f| \le |v|\} \subset \{|f| < 1\}$. The selected value v need not attain the minimum defining μ. □

Taking $w_0 = 1$ and $S = 2$ gives the earlier criterion $|1 - f(d)/v| \ge 2$ in every degree $n \ge 3$. If c attains the minimum modulus among *all* critical values and $v \ne 0$, the Fekete–resultant bound gives $|v| \le R^n < 1$ when all roots lie in a disc of radius $R < 1$. Deleting zero critical values before taking the minimum would invalidate that inference. For example, $(z-R)^2(z+R)$ with $0 < R < 1$ has critical values 0 and $32R^3/27$, so its least nonzero critical-value modulus exceeds R^3. The repeated root already gives a constant path, but it does not give this nonzero value the required bound. For a critical point chosen in another way, $|v| < 1$ must remain an explicit hypothesis. In degree three the choice $w_0 = 1$ already works at $S = 6/5$, since

$$\left(\frac{3}{5}\right)^{2/3} \log 11 < 2.$$

Lean checks this sharper cubic constant.

Remark 7.1 imposes separation in the value plane. Separation of the roots or critical points alone does not imply its hypothesis. To distinguish these conditions, consider the family

$$g_\varepsilon(z) = z^4 - \varepsilon z^3 - 2z^2 + 3\varepsilon z + \tfrac{1}{2}, \qquad 0 < \varepsilon \le \tfrac{1}{32}.$$

Its derivative and two critical values are

$$g_\varepsilon'(z) = (z^2-1)(4z - 3\varepsilon), \qquad g_\varepsilon(\pm 1) = -\tfrac{1}{2} \pm 2\varepsilon.$$

At the endpoints of each of the intervals $(-3/2, -5/4)$, $(-3/4, -1/2)$, $(1/2, 3/4)$ and $(5/4, 3/2)$, the values of g_ε have opposite signs throughout this parameter range. Each interval therefore contains exactly one root. Together with the critical points $-1, 3\varepsilon/4, 1$, these give seven points at pairwise distances greater than $1/4$, uniformly in ε. Nevertheless the two displayed critical values differ by $4\varepsilon \to 0$. The earlier numerical choice is the exact member $\varepsilon = 1/262144$:

$$g(z) = z^4 - \frac{z^3}{262144} - 2z^2 + \frac{3z}{262144} + \frac{1}{2}, \qquad g'(z) = 4(z+1)(z - 3/1048576)(z-1),$$

for which that difference is $1/65536$. The monic polynomial $2^{-4}g_\varepsilon(2z)$ puts every root inside $|z| < 3/4$ and retains a spatial separation greater than $1/8$. For the critical points $\pm 1/2$ of the scaled polynomial, the ratio of the two values is unchanged and obeys

$$\frac{g_\varepsilon(-1)}{g_\varepsilon(1)} = \frac{1+4\varepsilon}{1-4\varepsilon} \le \frac{9}{7} < \frac{4}{3}.$$

The reciprocal ratio lies between 0 and 1. Hence, for any centre $w_0 \in [0,1]$, neither selected value satisfies the theorem's radius-4/3 isolation condition: the other normalised value is less than $4/3$ away. This does not exclude a suitable isolated value at a *different* critical point.

Sources and scope. The two-sheeted component, its square-root uniformisation, the Bergman segment inequality, the component-capacity estimate, Pólya's area–capacity inequality and the exhaustion are ordinary arguments. They are collected in the proof using an isolated critical value, whose capacity step is cited separately above.

The proof requires both a simple critical point and a disc centred on $[0,1]$ that isolates its normalised value. For example, the trinomial $z^n - b$ with $n > 2$ and $0 < |b| < 1$ has a short path by Theorem 3.1, but its only critical point is not simple, so this method does not apply. The restriction is on the chosen critical point and on the distance of the other critical values from the chosen disc. The other critical points may be multiple, and their values need not be separated from one another. No sharpness is claimed for $S = 4/3$, and no assertion is made that every polynomial has a suitable isolated simple value. In particular, this is not a proof of the unrestricted assertion of Erdős #1041.

A related area comparison is Dubinin's Theorem 1. The source is [12], Theorem 1 on printed page 85 of the POMI original. It concerns a holomorphic function that gives a full n-fold covering of an annulus $t_1 < |w| < t_2$, and, in the notation of that theorem, with E its explicitly defined complementary set, it states

$$\Big(\frac{t_2}{t_1}\Big)^{2/n} \le \frac{m(E \cup D)}{m(E)}.$$

The full covering and the explicit complementary set are hypotheses of the theorem. Tao cited it on the Erdős Problem #1041 discussion page on 25 March 2026 for the relative scaling factor $s^{2/n}$ between the areas of two nested sublevel sets, which is the reading of the displayed inequality in which E and $E \cup D$ are those two sublevel sets and the covering hypothesis holds. No step of this note uses that relative inequality. The disk-family theorem instead uses Pólya's absolute area–capacity inequality on a component containing fewer than n roots, after the component-capacity estimate supplies its strict capacity gap. The absolute sublevel inequality $\mathrm{Area}\{|P| < T\} \le \pi T^{2/n}$ remains the area input to Remark 4.1 and to Remark 5.1. Dubinin's theorem is a neighbouring result under a full covering hypothesis and supplies no curve joining two roots.

8 Examples separating the geometric hypotheses

The two critical-value hypotheses leave different restrictions on a polynomial. We first give a cubic satisfying neither of them. A second cubic shows why the root count at the first merger cannot determine capacity at a later level.

Proposition 8.1 (failure of two critical-value criteria to cover all polynomials)**.** *Let* $f(z) = z^3 + (3/100)z - 3/4$. *Every root lies in the open unit disc, both critical points are simple, the critical values lie on distinct positive rays,* $\mu > 13/25$, *and*

$$|1 - f(c_-)/f(c_+)| < 2/375 < 2.$$

Proof. On $|z| = 1$, $|3z/100 - 3/4| \le 78/100 < 1$, so Rouché's theorem puts all three roots in the disc. The critical points are $c_\pm = \pm i/10$, and $f(c_\pm) = -3/4 \pm i/500$. They are nonzero, have distinct arguments and modulus greater than $3/4$. Moreover

$$\left|1 - \frac{f(c_-)}{f(c_+)}\right| = \frac{1/250}{|f(c_+)|} < \frac{2}{375}.$$

For any $w_0 \in [0,1]$, the other normalised value is at distance less than $1 + 2/375 < 4/3$ from w_0, so the radius-4/3 hypothesis fails. □

A component with two roots immediately after the first merger may contain more roots at a later level. Thus its first root count alone does not give a capacity saving at level 2μ.

Proposition 8.2 (root count does not force a capacity gap)**.** *Let $g(z) = z^3 - (3/400)z - 3/32$. All roots lie in the open unit disc and $\mu = 187/2000 \le 1/2$. The first merger joins two root components, so $k_0 = 2$, but the component at level 2μ containing that pair has normalised capacity* 1.

The capacity clause uses the classical monic-lemniscate formula $\operatorname{cap}\{|p| \le t\} = t^{1/n}$ [2, §2], [28, Theorem 5.2.5], with capacity equal to transfinite diameter by Fekete–Szegő [28, p. 153]. This is the named input `LemniscateTransfiniteDiameter` in Lean; the root count $k_0 = 2$ is proved without it.

Proof. Rouché's theorem on the unit circle applies because $3/400 + 3/32 < 1$. The critical points are $\pm 1/20$, with critical values $-187/2000$ and $-188/2000$. The first merger therefore has one simple critical point and joins two root components. At level $2\mu = 374/2000$, both critical points are inside the filled lemniscate. The component count $n - (n-1) = 1$ from Riemann–Hurwitz shows that it is connected. Its capacity is $(2\mu)^{1/3}$ [28, Theorem 5.2.5], so division by that scale gives one. The use of logarithmic capacity agrees with transfinite diameter by the Fekete–Szegő theorem [28, p. 153]. □

The degree-eight perimeter example, Proposition 5.5, requires no root-disc hypothesis. It refutes the proposed Gamma constant, while the existence of a uniform one-root perimeter bound remains open in Problem 5.4. The Runge construction reported in [26, Proposition 2] has no supplied proof and is not used.

9 Collinear roots and two sparse polynomial families

The next constructions use the roots or coefficients directly. For collinear roots we compare with a rescaled Chebyshev polynomial. In a quintic with two missing coefficients, the resulting moment identities select two radial segments. A cubic composed with a power map provides a different use of a single contained segment: its rotational copies join through the centre of a fibre.

9.1 Collinear roots and Chebyshev comparison

We scale the extreme zeros of T_n to ± 1 and make the polynomial monic. Its endpoint zeros then agree with those of the polynomial under study, while subtraction lowers the degree. This is the comparison needed for the alternation argument. Put

$$r_n = \cos\frac{\pi}{2n}, \qquad C_n = \frac{1}{2^{n-1} r_n^n}.$$

Theorem 9.1 (Chebyshev comparison)**.** *Let $m \geq 0$, let $p \in \mathbb{R}[X]$ be monic of degree $m+2$, and let*

$$-1 < c_0 < \cdots < c_m < 1, \qquad |c_i| \leq 1.$$

Suppose $p(-1) = p(1) = 0$ and $p(c_i)p(c_{i+1}) < 0$ for $0 \leq i < m$. Then

$$\min_{0 \leq i \leq m} |p(c_i)| \leq C_{m+2}.$$

This theorem is the finite alternation inequality; it does not construct the segment between two roots.

All real-rooted polynomials, and their images under rotations and translations, satisfy the collinearity hypothesis. Three noncollinear roots already fall outside it. The gain is an explicit sharp bound for the modulus along one adjacent-root segment, not a bound obtained by assuming that segment is contained.

Theorem 9.2 (a sharp bound for collinear roots)**.** *Let f be a monic polynomial of degree $n \geq 2$ whose zero occurrences are collinear, and let D be their diameter. Some two adjacent zero occurrences are joined by a segment of length at most D on which*

$$|f(z)| \leq \frac{(D/2)^n}{2^{n-1}\cos^n(\pi/(2n))}. \tag{9}$$

The constant in (9) is best possible in every degree. Equality is attained by affine images of the zeros of T_n whose extreme zeros have distance D.

Proof. A repeated zero gives the constant path, so assume the zero values are distinct. Let m be the midpoint of the two extreme zeros, and choose $\theta \in \mathbb{R}$ so that their line is $m + e^{i\theta}\mathbb{R}$. Put $R = D/2$. Translation by m, rotation by $e^{i\theta}$ and scaling by R give the polynomial

$$q(w) = R^{-n}e^{-in\theta}f(m + Re^{i\theta}w),$$

which is monic with real zeros $-1 = y_1 < \cdots < y_n = 1$, and $|f(m + Re^{i\theta}w)| = R^n|q(w)|$.

Compare q with the monic endpoint-normalized Chebyshev polynomial

$$q_*(x) = \frac{T_n(r_n x)}{2^{n-1}r_n^n}.$$

Both q and q_* vanish at ± 1, and $|q_*| \leq C_n$ on $[-1,1]$. For each gap $[y_i, y_{i+1}]$, choose c_i at which $|q|$ is maximal. The maximum lies in the interior: q vanishes at the endpoints and nowhere inside the gap. Since every root is simple, the signs at the $n-1$ chosen points alternate. If every gap maximum were larger than C_n, then $q - q_*$ would have the same nonzero signs there. The intermediate value theorem would give $n-2$ zeros between consecutive c_i, in addition to the two zeros at ± 1. These n zeros are distinct, but $q - q_*$ is nonzero and has degree at most $n-1$ because its leading terms cancel. This contradiction selects a gap on which $|q| \leq C_n$. Scaling back proves (9).

For sharpness, take

$$y_k = \frac{\cos((2k-1)\pi/(2n))}{r_n}, \qquad 1 \leq k \leq n.$$

These are the zeros of q_*, have extremes ± 1, and every adjacent gap contains a scaled Chebyshev extremum where $|q_*| = C_n$. No smaller universal constant can work. □

Corollary 9.3 (collinear Erdős case)**.** *If the zero occurrences of a monic polynomial of degree $n \geq 2$ lie on one line in the open unit disc, two of them are joined by a curve of length strictly below* 2 *inside* $\{|f| < 1\}$.

Proof. A repeated zero gives the constant path, so assume the zeros are distinct. Their diameter satisfies $0 < D < 2$. Since $\cos(\pi/(2n)) \geq 1/\sqrt{2}$, the defining formula gives $C_n \leq 2^{1-n/2} \leq 1$, with equality possible only at $n = 2$. Thus (9) is strictly below one, and the selected segment has length at most $D < 2$. □

Corollary 9.3 already follows from a theorem of Erdős, Herzog and Piranian [4, Theorem 1, p. 126]: if the zeros of a monic polynomial of degree n are real, lie in $[-1, 1]$ and have centroid in $[0, 1]$, then $\{|f| < 1\} \cap \mathbb{R}$ contains an interval holding at least $n/2$ of the zeros. Apply it to the normalized polynomial q, replacing it by the monic polynomial $(-1)^n q(-w)$ if its centroid is negative. This reverses the root line without changing modulus bounds. For $n \geq 3$ two consecutive zeros in that interval are joined by a segment of length at most D inside $\{|f| < (D/2)^n\}$, and for $n = 2$ the segment between the zeros lies in $\{|f| \leq (D/2)^2\}$. Theorem 9.2 adds the sharp level and its equality configurations.

Critical sequences and the extremal configuration. For distinct real zeros, each gap between consecutive zeros contains exactly one zero of q', and $|q|$ attains its maximum over the gap only there. The points c_i chosen in the proof of Theorem 9.2 are therefore the critical points of q, and the gap maxima are the moduli of its ordered critical sequence $(q(c_1), \dots, q(c_{n-1}))$, with $c_1 < \cdots < c_{n-1}$. In this distinct-root setting, the ordered sequence determines the real polynomial up to an increasing real affine change of variable [17, Theorem 1]. The proof above uses only the alternation count, not this classification theorem. In this language, (9) says that a real polynomial of degree n with n distinct real zeros, leading coefficient a and zero diameter D has a critical value of modulus at most $C_n|a|(D/2)^n$.

The comparison polynomial q_* is the monic polynomial of least deviation from zero on $[-1/r_n, 1/r_n]$, characterised by the equioscillation conditions recalled in [17, §1]. All $n - 1$ of its critical values have modulus C_n, so every critical point of q_* lies on the level curve $\{|q_*| = C_n\}$. Extremals with this property also occur in the level-curve length problem, where some extremal polynomial has all its critical points on $\{|p| = 1\}$ [14, Lemma 5], and in the sharp bound for $|f'|$ on a connected sublevel set $\{|f| \leq 1\}$ of a monic polynomial, proved by Eremenko and Lempert [18, Theorem 1], whose equality cases are $e^{-in\theta}T_n(2^{1/n-1}e^{i\theta}z + b)$ with θ real [19, Theorem A].

Scope. The sharpness assertion concerns the maximum of $|f|$ on the selected adjacent-root segment, and the displayed configurations attain it; no claim is made that they exhaust the equality cases, and no claim is made that the selected segment is a shortest path in $\{|f| < 1\}$.

9.2 Quintics with two missing coefficients

The missing cubic and quadratic terms of $z^5 + az^4 + bz + c$ give three identities among the real and radial coordinates of its roots. We first use those identities to select two indices

whose radial segments will be contained. For $0 < r < 2$ and $0 \le s_i \le 1$, $x_i^2 \le s_i$, put

$$E_i = s_i^4(s_i + r^2 + 2rx_i).$$

Theorem 9.4 (a consequence of three moment identities)**.** *Suppose*

$$\sum_{i=0}^{4} x_i = -r, \qquad \sum_{i=0}^{4} (2x_i^2 - s_i) = r^2, \qquad \sum_{i=0}^{4} (4x_i^3 - 3s_i x_i) = -r^3. \tag{10}$$

Then at least one of the ten pairs $0 \le i < j \le 4$ satisfies $E_i < 1$ and $E_j < 1$.

The theorem selects two distinct indices. It does not assert that the corresponding complex root values are distinct.

The polynomial must have no z^3 or z^2 term in the coordinates used here. For example, $z^5 + bz + c$ is included, but a general quintic with all coefficients nonzero is not. The root-location assumption alone does not imply the three moment identities. Those identities are precisely what lets the finite inequality find two radial segments.

Theorem 9.5 (a quintic with two missing coefficients)**.** *Let*

$$p(z) = z^5 + az^4 + bz + c$$

and suppose its five zero occurrences $w_0, \dots, w_4$ lie in the closed unit disc. At least two distinct indices satisfy

$$|bw_i + c| \le 1. \tag{11}$$

If $a \neq 0$, two indices can be chosen with strict inequalities. If $a = 0$, every index satisfies (11), and equality holds exactly when $|w_i| = 1$.

For open-disc zeros, two zero occurrences are joined inside $\{|p| < 1\}$ by a curve of length below 2: use the two radial spokes through 0 when their values are distinct, and the constant path when the selected occurrences have the same value.

Proof. If $a = 0$, the root equation gives $|bw_i + c| = |w_i|^5$ for every index, proving (11) and its equality clause. For $a \neq 0$, write $a = re^{i\phi}$ with $r > 0$ and set $z_i = e^{-i\phi} w_i$. The missing z^3 and z^2 coefficients and Newton's identities give

$$\sum z_i = -r, \qquad \sum z_i^2 = r^2, \qquad \sum z_i^3 = -r^3. \tag{12}$$

At a zero,

$$|bw_i + c| = |w_i|^4 |w_i + a|.$$

Thus, for $x_i = \Re z_i$ and $s_i = |z_i|^2$, the squared modulus $|bw_i + c|^2$ is exactly E_i and (12) becomes (10). The third moment gives $r^3 = |\sum_i z_i^3| \le 5 < 8$, so $0 < r < 2$ as required by the finite inequality.

We need to rule out four indices with squared modulus at least 1. The three known power sums determine the sum of any real harmonic polynomial of degree at most three over the roots. On the unit circle, $s^4(s + r^2 + 2rx) \ge 1$ reduces to $x \ge -r/2$. We therefore

use the harmonic extension of $(-r/2-x)(1-x)^2$, which is nonpositive on that part of the circle. In the variables $x=\Re z$ and $s=|z|^2$, this extension is

$$H_r(x,s)=(-r/2-x)(1-x)^2+(1-s)\left(1-\frac{r}{4}-\frac{3x}{4}\right).$$

The useful feature of this choice is the expansion

$$H_r(\Re z,|z|^2)=1-\frac{3r}{4}+\Re\left[\left(r-\frac{7}{4}\right)z+\left(1-\frac{r}{4}\right)z^2-\frac{z^3}{4}\right].$$

It is harmonic, and summing this expression with the three moments in (*12*) gives

$$\sum_{i=0}^{4}H_r(x_i,s_i)=5-2r. \tag{13}$$

On the unit circle its value is $(-r/2-x)(1-x)^2$, and

$$4-2r-(-r/2-x)(1-x)^2=(x+1)\left(\frac{(2-r)(3-x)}{2}+(1-x)^2\right)\geq 0.$$

The maximum principle therefore gives $H_r\leq 4-2r$ throughout the disc. For an index with $s^4(s+r^2+2rx)\geq 1$, one has $s>0$ and $s^{-4}\geq 1+4(1-s)$ by convexity on $(0,1]$. Consequently

$$2r(x+r/2)\geq s^{-4}-s\geq 5(1-s).$$

Put $d=x+r/2$, $u=1-x$, $A=1-r/4-3x/4$, and $P=-u^2+(2r/5)A$. The assumption $s^4(s+r^2+2rx)\geq 1$ gives $0\leq d<2$. Since $H_r=-du^2+(1-s)A$, it is nonpositive when $A\leq 0$; when $A>0$, the preceding bound gives $H_r\leq dP$. The remaining estimate is the exact sum-of-squares identity

$$\frac{1}{31}-P=\frac{2}{5}\left(d-\frac{38}{31}+\frac{1}{4}(u-\frac{3}{31})\right)^2+\frac{31}{40}\left(u-\frac{3}{31}\right)^2\geq 0,$$

which gives $P\leq 1/31$, and hence $H_r(x,s)\leq 2/31$ whether P is positive or nonpositive. Thus $H_r\leq 2/31$ whenever the squared modulus is at least 1. If four indices had squared modulus at least 1, (*13*) would give

$$5-2r\leq 4-2r+\frac{8}{31}<5-2r,$$

a contradiction. This proves the two-index conclusion when $a\neq 0$.

For either value of a, at a zero w,

$$p(tw)=(1-t)c+(t-t^4)(bw+c)-t^4(1-t)w^5.$$

For $0\leq t\leq 1$ the three nonnegative weights sum to $1-t^5$. The tail bound, $|w|\leq 1$, and $|c|\leq 1$ therefore keep each selected radial segment in $\{|p|\leq 1\}$. Under the open-disc hypothesis, $|c|<1$. Its coefficient $1-t$ is positive for $0\leq t<1$, so

$$|p(tw)|<(1-t)+(t-t^4)+t^4(1-t)=1-t^5\leq 1.$$

At $t=1$ the value is zero. Thus both segments lie in $\{|p|<1\}$, and their total length is $|w_i|+|w_j|<2$. □

9.3 Polynomials obtained from a cubic by a power substitution

For $f(z) = P((z-h)^q)$, each nonzero root of P gives a regular q-gon of roots centred at h. We exploit this symmetry by lifting one contained segment for the cubic into two segments through h. A generic polynomial of degree $3q$ has no such fibre structure. The first theorem supplies the segment to be lifted.

Theorem 9.6 (a contained radial segment for a cubic)**.** *If $r, s, v \in \mathbb{C}$ have modulus below one, at least one $u \in \{r, s, v\}$ satisfies*

$$|(tu-r)(tu-s)(tu-v)| \le 1 \qquad (0 \le t \le 1).$$

Theorem 9.7 (a cubic composed with a power map)**.** *Let $q \ge 2$, $h \in \mathbb{C}$, P be monic cubic, and*

$$f(z) = P((z-h)^q).$$

If every zero of f lies in the open unit disc and f has at least two distinct zero values, then two zeros are joined through h by a two-segment path of length below 2 inside $\{|f| < 1\}$. Equivalently, under the same hypotheses the conclusion holds for polynomials of the form

$$(z-h)^{3q} + A(z-h)^{2q} + B(z-h)^q + C$$

in every degree $3q \ge 6$.

Proof. One nonzero root of P will supply both endpoints of the path through h. Before selecting it, we check that every zero of P lies in the open unit disc. Fix a qth root y of a zero of P and a primitive qth root of unity ζ. The full fibre consists of $h + y\zeta^k$. Since every fibre point lies in the open unit disc,

$$\frac{1}{q}\sum_{k=0}^{q-1} |h + y\zeta^k|^2 = |h|^2 + |y|^2 < 1. \tag{14}$$

Thus $|y| < 1$, and the corresponding quotient root has modulus $|y|^q < 1$. This verifies the open-disc hypothesis for all quotient roots before selecting a radial segment for the cubic. The identity retains more information than these separate bounds: $|h| < 1$ and every fibre radius is strictly below $\sqrt{1-|h|^2}$. Consequently, any two contained fibre segments constructed below have total length strictly less than $2\sqrt{1-|h|^2}$. The position of the symmetry centre therefore improves the length bound; replacing it by 2 discards this information.

Write $P(w) = (w-r)(w-s)(w-v)$ and associate to r the real number $A_r = \Re(r\overline{s+v})$, cyclically. The exact sum is

$$A_r + A_s + A_v = |r+s+v|^2 - (|r|^2 + |s|^2 + |v|^2) > -3,$$

so one of these numbers, say A_r, exceeds -1. For $0 \le t \le 1$,

$$\frac{|tr-s|^2 + |tr-v|^2}{2} = t^2|r|^2 + \frac{|s|^2+|v|^2}{2} - tA_r < 1 + t + t^2.$$

AM–GM now gives $|tr - s|\,|tr - v| < 1 + t + t^2$. Consequently, for $0 \le t < 1$,

$$|P(tr)| < (1 - t)(1 + t + t^2) = 1 - t^3 \le 1. \tag{15}$$

At $t = 1$ the polynomial vanishes. Hence the entire segment, including the origin, lies in the open unit lemniscate.

For a nonzero quotient root satisfying the segment bound from (15), two distinct fibre points satisfy

$$f(h + ty\zeta^k) = P(t^q r),$$

and the two segments through h have total length $2|y| < 2\sqrt{1 - |h|^2} \le 2$. If the selected quotient root is zero, choose a nonzero quotient root s and write $P(w) = w(w-s)(w-v)$. For $0 < t < 1$,

$$|P(ts)| = t(1 - t)|s|^2|ts - v| < 2t(1 - t) \le \frac{1}{2}.$$

The values at both endpoints are zero, so this nonzero root supplies two distinct fibre points. If no nonzero quotient root exists, f has only one distinct zero value, contrary to the hypothesis. □

Scope. The theorem gives length below 2. Lean does not check the sharper bound $2\sqrt{1 - |h|^2}$ obtained in the proof.

10 Further families and counterexamples to proposed proof steps

Some further uses of the radial and conformal arguments are collected here, together with examples where a prescribed construction fails. The numbered results retain their linked formal statements. The additional analytic arguments appear as unformalised remarks, even when an ordinary proof is supplied in full. These qualifications specify the available formal evidence without changing the finite calculations.

The underlying research notes are retained at commit `f214a6b4` and summarised in the dated overview of these arguments. Numerical observations are identified separately. These are restricted questions about polynomial classes and constructions; they do not supply a programme for proving the refuted unrestricted assertion.

Unformalised remark: roots close to a regular polygon

Let $f(z) = \prod_{i=1}^{n}(z - a_i)$, $n \ge 2$, with all $|a_i| \le 1$. Put $\rho_i = |a_i|$ and $D = |\operatorname{disc}(f)|/n^n$. The Vandermonde determinant estimate says that, when $D \ge 1 - \eta$ and $0 \le \eta \le 1/(80n^2)$,

$$1 - \rho_i^2 \le \frac{n\eta}{n-1}, \qquad |a_i - a_j| \ge \frac{2 - 2\sqrt{\eta} - n\eta}{n-1},$$

where the second inequality is for $i \ne j$. Moreover, the roots lie within $7\sqrt{\eta}$ of a rotated regular n-gon, after a bijection. These estimates come from near equality in Hadamard's determinant inequality, as follows.

Take the Vandermonde matrix $V_{im} = a_i^m$, $0 \le m < n$, and let $G = VV^*$, $s_i = G_{ii} = \sum_{m=0}^{n-1} \rho_i^{2m}$ and $H = \operatorname{diag}(s_i^{-1/2}) G \operatorname{diag}(s_i^{-1/2})$. The normalisation makes H positive semidefinite with diagonal entries one, and

$$D = \det H \prod_i \frac{s_i}{n}.$$

Every factor is at most one by Hadamard's inequality, so $D \ge 1 - \eta$ forces each factor to be at least $1 - \eta$. In particular,

$$(n-1)(1-\rho_i^2) \le n - s_i \le n\eta.$$

Hadamard–Fischer applied to the $\{i, j\}$ principal block gives $\det H \le 1 - |H_{ij}|^2$, hence $|G_{ij}| \le n\sqrt{\eta}$. For $t = a_i \overline{a_j}$, the finite geometric sum then gives

$$n\sqrt{\eta} \ge \left| \sum_{m=0}^{n-1} t^m \right| \ge n - \frac{n(n-1)}{2} |1 - t|.$$

Together with $1 - t = 1 - \rho_j^2 + \overline{a_j}(a_j - a_i)$, this proves the displayed separation bound. These two bounds need only $0 \le \eta < 1$; the smaller range is used for the matching with a regular polygon.

For that matching, put $\kappa = n^2\eta/(n-1) \le 1/80$ and $u_i = a_i/\rho_i$. The radial bound gives $\rho_i^{2n} \ge 1 - \kappa$, while $(1-t)G_{ij} = 1 - t^n$ gives $|1 - a_i^n \overline{a_j^n}| \le 2n\sqrt{\eta}$ for $i \ne j$. Using $|1 - be^{i\theta}|^2 = (1-b)^2 + b|1 - e^{i\theta}|^2$ with $b = \rho_i^n \rho_1^n \ge 1 - \kappa$, we obtain

$$|u_i^n - u_1^n| \le \frac{2n\sqrt{\eta}}{\sqrt{1-\kappa}}.$$

Choose the nearest nth-root phase to u_i among $u_1 e^{2\pi i j/n}$. The inequality $|e^{i\theta} - 1| \ge 2|\theta|/\pi$ for $|\theta| \le \pi$ shows that its distance from a_i is at most

$$\frac{n\eta}{n-1} + \frac{\pi\sqrt{\eta}}{\sqrt{1-\kappa}} \le 7\sqrt{\eta}.$$

The separation bound exceeds $14\sqrt{\eta}$ in the stated range, so two roots cannot be assigned the same phase. There are n roots and n phases, giving the required bijection. This ordinary argument uses standard determinant inequalities. It has no new Lean declaration, and no priority is claimed for the stability estimate.

The discriminant hypothesis $D \ge 1 - \eta$ with $\eta \le 1/(80n^2)$ is stronger than closeness to the unit circle alone. The regular n-gon has $D = 1$, whereas a configuration with a repeated root has $D = 0$. In particular, radial information alone cannot supply this discriminant bound.

We next compare moduli pointwise on a specified segment. No ordering of complex polynomial values is involved. Write $a_k = \rho_k u_k$, with $|u_k| = 1$. Under $D \ge 1 - \eta$ and $0 \le \eta \le 1/(10n^4)$, the same source proves

$$|f(su_i)| \le \prod_k |su_i - u_k| \qquad (0 \le s \le \rho_i).$$

The elementary step is

$$|su_i - \rho_k u_k|^2 - |su_i - u_k|^2 = (1-\rho_k)(2s\Re(u_i\overline{u_k}) - 1 - \rho_k).$$

Here is where the discriminant hypothesis enters. The two bounds just proved imply

$$|a_i - a_k| > \frac{1}{n-1}\ (i \neq k), \qquad 1-\rho_k \leq \frac{n\eta}{n-1} < \frac{1}{(n-1)^2}.$$

For $k = i$, the required inequality is $2s \leq 1+\rho_i$, which follows from $s \leq \rho_i \leq 1$. For $k \neq i$, it is immediate if $\Re(u_i\overline{u_k}) \leq 0$. Otherwise $s \leq \rho_i$ gives

$$\begin{aligned}\rho_k(2s\Re(u_i\overline{u_k}) - 1 - \rho_k) &\leq \rho_i^2 - \rho_k - |a_i - a_k|^2 \\ &\leq 1 - \rho_k - |a_i - a_k|^2 < 0.\end{aligned}$$

Multiplying the factorwise inequalities proves the comparison. This argument uses the two elementary bounds, not the polygon-matching range; it therefore also covers $n = 2$, when $1/(10n^4) > 1/(80n^2)$. Moving the roots inward reduces this modulus bound, but does not select the angular directions of two contained segments.

For exact regular-polygon directions, write $a_i = \rho_i\omega\zeta^i$, with $|\omega| = 1$ and $\zeta = e^{2\pi i/n}$. In any degree $n \geq 2$, the sufficient condition

$$0 < \rho_k \leq 1, \qquad \rho_k \geq 2\cos(2\pi/n) - 1 \quad \text{for every } k$$

makes the same factorwise comparison hold. Indeed, for $k \neq i$ with positive cosine, $s \leq 1$ gives $2s\Re(\zeta^{i-k}) \leq 2\cos(2\pi/n) \leq 1 + \rho_k$; nonpositive cosines and $k = i$ are handled as above. Since $\prod_k (z - \omega\zeta^k) = z^n - \omega^n$, we obtain

$$|f(s\,\omega\zeta^i)| \leq 1 - s^n \quad (0 \leq s \leq \rho_i).$$

Every root-to-origin segment is therefore contained in $\{|f| \leq 1\}$. If all $\rho_i < 1$, containment is strict: the displayed bound is below 1 for $s > 0$, and $|f(0)| = \prod_i \rho_i < 1$. Any two roots can then be joined through the origin with length $\rho_i + \rho_j < 2$. For $2 \leq n \leq 6$ the lower bound on ρ_k is nonpositive, so every choice of positive radii at most one is allowed. For $n \geq 7$ the argument instead requires each radius to be at least $2\cos(2\pi/n) - 1$. This is a sufficient condition for the factorwise comparison, not a claimed necessary condition for a contained path. The favourable index in the general unit-circle averaging identity may move with s, however; an existence statement at one level does not supply a fixed pair of roots.

The exact binomial chord calculation

For $n \geq 2$ and $0 < |a| < 1$, put $r = |a|^{1/n}$ and $r_* = (1 + \cos^n(\pi/n))^{-1/n}$. A rotation reduces $z^n - a$ to $z^n - r^n$ without changing lengths or moduli. For $n = 2$, $r_* = 1$ and only the outer-diameter case occurs.

Theorem 10.1 (complementary binomial chords)**.** *Two adjacent zeros of $z^n - a$ can be joined by an explicit polygonal path inside $\{|z^n - a| < 1\}$ of length strictly below 2. For $r < r_*$ the adjacent-root chord itself works. For $r \geq r_*$, two radial legs and an inner adjacent crossing chord work after an arbitrarily small radial contraction. These two constructions meet at $r = r_*$, where the outer chord attains $|f| = 1$ at its midpoint and therefore lies only in the closed lemniscate. Open containment at and above the switch uses the inner chord after a radial contraction.*

For $f(z) = z^n - r^n$, $0 < r < 1$ and $n \geq 2$, the two constructions depend on the maximum over an entire chord. The endpoint values alone do not give that maximum. Put $\omega = e^{2\pi i/n}$ and $c = \cos(\pi/n)$. When $n \geq 3$, define $r_* = (1 + c^n)^{-1/n}$ and $\varepsilon = (1 - r^n)^{1/n}$; in the inner-chord case $r \geq r_*$, set $t = \varepsilon/c \leq r$.

We prove the chord estimate also at the smaller radii needed for the second construction. For $n \geq 3$, let $0 < s \leq r$. A point on the chord from s to $s\omega$ can be written

$$z = \frac{sc}{\cos\theta} e^{i(\pi/n+\theta)}, \qquad |\theta| \leq \pi/n.$$

Concavity of $\log\cos x$ on $[0, \pi/2)$ gives $\cos(n\theta/2) \leq \cos^{n/2}\theta$ for $0 \leq \theta < \pi/n$. By symmetry and continuity,

$$1 + \cos(n\theta) \leq 2\cos^n\theta \qquad (|\theta| \leq \pi/n).$$

Set $u = \cos^{-n}\theta \geq 1$. Since $z^n = -(sc)^n u e^{in\theta}$, this inequality gives

$$|z^n - r^n|^2 - \left(r^n + (sc)^n\right)^2 \leq (u-1)(sc)^n\left((sc)^n(u+1) - 2r^n\right) \leq 0.$$

The last step uses $(sc)^n u \leq s^n \leq r^n$ and $(sc)^n \leq r^n$. Equality holds at the midpoint, where $\theta = 0$. Consequently the maximum on every such chord is exactly $r^n + (sc)^n$. Taking $s = r$ proves the outer threshold; taking $s = t = \varepsilon/c \leq r$ proves the inner bound, and taking $s = \lambda t < t$ makes it strict. Every radial leg between s and r has modulus $r^n - |z|^n < 1$. Its two legs and crossing chord have length $2r - 2s(1 - \sin(\pi/n)) < 2r$, which proves the stated length formula and shows that contraction preserves the strict length bound.

For $n = 2$, the chord is a diameter and its maximum is $r^2 < 1$; its length is $2r < 2$. For $n \geq 3$, the outer chord works exactly when $r < r_*$, and its length is $2r\sin(\pi/n)$. At $r = r_*$ it reaches level one at its midpoint. If $r \geq r_*$, the two radial legs and inner crossing have length

$$2r - 2\varepsilon\tan\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) < 2r < 2.$$

Contracting the crossing radius makes the level bound strict, as proved above. The comparison concerns these two constructions only; it makes no claim about shortest paths among all admissible curves.

Unformalised remark: a bounded-radius concyclic class

When the common root circle is small enough, its self-inversive polynomial gives a contained adjacent-root chord. This ordinary argument has no complete formal counterpart in the supplied record. Let f be monic of degree $n \geq 3$ with distinct zeros on a circle of radius ρ. If $2\rho^n \leq 1$, two adjacent zeros are joined by their straight chord, whose length is at most $2\rho\sin(\pi/n) < 2$, and the chord lies in $\{|f| < 1\}$. (If a zero is repeated, the short-connection conclusion is immediate; the distinct case is the substantive one.)

For each fixed degree n, the radius condition is $\rho \leq 2^{-1/n}$. It allows arbitrary spacing on that circle, but does not cover every radius $\rho < 1$. No restriction on the circle's centre is imposed.

After translating and scaling, let g be the monic polynomial with roots $w_1, \dots, w_n$ on the unit circle. Such a polynomial is *self-inversive*: $z^n\overline{g(1/\bar{z})}$ is a constant multiple of $g(z)$, with the constant of modulus one. Equivalently, $e^{-in\theta/2}g(e^{i\theta})$ is real up to a fixed phase. Choose

$$c = (-1)^{n+1}\prod_j w_j, \qquad q(z) = z^n - c.$$

Then g and q have the same leading and constant coefficients, and their boundary values have a common real phase. If every root gap contained a point with $|g| > |q|$, the real boundary function of $g - q$ would have alternating signs at these points. Advancing the angle by 2π multiplies that function by $(-1)^n$, so the alternation also holds across the final gap. The intermediate value theorem would give n distinct unit-circle zeros of $g - q$. This is impossible: its degree is at most $n-1$, and $g = q$ cannot satisfy the assumed strict inequalities. Thus some adjacent-root arc satisfies $|g| \le |q| \le 2$. The interior of that arc contains no zero of q, since g is nonzero there. As consecutive zeros of q are separated by angle $2\pi/n$, the selected arc has angular width at most $2\pi/n$. This explains why the same selected pair also has the required short chord. To pass from this arc to its chord $[a, b]$, let ν be the unit normal pointing into the circular segment bounded by them. On the open chord, logarithmic differentiation gives

$$\partial_\nu \log|g(z)| = \sum_{w_j \notin \{a,b\}} \frac{\mathrm{Re}(\nu\overline{z - w_j})}{|z - w_j|^2} > 0.$$

The endpoint terms vanish because their directions are tangent to the chord. Every remaining numerator is positive because the other roots lie strictly on the far side of its line; $n \ge 3$ ensures at least one such term. The harmonic function $\log|g|$ has no interior maximum and tends to $-\infty$ at a, b. A maximum cannot lie on the open chord either, since moving into the circular segment increases the function. It is therefore attained on the arc, and the maximum on the chord is strictly smaller. Scaling back gives $|f| < 2\rho^n \le 1$ throughout the chord, with $f = 0$ at its endpoints. These are the arguments of Theorem C and Lemma S, Sections 3–4 of the concyclic note.

The argument is an ordinary proof outside Lean. The exact-rational checker checks finitely many identities used in the proof and configurations, while the numerical checker is regression and stress-test evidence. The arc constant 2 is attained by the regular n-gon. Thus the bound obtained by passing through the arc is $\rho \le 2^{-1/n}$, although this cutoff tends to 1 as $n \to \infty$. This does not assert optimality of the chord bound or exclude another contained path at a larger radius. Nor does arc sharpness make the pair of radial segments shortest. For example, for $g(z) = z^3 - 1/8$ the chord between $1/2$ and $e^{2\pi i/3}/2$ has length $\sqrt{3}/2 < 1$, whereas the two radii have total length 1. The binomial calculation above gives a maximum of $9/64 < 1$ on that chord. The length-2 equality at $z^n - 1$, proved below, concerns the uncontracted closed-level problem.

Power substitutions in trinomials

The segment identity survives a power substitution. We record the lifting argument as an unformalised remark: the source checks the segment inequality and finite length

calculation, but not selection and lifting of a suitable quotient root. Fix integers $1 \le r < m$ and $q \ge 1$, and consider the translated monic polynomial

$$f(z) = (z-h)^{qm} + a(z-h)^{qr} + c.$$

The strict exponent inequality keeps the leading monomial separate. The algebraic identity below also holds when $r = m$, but then the two leading terms merge and the displayed polynomial need not be monic or have degree qm. Writing $w = (z-h)^q$ reduces the root equation to $w^m + aw^r + c = 0$. At such a quotient root the middle coefficient can be eliminated exactly: for $0 \le u \le 1$,

$$u^m w^m + au^r w^r + c = (1-u^r)c - (u^r - u^m)w^m.$$

This is a nonnegative combination of c and $-w^m$, with weights $1-u^r$ and $u^r - u^m$ summing to $1-u^m$. The formal radial-segment estimate gives strict containment when $|w| < 1$ and $|c| < 1$.

For $q \ge 2$ and a nonzero quotient root satisfying these bounds, choose two distinct solutions y_1, y_2 of $y^q = w$. Their common modulus is $|w|^{1/q} < 1$. To lift a quotient radial segment, set $u = t^q$:

$$|f(h + ty_i)| \le (1 - t^{qm}) \max\{|c|, |w|^m\} < 1 \qquad (0 \le t \le 1).$$

Thus the two lifted segments join $h + y_1$ to $h + y_2$ through h with length $2|w|^{1/q} < 2$. The construction uses any quotient root with the stated modulus bound; it does not prove that such a root exists without an additional hypothesis. A zero quotient root instead gives a root of multiplicity at least q at h and hence a constant path between two occurrences. The formal source checks the factorization, radial-segment estimate and finite length inequality. It does not establish the existence of a suitable quotient root, formalise the selection and lifting just described, or prove unrestricted Erdős #1041.

The radial-segment identity and strict bound are formalised in cancellation along a trinomial root segment and containment of a trinomial root segment.

Unformalised remark: a coefficient condition for four-term polynomials

One more monomial introduces a partial sum that need not lie in the unit disc. Bounding that sum gives the following ordinary coefficient criterion. Let

$$g(w) = w^m + aw^r + bw^s + c, \qquad m > r > s \ge 1,$$

assume all roots of g lie in the open unit disk, and let w_1, w_2 be roots of the two smallest moduli. If

$$|c| + |b|\,|w_2|^s < 1,$$

then the complete radial spokes from w_1 and w_2 to the origin lie in $\{|g| < 1\}$, and their broken line has length strictly below 2. The simpler coefficient condition

$$|b| + |c| \le 1$$

makes every root-to-origin segment contained; the coefficient a has no additional restriction beyond the assumed root locations. This coefficient condition includes $b = 0$ and $|c| \le 1$, but is not automatic for all four-term polynomials with roots in the disc.

The mechanism is visible in one identity. At a root w and for $0 \le u \le 1$,

$$g(uw) = (1 - u^s)c + (u^s - u^r)(c + bw^s) + (u^r - u^m)(-w^m).$$

Here the root equation has eliminated aw^r before taking absolute values. The three weights are nonnegative and sum to $1 - u^m$. The controlled terms are c, $c + bw^s$ and $-w^m$; their moduli are strictly below one under the root-dependent condition. The same identity explains why no bound on a is needed. Lean checks the factorization and the resulting strict radial-segment theorem under the exact weak exponent hypotheses $1 \le s \le r \le m$; it also checks the coefficient-only corollary above. Its weak bound $|b| + |c| \le 1$ still gives strict containment: the root-disc hypothesis implies $|c| < 1$, and, if $b \ne 0$, $|w|^s < 1$ gives $|c| + |b||w|^s < |c| + |b| \le 1$. If $b = 0$, the required bound is simply $|c| < 1$.

An alternative criterion uses a complex power sum to select two indices. Let S index a finite family of roots, $N = |S| \ge 2$, and

$$M = \sum_{i \in S} w_i^s.$$

The squared-modulus identity gives

$$\sum_{i \in S} |c + bw_i^s|^2 = N|c|^2 + |b|^2 \sum_{i \in S} |w_i^s|^2 + 2\,\mathrm{Re}(\bar{c}\, bM).$$

Consequently, if every $|w_i| < 1$, $|c| < 1$, and

$$N(|b|^2 + |c|^2) + 2\,\mathrm{Re}(\bar{c}\, bM) < N - 1,$$

then two distinct indices $i, j \in S$ satisfy $|c + bw_i^s| < 1$ and $|c + bw_j^s| < 1$. Indeed, the displayed identity bounds the sum of squared moduli strictly below $N - 1$. If at most one index had modulus below one, the other $N - 1$ terms would each contribute at least one. This explains both the threshold $N - 1$ and why the criterion selects two indices. Applying the resulting contained-segment theorem to these inequalities proves that both root segments lie strictly in $\{|g| < 1\}$. Retaining the signed cross term permits the inequality to hold for configurations outside the coefficient-only triangle $|b| + |c| \le 1$. The formal hypotheses do not require $i \mapsto w_i$ to be injective, so distinct indices need not denote distinct root values. For the polynomial consequence, the family must be a submultiset of the zeros, respecting their algebraic multiplicities. A repeated zero then already gives a constant path between two occurrences. Repeating a simple zero in an arbitrary indexed family cannot supply a second occurrence. The two sufficient conditions are not ordered by strength. For example, $g(w) = w^5 + 19/20$ has all roots in the open unit disc and satisfies the coefficient condition with $b = 0$. For its full five-root list, however, the left side of the power-sum condition is $5(19/20)^2 = 361/80 > 4 = N-1$, so that condition fails. Its benefit is the use of cancellation in the cross term, not inclusion of every case covered by the coefficient condition. Conversely,

$$g(w) = (w^2 - 1/2)(w^3 + 4/5) = w^5 - \tfrac{1}{2}w^3 + \tfrac{4}{5}w^2 - \tfrac{2}{5}$$

has all roots of modulus $1/\sqrt{2}$ or $(4/5)^{1/3}$, hence below one, but $|b|+|c| = 6/5 > 1$. Its full root list has $M = \sum_i w_i^2 = 1$: the two quadratic roots contribute 1, and the three cubic roots contribute 0. The power-sum left side is $5(16/25+4/25)-16/25 = 84/25 < 4$. Thus this criterion does cover root-disc polynomials excluded by the coefficient condition.

For $q \geq 2$ and $f(z) = g((z-h)^q)$, assume also that all roots of f lie in the open unit disc. If $y^q = w$ and $\omega = e^{2\pi i/q}$, the roots $h + \omega^j y$ satisfy

$$\frac{1}{q}\sum_{j=0}^{q-1} |h + \omega^j y|^2 = |h|^2 + |y|^2 < 1.$$

Thus every quotient root has $|w| = |y|^q < 1$, which supplies the hypothesis on the zeros used above, and every lifted segment has length $|y| < 1$. Two selected spokes join through h with total length below 2; repeated root occurrences instead permit the constant path. The Lean theorem assumes the finite root family, its complex power sum M, and the displayed inequality. It neither derives that inequality from arbitrary coefficients nor verifies the family's polynomial multiplicities, the cyclic lifting, or the final path construction. Completeness of the root list is not required for the selection argument: a submultiset respecting multiplicity suffices. Without one of the power-sum, root-dependent, or coefficient-only inequalities, the tetranomial case is not covered by this argument; this family does not solve unrestricted Erdős #1041.

Unformalised remark: a quartic composed with a power map

We next examine the ordinary lifting argument based on Pendyala's degree-four theorem [11, Theorem 1] and four-point radial lemma [11, Lemma 1]. The cited quartic result alone does not formalise this higher-degree construction. Let P be a monic quartic, let $q \geq 2$ be an integer, and set

$$f(z) = P((z-h)^q).$$

Assume that every root of f lies in the open unit disc. A repeated root of f gives the constant path, so suppose f is squarefree. Then the roots of P are distinct and nonzero. For a root $w = y^q$ of P, all points $h + \zeta^j y$, with $\zeta = e^{2\pi i/q}$, are roots of f. Hence the open-disc hypothesis gives

$$\frac{1}{q}\sum_{j=0}^{q-1} |h + \zeta^j y|^2 = |h|^2 + |y|^2 < 1.$$

In particular $|w|^{1/q} = |y| < 1$, which is the bound needed for the lifted length, and also $|w| < 1$. Pendyala's chord-or-radial argument supplies the quotient geometry in two cases. If two quotient roots satisfy $|w_i - w_j| < 1$, every point w of the chord $[w_i, w_j]$ lies in the unit disc; the two endpoint factors of $|P(w)|$ have product at most $|w_i - w_j|^2/4$ and the other two factors are below 2, so $|P(w)| \leq |w_i - w_j|^2 < 1$. Otherwise the quotient roots are pairwise at distance at least 1. Choosing R with $\max_k |w_k| < R < 1$ and applying the four-point radial lemma to the points w_k/R gives distinct i, j with $|P(tw_i)| \leq R^4$ and $|P(tw_j)| \leq R^4$ for $0 \leq t \leq 1$, so the radial segments $[0, w_i]$ and $[0, w_j]$ lie in $\{|P| < 1\}$. The

extra issue is metric: a short chord in the w-plane need not lift isometrically through $y \mapsto y^q$.

Put $\alpha = 1/q$. On a chord avoiding 0, choose a continuous argument and hence a root lift y with $y^q = w$. Differentiation gives $|dy| = \alpha|w|^{\alpha-1}|dw|$, independently of the chosen root. If the supporting line has distance d from 0, and x is distance along that line from the perpendicular foot, then

$$(\sqrt{d^2 + x^2})^{\alpha-1} \le x^{\alpha-1} \qquad (x > 0).$$

The negative exponent is why replacing $|w|$ by x gives an upper length bound. Lean checks the integral of the power-map derivative

$$\alpha \int_0^A x^{\alpha-1}\, dx = A^\alpha.$$

If the foot lies on the chord, split there: the two integrals are bounded by the α-powers of the distances from the foot to the endpoints, hence by $|a|^\alpha + |b|^\alpha$. If the foot lies outside the chord, integrate on one side of it; extending the interval to the foot bounds the result by the α-power of the farther endpoint's modulus. When the chord passes through 0, choose a continuous root on each half and join the lifts at 0; the integral is finite because $\alpha > 0$. This proves the chord-lift estimate

$$\mathrm{length}(\widetilde{[a,b]}) \le |a|^{1/q} + |b|^{1/q}.$$

The strict endpoint-length estimate also proves that if $0 \le a, b < 1, \alpha > 0$, and a candidate length L is at most $a^\alpha + b^\alpha$, then $L < 2$. The close-pair chord therefore has a lift of length below 2. In the other case the quotient arms meet at 0, so their lifts are radial segments meeting at h, with total length $|w_i|^{1/q} + |w_j|^{1/q} < 2$. In both cases $f = P \circ ((z-h)^q)$ transfers containment from the quotient curve. Distinct quotient endpoints give distinct roots of f. This proves the path assertion in every degree $4q \ge 8$. The fibre average gives a little more: $|h| < 1$ and $|w_k|^{1/q} < \sqrt{1 - |h|^2}$ for every quotient root. Consequently the same constructed path has length

$$\mathrm{length} < 2\sqrt{1 - |h|^2} \le 2.$$

This refinement concerns the translated power-substitution family, not arbitrary degree-$4q$ polynomials.

Pendyala proves the quartic geometric theorem and its four-point radial lemma. The local Lean module checks the antitone density inequality, its integral, the strict powered endpoint-length bound, and the final length estimate. It does not formalize Pendyala's geometric lemma, the continuous covering-space construction of the root lift, or the ordinary chord/radial case assembly. This is a family defined by a power substitution, not a proof of unrestricted Erdős #1041.

The joining point in the quartic construction. Pendyala's proof of [11, Theorem 1, pp. 2–3] uses a smallest enclosing disc and joins the selected roots through its centre. For the power substitution we instead apply [11, Lemma 1] to the centred disc $|w| \le R$.

That lemma requires only the disc and pairwise-distance hypotheses, not minimality of the disc. Keeping the joining point at 0 is what makes the radial lift above valid. The endpoint-length bound for a single chord would not justify lifting an arbitrary broken line through a nonzero quotient vertex.

A mean bound for critical values in every degree

The Poisson argument first gives a quadratic mean bound, allowing a disc of any centre and radius and counting all multiplicities. This is the formally supported theorem below. The subsequent fourth-power refinement, and the higher exponents obtained from vanishing complex power sums, are unformalised analytic remarks. A critical-value moment bound supplies no estimate for the lengths of inverse paths ending at those points.

Dubinin [21, Theorem 2, (9), pp. 1172–1173] proves the sharp unit-disc critical-value product inequality using the resultant identity, the maximum-modulus principle and Schur's Vandermonde inequality. The positive-moment estimate below implies his product bound by AM–GM. When $n \geq 3$, the product estimate alone cannot control a positive moment: a list $T, T^{-1}, 1, \dots, 1$ has product one, but its sum of pth powers is unbounded as $T \to \infty$ for every $p > 0$. This compares the information in two inequalities for nonnegative lists; it does not claim that these lists occur as critical values of polynomials in the theorem. In degree two there is just one critical value, and the product and positive-moment bounds are equivalent. For a zero prescribed at the origin, his Theorem 3 (p. 1174) gives

$$\left(\prod_j |f(c_j)| \right)^{1/(n-1)} \leq (n-1) \left(\frac{|f'(0)|^2}{n^n} \right)^{1/(n-1)}.$$

Dubinin explicitly identifies Tischler's earlier contribution [25, p. 444, as reported in Dubinin]. His Theorem 1 uses dissymmetrisation; that is not the proof mechanism of Theorem 2. Schur's original paper [27] is the historical antecedent cited there, not a separately re-proved source in this record. A sharp marked-zero positive-moment bound remains a separate question. The addendum [26, Section 3] reports a reduction to boundary-root configurations, without solving the resulting angular optimisation. That unavailable addendum is not a proof dependency of the critical-value theorem below.

Theorem 10.2 (a mean bound for critical values)**.** *Let f be monic of degree $n \geq 2$, with roots in a closed disc of radius $R \geq 0$. If $c_1, \dots, c_{n-1}$ are its critical points counted with multiplicity, then*

$$\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1) R^{2n/(n-1)}. \tag{11}$$

Consequently the lower exponents used elsewhere in the record satisfy

$$\sum_{j=1}^{n-1} |f(c_j)|^{1/(n-1)} \leq (n-1) R^{n/(n-1)} \tag{12}$$

and

$$\sum_{j=1}^{n-1} |f(c_j)|^{1/n} \le (n-1)R.$$

The constant is attained by $f(z) = (z-\tau)^n - \lambda$ *with enclosing disk centred at* τ *and radius* $R = |\lambda|^{1/n}$.

The finite inequality behind the theorem concerns arbitrary points of the disk, without asking them to arise as critical points. Its quadratic form is

$$\sum_{j=1}^{m} \left(\prod_{k=1}^{m} |1 - \overline{c_j} c_k| \right)^{2/m} \le m, \qquad |c_j| \le 1.$$

The following pointwise bound explains why these finite inequalities control critical values in every degree.

Lemma 10.3 (reflected-derivative bound)**.** *If* f *is monic of degree* $n \ge 2$ *with roots in the closed unit disk, and* $c_1, \dots, c_{n-1}$ *list its critical points with multiplicity, then*

$$|f(c_j)| \le \prod_{k=1}^{n-1} |1 - \overline{c_j} c_k|. \tag{13}$$

Proof. First put all roots a_i strictly inside the disk. We want a numerator that equals nf at the critical points and can be compared with f' on the circle. Set

$$N(z) = nf(z) - zf'(z), \qquad G(z) = n \prod_{k} (1 - \overline{c_k} z).$$

The numerator is the polar derivative at 0: the conventional polar derivative at α is $nf(z) + (\alpha - z)f'(z)$; see Rather–Gulzar, Section 1 for this notation. Its leading term cancels, and $N(c_j) = nf(c_j)$. Reflection gives $|G| = |f'|$ on the circle. Gauss–Lucas puts every c_k inside the disc, so G has no zero on its closure. Thus N/G is a holomorphic quotient to which the maximum-modulus principle applies. On $|\zeta| = 1$,

$$\operatorname{Re} \frac{\zeta f'(\zeta)}{f(\zeta)} = \sum_{i} \left(\frac{1}{2} + \frac{1 - |a_i|^2}{2|1 - a_i \bar{\zeta}|^2} \right) \ge \frac{n}{2}.$$

Writing the logarithmic derivative as w, the identity $|n - w|^2 - |w|^2 = n^2 - 2n \operatorname{Re} w \le 0$ gives $|N| \le |G|$ on the circle. The maximum-modulus principle applied to N/G gives the same comparison inside. At c_j, $N(c_j) = nf(c_j)$; dividing by n and conjugating each factor proves (13). For closed-disk roots apply this result to $f_r(z) = r^n f(z/r)$, $0 < r < 1$, whose critical points are rc_k:

$$r^n |f(c_j)| \le \prod_{k} |1 - r^2 \overline{c_j} c_k|.$$

Let $r \uparrow 1$. This also handles boundary critical points and all multiplicities without selecting local branches. □

First suppose the zeros of f lie in the closed unit disc, and set $m = n-1$. Gauss–Lucas puts the critical points in the same disc. Lemma 10.3 gives the first inequality below; the weighted Poisson inequality, proved next and applied with equal weights $1/m$, gives the second:

$$\sum_{j=1}^{m} |f(c_j)|^{2/m} \le \sum_{j=1}^{m} \left(\prod_{k=1}^{m} |1 - \overline{c_j} c_k| \right)^{2/m} \le m.$$

Applying this to $R^{-n} f(h+Rz)$ and scaling back proves (11) for $R > 0$; if $R = 0$, all critical values vanish.

The two displayed consequences are the lower power means of the same nonnegative m-tuple. Equivalently, after unit-disc normalization, apply the monotonicity of normalized L^p means from exponent $2/m$ first to $1/m$ and then to $1/(m+1) = 1/n$. Scaling back contributes respectively $R^{n/m}$ and R. These are consequences of the quadratic case; the higher-power argument below strengthens that case without changing these lower-exponent applications.

The quadratic moment controls the number of large critical values. For $R > 0$, write $r_j = |f(c_j)|^{1/n}$. Then

$$\sum_j (r_j/R)^{2n/(n-1)} \le n-1, \qquad \#\{j : r_j \ge tR\} \le \frac{n-1}{t^{2n/(n-1)}} \quad (t > 0).$$

The counting bound follows by retaining just the indicated summands. The fourth-power refinement proved below replaces the exponent $2n/(n-1)$ here by $4n/(n-1)$, improving the count for $t > 1$. For $0 < t \le 1$ the trivial bound $n-1$ is at least as good. The binomial equality family has every $r_j = R$, but these distributional bounds still leave the geometry of the joining paths to be supplied.

The weighted Poisson inequality and its equality case.

Theorem 10.4 (a weighted inequality for points in a disc)**.** *Let $c_1, \dots, c_m \in \overline{\mathbb{D}}$ and let $w_j > 0$ satisfy $\sum_j w_j = 1$. Set*

$$G(z) = \prod_k |1 - \overline{c_k} z|^{w_k}.$$

Then

$$\sum_j w_j G(c_j)^2 \le 1,$$

with equality if and only if every $c_j = 0$. Equal weights therefore give $\sum_j (\prod_k |1 - \overline{c_k} c_j|)^{1/m} \le m$ for every m.

For one point c of weight 1, the quantity to bound is $(1-|c|^2)^2 \le 1$; equality holds only at $c = 0$. For two points $c, -c$ of equal weight, it is $1 - |c|^4 \le 1$. These examples suggest why a nonzero configuration should lose from equality. Now let $w_j \ge 0$, $\sum_j w_j = 1$, and $|c_j| \le 1$, with repetitions allowed. Write $G(z) = \prod_k |1-\overline{c_k} z|^{w_k}$, taking a factor of exponent zero to be one. We seek an analytic function with modulus G: its logarithmic derivative will express the weighted Poisson kernel, and its Taylor coefficients will measure the loss from equality. For $|c| < 1$ and $|\zeta| = 1$, let $P_c(\zeta) = (1 - |c|^2)/|\zeta - c|^2$, and write

$dm = dt/(2\pi)$ on $\zeta = e^{it}$. When every centre is strictly inside the disc, choose analytic logarithms zero at the origin and put

$$g(z) = \exp \sum_j w_j \log(1 - \overline{c_j} z) = 1 + \sum_{\nu \geq 1} a_\nu z^\nu, \qquad P = \sum_j w_j P_{c_j}.$$

On the unit circle $P = 1 - 2\Re(\zeta g'/g)$, by logarithmic differentiation. The Poisson formula reproduces $g(c)$ from its boundary values and gives $\int P_c \, dm = 1$ for $|c| < 1$. Expanding a square therefore yields

$$\int |g|^2 P_c \, dm - |g(c)|^2 = \int |g - g(c)|^2 P_c \, dm \geq 0.$$

This explains why the same kernel bounds the value at each centre. Take $c = c_j$, multiply by w_j and sum to obtain

$$\sum_j w_j G(c_j)^2 \leq \int |g|^2 P \, dm = 1 - \sum_{\nu \geq 1} (2\nu - 1)|a_\nu|^2.$$

To see the last equality, expand $|g|^2$ and $\zeta g' \overline{g}$ and integrate on the circle. Orthogonality leaves $1 + \sum_{\nu \geq 1} |a_\nu|^2$ and $\sum_{\nu \geq 1} \nu |a_\nu|^2$, respectively. The analytic function g extends to a neighbourhood of the closed disc in this interior case, which justifies the termwise integrations. For closed-disc centres replace c_j by $r c_j$, $0 < r < 1$. The actual coefficient of degree ν becomes $r^\nu a_\nu$, where the fixed coefficients are defined from the analytic function near zero. Pass to $r \uparrow 1$ first with an arbitrary finite coefficient sum. Nonnegative finite deficits then give summability and the infinite inequality; no boundary holomorphic logarithm is required. Lean checks the exact radial transport of each Cauchy coefficient, the finite boundary-deficit inequality, and the summable full boundary deficit.

If equality holds, all positive-degree coefficients vanish, so $g = 1$ near zero and $\sum_j w_j \overline{c_j}/(1 - \overline{c_j} z) = 0$ there. Group equal centres before clearing denominators. A nonzero centre of positive total weight gives a nonzero pole coefficient, which is impossible. Thus equality is equivalent to $c_j = 0$ on positive support. In particular, strictly positive weights force all centres zero. A point of weight zero is unconstrained: weight one at 0 and weight zero at 1 still give equality. Grouping equal centres is necessary: an argument that assumed they were distinct would not prove the stated equality case.

Equal weights in $\sum_j w_j G(c_j)^2 \leq 1$ give the quadratic finite inequality above for every $m \geq 1$, with equality only at the zero configuration. The linear inequality follows separately. For $M = \sum_j w_j G(c_j)$, the variance identity

$$M^2 + \sum_j w_j (G(c_j) - M)^2 = \sum_j w_j G(c_j)^2$$

gives $M \leq 1$. Equality in this Cauchy–Schwarz step means that the values $G(c_j)$ are constant at all indices of positive weight; equality in the final bound also requires equality in the quadratic inequality. The three-point and four-point specialisations follow with their equality cases. Lean checks the quadratic weighted inequality. The hypotheses permit repeated centres and zero weights; the equality condition refers only to centres of positive weight.

This equality condition also identifies all equality cases of Theorem 10.2 for a fixed containing disc $\overline{D}(h,R)$ with $R>0$. Equality in the critical-value bound forces equality in the quadratic finite inequality after normalisation, so every critical point is 0. Thus the normalised derivative is nz^{n-1} and the polynomial is $z^n-\lambda$, with $|\lambda|=1$ forced by equality. Restoring the centre and radius gives exactly $f(z)=(z-h)^n-\lambda$ with $|\lambda|=R^n$. Equality in either lower-power consequence also forces equality in the quadratic bound, so it has the same family. For $R=0$ the only possible polynomial is $(z-h)^n$. Lean does not check this classification.

The earlier three-point Hölder argument and four-point matching argument are not used here. Their scalar identities do not justify the omitted inequalities; the proof is the weighted Poisson calculation above.

Higher powers and vanishing complex power sums. Lean checks only the quadratic case; the analytic argument reaches further. To estimate G^p for $p>0$, we need an analytic function whose squared modulus is G^p. Using the same logarithms as above, define

$$H_p=g^{p/2}=1+\sum_{\nu\ge1}b_\nu z^\nu.$$

For interior centres, logarithmic differentiation and the Poisson majorant give

$$P=1-\frac{4}{p}\Re\frac{\zeta H_p'}{H_p},$$

$$\sum_j w_jG(c_j)^p\le\int|H_p|^2P\,dm=1-\sum_{\nu\ge1}\left(\frac{4\nu}{p}-1\right)|b_\nu|^2.$$

The identity follows from the same two orthogonality calculations used for g; only the coefficient of the derivative term changes. For $0<p\le4$ every displayed coefficient is nonnegative. Hence

$$\sum_j w_jG(c_j)^p\le1\qquad(0<p\le4).$$

Boundary centres follow by replacing c_j by rc_j: the degree-ν coefficient becomes $r^\nu b_\nu$, and the values on the left are $G_r(rc_j)=\prod_k|1-r^2\overline{c_k}c_j|^{w_k}$. First retain finitely many nonnegative terms, then let $r\uparrow1$. Zero weights can be omitted throughout. This proves the closed-disc inequality without differentiating a boundary logarithm. The argument is the power-parameter Poisson identity; Lean does not check it.

In particular, equal weights and Lemma 10.3 give, with $m=n-1$,

$$\frac{1}{m}\sum_j|f(c_j)|^{4/m}\le\frac{1}{m}\sum_j G(c_j)^4\le1$$

on the unit disc. For roots in $\overline{D}(h,R)$ this becomes

$$\sum_j|f(c_j)|^{4/(n-1)}\le(n-1)R^{4n/(n-1)}.$$

As before, $R>0$ is handled by $R^{-n}f(h+Rz)$, and $R=0$ by the vanishing of all critical values. The constant is attained by $(z-h)^n-\lambda$ with $|\lambda|=R^n$. Thus the constant is sharp. Optimality of the exponent at each fixed degree is not asserted.

At $p = 4$ the coefficient of $|b_1|^2$ vanishes, so the previous quadratic equality proof cannot simply be reused. If equality holds, the finite-deficit bounds imply $H_4 = 1 + b_1 z$. If $b_1 \neq 0$, the rational identity $H_4'/H_4 = 2g'/g$ shows, by comparing poles and their residues, that all nonzero centres coincide at $a = -\overline{b_1}$ and have total weight $1/2$. The other half of the weight is at zero. Their fourth-power average is

$$\tfrac{1}{2}(1 + (1 - |a|^2)^2) = 1 - |a|^2 + \tfrac{1}{2}|a|^4 < 1 \qquad (0 < |a| \leq 1),$$

a contradiction. Thus $H_4 = 1$, and the pole argument already used for g forces every centre of positive weight to be zero. Conversely that configuration gives equality. It follows that the fourth-power critical-value estimate has the same equality family as the quadratic one for a fixed containing disc.

For $p > 4$, the coefficient $4/p - 1$ is negative, so the nonnegative-deficit argument no longer applies. To extend its range we can force the first Taylor coefficients to vanish. Let $s \geq 1$ be an integer and suppose

$$\sum_j w_j c_j^k = 0 \qquad (1 \leq k < s).$$

The expansion

$$\log H_p(z) = -\frac{p}{2} \sum_{k \geq 1} \frac{\overline{\sum_j w_j c_j^k}}{k} z^k$$

then has no terms of degree below s, so $b_1 = \cdots = b_{s-1} = 0$. All remaining coefficient weights are nonnegative for $0 < p \leq 4s$. The same proof therefore gives $\sum_j w_j G(c_j)^p \leq 1$ throughout that range. For a monic polynomial with roots in $\overline{D}(h, R)$, $R > 0$, the resulting endpoint is

$$\sum_j |f(c_j)|^{4s/(n-1)} \leq (n-1)R^{4sn/(n-1)}, \qquad \sum_j (c_j - h)^k = 0 \quad (1 \leq k < s).$$

There is no cancellation hypothesis when $s = 1$. For $s = 2$, the condition says that the critical-point centroid is the disc centre. Comparing the coefficients of f and f' shows

$$\frac{1}{n-1} \sum_j c_j = \frac{1}{n} \sum_{i=1}^{n} z_i,$$

so this is also the root centroid. Recentring can increase the required radius: for $0 < r < 1$, the roots of $(z - r)^2(z + r)$ lie in $\overline{D}(0, r)$, but their centroid is $r/3$, and a disc centred there needs radius $4r/3$ to contain the root $-r$. Thus the exponent-$8/(n-1)$ statement cannot be applied at the old centre with the old radius. The hypotheses require exact cancellation of the moments; upper bounds for their moduli are insufficient. Lean does not check the higher-power conclusions in this paragraph; the Lean proof of Theorem 10.2 has exponent $2/(n-1)$.

Lean proofs of the Poisson and integration steps. The square-expansion proof above applies whenever g is holomorphic on the disc and continuous on its closure, with $|c| < 1$. The Lean norm-square majorization uses the Poisson formula for a real part and the nonnegativity of a square. The weighted kernel identity identifies the pointwise Poisson mixture. Separately, absolute series transport justifies termwise circle integration when the terms are circle integrable and admit a summable uniform norm bound; its Taylor double-product form assumes absolute summability of both coefficient sequences.

A uniform central region in every degree. For every $m \geq 1$, the inequality for points in a smaller disc proves

$$|c_j| \leq \sqrt{1-e^{-2}} \quad (1 \leq j \leq m) \quad \Longrightarrow \quad \sum_{j=1}^{m} \left(\prod_{k=1}^{m} |1 - \bar{c}_j c_k| \right)^{1/m} \leq m.$$

To see the mechanism, set $h_j = m^{-1} \sum_k \log|1 - \bar{c}_j c_k|$ and $D_j = -\log(1 - |c_j|^2)$. Write $M_\nu = m^{-1} \sum_k c_k^\nu$ for the normalised complex power sums used in this calculation. The absolutely convergent series for $\log(1-z)$ gives

$$h_j = -\Re \sum_{\nu \geq 1} \frac{\bar{c}_j^{\,\nu} M_\nu}{\nu}, \qquad -\frac{1}{m} \sum_j h_j = \sum_{\nu \geq 1} \frac{|M_\nu|^2}{\nu}.$$

In particular, $\sum_j h_j \leq 0$. Cauchy–Schwarz applied to the first series, using $\sum_{\nu \geq 1} |c_j|^{2\nu}/\nu = D_j$, gives $h_j^2 \leq (-m^{-1} \sum_i h_i) D_j$. The radius bound gives $D_j \leq 2$. Put $r = \sqrt{-m^{-1} \sum_i h_i}$. If $r = 0$, every h_j vanishes. For $r > 0$, convexity bounds e^h by the chord of its graph over $[-\sqrt{2}r, \sqrt{2}r]$. Averaging and using $m^{-1} \sum_j h_j = -r^2$ gives

$$\frac{1}{m} \sum_j e^{h_j} \leq \cosh(\sqrt{2}r) - \frac{r}{\sqrt{2}} \sinh(\sqrt{2}r) \leq 1.$$

For the last inequality, the function $\cosh u - (u/2)\sinh u$ equals 1 at $u = 0$ and has derivative $(\sinh u - u \cosh u)/2 \leq 0$ for $u \geq 0$. Thus the radius bound supplies the estimates needed for the geometric-mean inequality; no further series hypotheses are imposed. This alternative proof includes the all-zero configuration but requires the smaller radius $\sqrt{1-e^{-2}}$. The weighted theorem above covers the whole closed unit disc. The next logarithmic argument imposes a different sufficient radius condition; neither restriction belongs to the weighted theorem.

For $m \geq 1$, let $|c_i|^2 \leq a < 1$, with $a \geq 0$, and put $L = -\log(1-a)$. The uniform-radius theorem gives the geometric-mean inequality

$$e^L \leq 1 + 2L \quad \Longrightarrow \quad \sum_{i=1}^{m} \left(\prod_{j=1}^{m} |1 - \bar{c}_i c_j| \right)^{1/m} \leq m.$$

For $a = 1/2$ the scalar hypothesis holds, since $2 \leq 1 + 2\log 2$; for $a = 3/4$ it fails, since $4 > 1 + 2\log 4$. Thus it supplies a sufficient restriction on the squared radii, not a necessary condition for the geometric-mean inequality.

The proof uses point-dependent logarithmic bounds. Write $H_i = m^{-1}\sum_j \log|1-\bar{c}_i c_j|$ and $D_i = -\log(1-|c_i|^2)$. The preceding logarithmic calculation gives $\sum_i H_i \le 0$ and $H_i^2 \le -(D_i/m)\sum_j H_j$. For nonnegative upper bounds $M_i \ge H_i$, Taylor's formula with integral remainder gives

$$e^{H_i} \le 1 + H_i + \Phi(M_i)H_i^2, \qquad \Phi(t) = \int_0^1 (1-s)e^{st}\,ds.$$

Since $\Phi(M_i) \ge 0$, summing and using the quadratic bounds yields

$$\sum_i e^{H_i} \le m + \left(1 - \frac{1}{m}\sum_i \Phi(M_i)D_i\right)\sum_j H_j \le m$$

whenever $\sum_i \Phi(M_i)D_i \le m$. This is the sufficient condition in the formal inequality with individual upper bounds. Taking $M_i = L$ gives $\Phi(L)L \le 1$ as a sufficient condition, since $H_i, D_i \le L$. For $L > 0$, integration gives $\Phi(L) = (e^L - 1 - L)/L^2$; hence this condition is exactly $e^L \le 1 + 2L$. At $L = 0$ all centres vanish, so the conclusion is immediate.

Four points without a matching or numerical split. The preceding weighted proof with $m = 4$ and $w_j = 1/4$ gives

$$\sum_{j=1}^{4}\left(\prod_{k=1}^{4}|1-\overline{c_k}c_j|\right)^{1/4} \le 4, \qquad |c_j| \le 1,$$

with equality if and only if all four centres vanish. Equality in this linear form forces equality in the quadratic inequality, by the variance identity above; Lean proves that equality case. In particular, the inequality is strict as soon as one centre is nonzero, even with repeated centres or centres on the unit circle.

This four-point consequence needs neither the former 21/25 split nor a perfect-matching Hölder step. The scalar stationary-point inequalities from that approach are not a proof of its missing steps. The two restricted arguments above remain independent alternatives.

The mean inequality gives $\mu \le R^n$. In the unit-disc normalisation $R = 1$, this does not force the smaller quintic threshold $\mu < 1/M_5$ below. For $f(z) = z^n - 1$, every critical value has modulus one, so no uniformly smaller bound holds on the closed-disc class. These modulus bounds still require a separate path selection or metric comparison before they yield an inverse-ray length estimate.

A sufficient critical-value threshold in degree five

The reciprocal balance at a critical point bounds the sum of the two nearest root distances. We begin with this formal geometric statement, then distinguish distance from containment in the degree-five argument.

Theorem 10.5 (a geometric-mean bound for distances to a critical point)**.** *Let $n \ge 2$, let $z_1, \dots, z_n, c \in \mathbb{C}$ with $c \ne z_k$ for every k, and suppose*

$$\sum_{k=1}^{n}\frac{1}{c - z_k} = 0.$$

If $r > 0$ is determined by

$$r^n = \prod_{k=1}^{n} |c - z_k|,$$

then there are distinct indices i, j such that

$$|c - z_i| + |c - z_j| \le 2r.$$

Proof. Relabel the points so that their distances satisfy $d_1 \le d_2 \le \cdots \le d_n$. The reciprocal balance and the triangle inequality give

$$\frac{1}{d_1} = \left| \sum_{j=2}^{n} \frac{1}{c - z_j} \right| \le \sum_{j=2}^{n} \frac{1}{d_j} \le \frac{n-1}{d_2}.$$

Thus $d_2 \le (n-1)d_1$. Also $d_1 d_2^{n-1} \le r^n$, since every remaining distance is at least d_2. For $t = d_2/d_1 \in [1, n-1]$,

$$\frac{d}{dt} \log \frac{2t^{(n-1)/n}}{1+t} = \frac{n-1-t}{nt(1+t)} \ge 0.$$

The logarithm vanishes at $t = 1$, so $d_1 + d_2 \le 2(d_1 d_2^{n-1})^{1/n} \le 2r$. □

The distance bound for the two nearest roots.

Corollary 10.6 (two nearest roots)**.** *If the roots lie in the open unit disc and c is a non-root critical point, the two nearest roots to c have total distance strictly below 2.*

Let f have degree $n \ge 2$ and all roots in the closed unit disc, and let c be a critical point with $f(c) \ne 0$. List roots with multiplicity. Its two nearest roots have total distance at most 2. The root-disc hypothesis, rather than a bound on $|f(c)|$, is what gives this estimate.

Rotate so that $c = t \ge 0$, and order the distances as $0 < d_1 \le d_2 \le \cdots \le d_n$. Logarithmic differentiation gives $\sum_j (z_j - c)^{-1} = -f'(c)/f(c) = 0$. Dividing $|z_j - c|^2 + 2t\Re(z_j - c) \le 1 - t^2$ by d_j^2, then summing and using the reciprocal balance, gives

$$n \le (1 - t^2) \sum_j d_j^{-2}.$$

Thus $t < 1$ and $d_1 \le \sqrt{1 - t^2} \le 1$. The same reciprocal balance, with the nearest term isolated, gives $d_2 \le (n-1)d_1$.

Suppose $d_1 + d_2 > 2$. Since $d_2 \le 1 + t$, we have $d_1 > 1 - t$, and hence

$$1 - t^2 < d_1(2 - d_1) < d_1 d_2.$$

The first strict inequality uses that $x(2 - x)$ is increasing on $[0, 1]$. It follows that

$$n \le (1 - t^2)(d_1^{-2} + (n-1)d_2^{-2}) < \frac{d_2}{d_1} + (n-1)\frac{d_1}{d_2} \le n.$$

For the last step, put $x = d_2/d_1 \in [1, n-1]$ and expand $(x - 1)(x - (n-1)) \le 0$. This contradiction proves the closed-disc bound. For roots in the open unit disc, scaling by

a containing radius $0 < R < 1$ gives $d_1 + d_2 \le 2R < 2$. A research note gives the same argument. Lean checks the strict distance bound from the critical polynomial, including the root-disc reduction and the choice of the two nearest indices.

This argument selects the two nearest roots but does not put their segments inside a polynomial sublevel set. The next calculation supplies that separate containment estimate under a bound on $|f(c)|$.

Two failures of straight segments.

Proposition 10.7 (two counterexamples to straight-path assertions)**.** *There is a monic quintic with all roots in the open unit disc and a non-root critical point c whose unique nearest root has a point on the straight segment to c outside $\{|f| < 1\}$. There is also a monic cubic with all roots in the open unit disc such that the midpoint of every pair of distinct roots lies outside $\{|f| < 1\}$.*

Proof. For the quintic, put $p = 999/1000$ and $a = (901/902)p$, and take

$$Q(z) = (z - a)(z^2 + p^2)\left(z^2 + \frac{902}{901}pz + p^2\right).$$

Its roots are

$$a, \quad ip, \quad -ip, \quad p\frac{-451 + 780i}{901}, \quad p\frac{-451 - 780i}{901}.$$

Since $451^2 + 780^2 = 901^2$, the last four roots have modulus $p < 1$, while $0 < a < p$. Also $Q'(0) = 0$ and $Q(0) = -ap^4 \neq 0$. Thus a is the uniquely nearest root to the non-root critical point 0. At one tenth of that segment, exact rational arithmetic gives

$$|Q(a/10)| = \frac{9a}{10}\left(\frac{a^2}{100} + p^2\right)\left(\frac{a^2}{100} + \frac{11p^2}{10}\right) > 1.$$

For the cubic, take $g(z) = z^3 - (99/100)^3$. If z, w are distinct roots, then $z + w$ is the negative of the third root, so

$$g\left(\frac{z+w}{2}\right) = -\frac{9}{8}\left(\frac{99}{100}\right)^3, \qquad \frac{9}{8}\left(\frac{99}{100}\right)^3 = \frac{8732691}{8000000} > 1.$$

All three roots have modulus $99/100$. □

Containment of the two selected segments. Let f be monic of degree $n \ge 3$ with roots in the closed unit disc. The following estimate controls the segments from a critical point to its two nearest roots. To specify the constant, put

$$M_n^2 = \max_{\substack{0 \le t \le 1 \\ 4-n \le y \le n}} (1-t)^2((1-t)^2 + 2ty + n(n-2)t^2) \cdot \left((1-t)^2 + \frac{2t(n-y)}{n-2}\right)^{n-2}.$$

Here t parametrises the segment. The product estimates below use $f'(c) = 0$ and the ordering of distances, not the root-disc hypothesis; the latter enters when bounding the

combined segment length. To explain the other parameter and the coefficient $n(n-2)$, suppose $f(c) \neq 0$, order the roots $a_1, \dots, a_n$ by distance from c, and set $v_j = (a_2 - c)/(a_j - c)$. Then $v_2 = 1$, $|v_j| \leq 1$ for $j \geq 2$, and $\sum_j v_j = 0$ by $f'(c) = 0$. With $y = 1 - \Re v_1$, this gives

$$y = 2 + \sum_{j=3}^{n} \Re v_j \in [4-n, n], \qquad |v_1| \leq n-1.$$

Consequently

$$|1 - tv_1|^2 \leq (1-t)^2 + 2ty + n(n-2)t^2,$$
$$\frac{1}{n-2} \sum_{j=3}^{n} |1 - tv_j|^2 \leq (1-t)^2 + \frac{2t(n-y)}{n-2}.$$

The factor $v_2 = 1$ contributes $(1-t)^2$. Applying AM–GM to the remaining $n-2$ squared factors in $f(c + t(a_2 - c))/f(c) = \prod_j (1 - tv_j)$ gives the defining expression for M_n^2. Thus the maximisation is an explicit bound for the second segment, not an additional optimisation hypothesis about the roots.

For the nearer segment, put $x_j = (a_1 - c)/(a_j - c)$. Now $x_1 = 1$, $|x_j| \leq 1$ for every j, and $\sum_j x_j = 0$. Hence

$$\sum_{j=2}^{n} |1 - tx_j|^2 \leq (n-1)(1+t^2) + 2t,$$

so AM–GM gives

$$\left| \frac{f(c + t(a_1 - c))}{f(c)} \right|^2 \leq (1-t)^2 \left((1-t)^2 + \frac{2tn}{n-1} \right)^{n-1}.$$

This bound is at most M_n^2 for every $n \geq 3$, not just for numerically tested degrees. Indeed, set $y = n/(n-1)$ in the defining maximum. This value is admissible because $n/(n-1) - (4-n) = (n-2)^2/(n-1) \geq 0$. The last factor then has base $(1-t)^2 + 2tn/(n-1)$, and the preceding factor has this same base plus $n(n-2)t^2 \geq 0$. Their product therefore dominates the displayed nearer-segment bound. Thus $|f(z)| \leq M_n |f(c)|$ on both segments, so $0 < |f(c)| \leq 1/M_n$ gives containment in $\{|f| \leq 1\}$; the critical-point distance estimate gives total length at most 2. If $f(c) = 0$, two root occurrences already coincide at c. The quadratic case is the direct root segment treated earlier. For degree five the maximum evaluates to

$$M_5 = (1 - t_*)(1 + t_*)^3 \sqrt{16t_*^2 - 4t_* + 1}, \qquad t_* = \frac{5}{16} + \frac{3\sqrt{105}}{80}, \qquad \frac{1}{M_5} = 0.2760461\ldots.$$

The breakpoint comes from the constraint on y, not from a numerical search. For fixed $0 < t < 1$, write the two bases in the definition of M_5 as

$$A = (1-t)^2 + 2ty + 15t^2, \qquad B = (1-t)^2 + \frac{2t(5-y)}{3}.$$

They are positive for $-1 \leq y \leq 5$, and $\partial_y \log(AB^3) = 2t(B - A)/(AB)$. Thus the unconstrained maximum occurs at $A = B$, or $y = 5/4 - 45t/8$. This reaches the lower endpoint

-1 at $t = 2/5$; thereafter the maximum is at $y = -1$. For $0 \le t \le 2/5$ the resulting expression is $(1-t)(1+t/2+19t^2/4)^2$, whose derivative is

$$\frac{5t(14-19t)(19t^2+2t+4)}{16} \ge 0.$$

For $2/5 \le t \le 1$, the square of the resulting expression is $(1-t)^2(1+t)^6(16t^2-4t+1)$, whose derivative is

$$-4t(1-t)(1+t)^5(40t^2-25t-2).$$

The quadratic changes sign once on this interval, at t_*, so the second branch increases from the common breakpoint to t_* and then decreases to zero. Since the first branch increases to that breakpoint, t_* gives the global maximum. This proves the displayed value without sampling either optimisation variable. As an existence criterion this is weaker than the 13/25 theorem. Its different content is the control of the two straight segments from the specified critical point, rather than a curve selected by area averaging. For $z^5 - b$ it requires $|b| \le 1/M_5$, so it excludes part of a binomial family already covered by the trinomial theorem. No claim is made that every quintic has a critical value below this level.

A near-regular pentagon illustrates why a prescribed joining point is restrictive. For sufficiently small $\epsilon > 0$, set

$$P_\epsilon(z) = z^5 + \epsilon^5 z^4 + \epsilon^3 z^3 - \epsilon^3 z^2 - \epsilon^5 z - 1, \quad F_\epsilon(z) = r^5 P_\epsilon(z/r), \quad r = 1 - \epsilon^9.$$

All five roots of F_ϵ lie in the open unit disc, approach a regular pentagon, and have escaping segments from the origin. Here is a proof of that assertion, including the effect of the radial contraction.

The roots of P_ϵ are simple perturbations $\zeta_\epsilon = \zeta_0 + O(\epsilon^3)$ of the fifth roots of unity. The identity

$$z^5 \overline{P_\epsilon(1/\bar{z})} = -P_\epsilon(z)$$

makes the roots invariant under conjugate inversion. Uniqueness of the root near each ζ_0 therefore forces $|\zeta_\epsilon| = 1$. The roots $r\zeta_\epsilon$ of F_ϵ consequently have modulus $r < 1$. The displacement estimate follows from the implicit function theorem at each simple root of $z^5 - 1$, since the first coefficient perturbations have order ϵ^3.

For $\zeta_0 = 1, e^{4\pi i/5}, e^{-4\pi i/5}$, take $t = \epsilon/4$. Substitution into the polynomial gives

$$|F_\epsilon(rt\zeta_\epsilon)|^2 = 1 + \left(\frac{\Re\zeta_0^2}{8} - \frac{1}{512}\right)\epsilon^5 + O(\epsilon^6).$$

The coefficient is positive, since $\Re\zeta_0^2 \ge (\sqrt{5}-1)/4$. For the remaining roots, with $\zeta_0 = e^{2\pi i/5}$ or $e^{-2\pi i/5}$, take $t = \epsilon^2/4$ instead. Then

$$|F_\epsilon(rt\zeta_\epsilon)|^2 = 1 + \frac{3\sqrt{5}-5}{32}\epsilon^7 + O(\epsilon^9) > 1$$

for sufficiently small ϵ. The two scales are needed because $\Re\zeta_0^2$ is negative for this second pair; its positive linear contribution dominates at the smaller scale. Multiplication

by $r^{10} = 1 + O(\epsilon^9)$ cannot remove either positive leading term. The remainders are uniform over the five root branches, so one sufficiently small ϵ works for every segment. This is an ordinary asymptotic proof; it gives no numerical upper bound for the permitted ϵ.

The addendum [26, Proposition 1] reports the stronger assertion that no critical point of this family supplies two contained straight arms of total length at most 2. Its proof is not available. The origin calculation above does not prove it: $P'_\epsilon(0) = -\epsilon^5$ shows that the origin is not a critical point. The stronger reported assertion is not used in any theorem here. Neither the proved failure at the origin nor the reported critical-point obstruction excludes curved inverse images of rays or a different joining point.

A different local question allows a freely chosen, noncritical vertex: for some $\delta > 0$, does every monic quintic with open-disc roots within δ of a rotated regular pentagon admit two contained straight arms through some $h \in \mathbb{C}$ of total length below 2? Neither the origin nor a critical point is prescribed in this question. The sufficient low-critical threshold preceding this paragraph is unchanged.

Area of a connected sublevel set. The following calculation applies in every degree $n \geq 2$, independently of the degree-five segment criterion. Let f be monic of degree n and let $t > 0$ with $K_t = \{|f| \leq t\}$ connected. Translate its root centroid to 0, so $f(z) = z^n + c_{n-2}z^{n-2} + \cdots$, and let ψ be the exterior map normalised at infinity with positive leading coefficient. For a regular connected lemniscate, the identity $f(\psi(\zeta)) = t\zeta^n$ is [8, (2.3)] after scaling the value level. The following argument also permits critical boundary points. The exterior Green functions $(\log|f| - \log t)/n$ and $\log|\psi^{-1}|$ agree: both vanish at the boundary and have a logarithmic pole of coefficient one at infinity. Comparison there gives the leading coefficient $t^{1/n}$ for ψ. The quotient $f(\psi(\zeta))/(t\zeta^n)$ has constant modulus one and limit one at infinity, proving the polynomial identity. Its ζ^{n-1} coefficient forces the constant Laurent coefficient of ψ to vanish. Thus

$$\psi(\zeta) = t^{1/n}\zeta + \sum_{k\geq 1} a_k \zeta^{-k}.$$

Grönwall's area identity takes the form

$$\operatorname{Area}(K_t) = \pi\left(t^{2/n} - \sum_{k\geq 1} k|a_k|^2\right).$$

One can obtain it directly without assuming that level t is regular. For $r > 1$, Green's area formula on the analytic Jordan curve $\psi(re^{i\theta})$ gives the enclosed area $\pi(t^{2/n}r^2 - \sum_{k\geq 1} k|a_k|^2 r^{-2k})$. These enclosed compact sets decrease to K_t as $r \downarrow 1$; continuity of area from above and monotone convergence of the series give the identity. Comparing the coefficient of ζ^{n-2} in the exterior identity now gives

$$nt^{(n-1)/n}a_1 + c_{n-2}t^{(n-2)/n} = 0, \qquad a_1 = -\frac{c_{n-2}}{n\,t^{1/n}}.$$

The nonnegative coefficient sum quantifies the loss from Pólya's upper bound. It does not by itself control an internal path length. No analogous coefficient formula for an individual component of a disconnected sublevel set is established here. The degree-five statements and their evidence classes are recorded in the degree-five estimates and their proofs.

Numerical tests of selection by the smallest critical value

We now ask whether a small critical value selects a short pair of inverse arms. The following records test that selection rule and a related sum over critical points. For a critical point c whose two inverse arms over $[0, f(c)]$ reach roots, write $L_f(c)$ for their total length. The first computation tests the rule of choosing the critical point with uniquely smallest value modulus. A computational record gives an unnormalised quintic f_0, a selected critical point c_0, and the numerical values

$$L_{f_0}(c_0) \approx 2.057343275393654508, \qquad R \approx 1.021393477405696164,$$

where R is the radius of a smallest disc containing the roots of f_0. The first number is not the length after normalisation to the unit disc. If h is the centre of that disc, set

$$F(z) = R^{-5} f_0(h + Rz).$$

This preserves monicity, sends the roots into the closed unit disc, and divides every corresponding curve length by R. The reported normalised values are therefore

$$L_F\left(\frac{c_0 - h}{R}\right) \approx 2.01425143287505, \qquad \left|F\left(\frac{c_0 - h}{R}\right)\right| \approx 0.899569245.$$

A further contraction uses $F_s(z) = s^5 F(z/s)$, with $0 < s < 1$: roots and curve lengths are multiplied by s, and critical values by s^5. Thus the reported excess would persist for s sufficiently close to 1, with roots in the open unit disc.

These observations do not yet constitute a certified counterexample. A small residual at a computed fibre root does not enclose that root or establish its membership in the tracked inverse branch. The inscribed-polyline comparison needs both facts before it gives a rigorous lower bound for the intended arms. Another critical point supplies a shorter pair in the reported computation, which establishes no universal selection rule.

Second, a second computational record reports numerical violations of $\sum_c L_f(c) \le 2(n-1)R$ in degrees four and five. Here R is the radius of a smallest disc containing the roots. The quartic computation gives $\sum_c L_f(c) = 6.006352157\ldots$ against the proposed bound 6. Several differential-equation solvers agree, and an inscribed-polyline calculation using numerically computed fibre roots gives a similar excess. These checks do not provide rigorous enclosures for those roots, their branch assignment or the resulting length lower bound. They suggest failure of the aggregate inequality, but do not certify an example or an open set of violating polynomials. Its algebraic factor

$$\sum_{k=1}^{n-1} |f(c_k)|^{1/n} \le (n-1)R$$

is proved for every $n \ge 2$ in Theorem 10.2. The value estimate alone supplies no bound for the associated path-length sum. The algebraic theorem therefore remains useful independently of how the numerical aggregate examples are resolved.

Third, all five origin segments can escape even when the roots are arbitrarily close to a regular pentagon. The two-scale calculation above proves this for the family F_ϵ

attributed to the earlier note [26, Proposition A of the earlier note]; an isolated numerical example would not prove the assertion for arbitrarily small defects. Failure for every critical joining point is a different, stronger claim, reported in the addendum but not established by the origin calculation. No conclusion about curved inverse-ray paths or an arbitrary connecting curve follows from either straight-segment assertion.

Unformalised remark: limits of contained paths at fixed degree

The compactness argument linked from the short paper concerns arbitrary curves in a closed lemniscate. We give its ordinary proof here. It has not been formalised, and it gives no attainment statement for paths confined to an open lemniscate.

Fix $n \geq 2$, and let $\mathcal{K}_n$ be the compact coefficient class of monic degree-n polynomials with roots in the closed unit disc. Define

$$\Lambda(f) = \inf_{\gamma} \operatorname{length}(\gamma),$$

where $\gamma \subset \{|f| \leq 1\}$ joins two different listed root occurrences. A repeated root permits a constant curve; an empty set of competitors has infimum $+\infty$.

Passing a length bound to a limit. Let $f_j \to f$ in $\mathcal{K}_n$. If the lower limit of $\Lambda(f_j)$ is finite, pass to a subsequence on which these infima converge to it and choose rectifiable curves γ_j with lengths at most $\Lambda(f_j) + 1/j$. Every closed unit sublevel lies in $|z| \leq 2$: for $|z| > 2$, the root factorization gives $|f_j(z)| \geq (|z| - 1)^n > 1$. Constant-speed parametrisation on $[0, 1]$ therefore gives uniformly bounded images and Lipschitz constants. Arzelà–Ascoli supplies a uniformly convergent subsequence.

To retain the endpoints with their multiplicities, list the two chosen occurrences first among the roots of f_j, and list the remaining roots arbitrarily. Compactness of the product of n closed unit discs gives a further subsequence on which this entire list converges. The limiting factorization is f, so its first two entries remain two occurrences, even if their locations coincide. Uniform convergence of the curves and of the polynomials on $|z| \leq 2$ gives $|f| \leq 1$ on the limiting curve γ. For any partition $0 = t_0 < \cdots < t_M = 1$,

$$\sum_{\ell=1}^{M} |\gamma(t_\ell) - \gamma(t_{\ell-1})| = \lim_j \sum_{\ell=1}^{M} |\gamma_j(t_\ell) - \gamma_j(t_{\ell-1})| \leq \liminf_j \operatorname{length}(\gamma_j).$$

Taking the supremum over partitions proves the required lower semicontinuity of length and hence $\Lambda(f) \leq \liminf_j \Lambda(f_j)$. An infinite lower limit needs no argument. With f fixed, the same compactness proof shows that a finite infimum is attained. Thus a bound $\Lambda \leq 2$ on a dense class gives an actual path of length at most 2 throughout $\mathcal{K}_n$. For an open-disc polynomial, choose a containing disc of radius $0 < R < 1$ and scale: the resulting path has length at most $2R < 2$ and lies in $\{|f| \leq R^n\} \subset \{|f| < 1\}$.

For $f(z) = z^n - 1$, the inequality $|z^n - 1| \leq 1$ implies $2\Re(z^n) \geq |z|^{2n} > 0$ whenever $z \neq 0$. Thus the punctured sublevel lies in n disjoint angular sectors, one per root. A path between different roots must pass through zero and has length at least 2 by the triangle inequality. The two radial segments attain that length. The corresponding historical assertion on the closed class is

$$\boxed{\Lambda(f) \leq \Lambda(z^n - 1) = 2 \qquad (f \in \mathcal{K}_n).}$$

The lower-semicontinuity argument is proved in the compactness proof.

The displayed universal inequality is false in degree seven. Indeed, if it held throughout $\mathcal{K}_7$, the scaling step above would give a path of length less than 2 inside the open lemniscate of the polynomial in Theorem 1.2. That theorem excludes such a path. Thus no estimate on a dense class can imply the displayed universal inequality by this compactness argument. The reduction remains valid for separately specified polynomial classes.

Unformalised remark: Blaschke products and varying degree

Powers of a Blaschke product give critical values close to one another while also admitting short connectors in a suitable asymptotic regime. The finite certificates and the ordinary limiting arguments have different scopes, which we keep separate below.

A research note on powers of Blaschke products uses the degree-$2N$ polynomials

$$F_N(z) = [z(z+b)]^N - (1+bz)^N, \qquad N \geq 1, \quad 0 < b < 1,$$

and their radial contractions. Its finite rational certificates and its asymptotic proofs have different roles. The degree-eight certificate excludes an unconditional critical-value exponent $8/7$. It does not contradict the ordinary exponent $4/7$ proved by the Poisson argument above, or the smaller exponent $2/7$ in Theorem 10.2. For $N \geq 2$, the root centroid of F_N is $-b/2$, as its z^{2N-1} coefficient is Nb. A radial contraction by $r > 0$ changes this to $-rb/2$, not zero. Thus the degree-eight example does not satisfy the additional centroid condition that permits numerator 8 above. The degree-twenty-four certificate gives critical values in $|v+1| < 1/12$, so $\mu > 11/12$ while every radius-$4/3$ separation test fails. These fixed-degree examples do not establish optimality uniformly in degree. That conclusion uses the limiting measures in Sections 2–3 of the note.

For $b = \lambda/N$ with fixed $0 < \lambda \leq 1$, Section 4 constructs connectors of length at most $[\log(2/\lambda) + \pi + o(1)]/N$. The two radial pieces reduce the oscillatory term; a short circular arc then joins them in a sector where the remaining factor has modulus below one. The unspecified starting degree depends on λ. For $b = e^{-\alpha N}/N$ with fixed $\alpha > 0$, the same section instead proves $\Lambda(F_N) \to 2(1 - e^{-\alpha/2})$ for the uncontracted polynomials F_N displayed above and the closed-level functional defined earlier. The second assertion needs its own argument: the first does not allow a varying parameter to be substituted. Both use ordinary asymptotic arguments. Neither follows from the finite certificates, and neither has a formally checked analytic proof. They concern varying degree and do not contradict fixed-degree compactness or prove the unrestricted assertion.

What the earlier inverse-ray approach would have required

The earlier proposed sufficient condition concerned *curved* inverse images of rays, not straight segments. On a generic ray-separated class let $L_f(c)$ be the length of the two-root descent arc at an admissible critical point, where $|f(c)| \leq 1$. Then

$$\min_{c \text{ admissible}} L_f(c) \leq 2$$

would imply the existence of a short curve joining two roots by the stated compactness argument. The unrestricted infimum over all contained curves can be smaller; equality

of these two optimisation problems is not assumed. The obstruction to two straight segments through a critical point does not by itself prove or disprove this curved-arc estimate. Examples in which another critical point gives a shorter curve do not prove the estimate universally. The critical value bound, containment and length bound must hold for the same curve. The displayed universal inequality is a historical sufficient condition, not an open assertion: together with the stated compactness and scaling argument it would contradict Theorem 1.2. The reduction remains useful on separately specified polynomial classes.

An integral for successive merger scales

Consider a component containing k roots that later joins a larger component as the level of $|f|$ increases. If $r \in (0,1]$ is the ratio of these two merger scales, one logarithmic contribution assigned to each of its roots is $k^{-1}\log(1/r)$. An earlier length estimate instead involves the integral

$$I_k(r) = \int_r^1 \frac{dq}{q\log((1+q^{2/k})/(1-q^{2/k}))}.$$

For nearby merger scales, r is close to 1. The integral is then much smaller than $k^{-1}\log(1/r)$. The following change of variables proves that no fixed positive multiple of this logarithmic contribution is a uniform lower bound.

Theorem 10.8 (the relation between two merger-scale integrals)**.** *Define*

$$\Phi(x) = \int_0^x \frac{dt}{\log(\coth t)} \qquad (x \geq 0).$$

At $t = 0$ the integrand is understood by its continuous limiting value 0 (equivalently, the integral is improper at that endpoint). For every integer $k \geq 1$ and $0 < r \leq 1$, with $x = k^{-1}\log(1/r)$,

$$I_k(r) = k\Phi(x). \tag{O1}$$

The function Φ is increasing and strictly convex on $(0,\infty)$, and

$$\frac{\Phi(x)}{x} \longrightarrow 0 \qquad (x \downarrow 0). \tag{O2}$$

Consequently, for every fixed $k \geq 1$ and every $c > 0$, some $0 < r < 1$ satisfies

$$I_k(r) < c\,\frac{1}{k}\log\frac{1}{r}. \tag{O3}$$

In particular, no positive universal constant bounds $I_k(r)$ below by that constant times $k^{-1}\log(1/r)$.

Proof. Set $q = e^{-kt}$. Then $dq/q = -k\,dt$, the endpoints $q = 1, r$ become $t = 0, x$, and

$$\frac{1+e^{-2t}}{1-e^{-2t}} = \coth t,$$

which proves (O1). Since $\coth t$, and hence $\log(\coth t)$, is strictly decreasing, the positive integrand $1/\log(\coth t)$ is strictly increasing. This proves monotonicity and strict convexity. Moreover

$$0 \le \frac{\Phi(x)}{x} \le \frac{1}{\log(\coth x)} \longrightarrow 0,$$

which gives (O2); taking $r = e^{-kx}$ gives (O3). □

As r approaches 1, two successive merger scales approach each other. The integral can then be arbitrarily small relative to the logarithmic contribution, even for fixed k. Any estimate along a chain of mergers must retain this nonlinear dependence. Related weighted Jensen estimates and a lemma selecting two roots with small total logarithmic contribution are given in a research note. Those inequalities do not by themselves construct pieces of a contained path that meet one another.

11 The Newton value equation

Along a Newton trajectory the polynomial value decays exponentially. We first derive this scalar equation, then use it to exclude a finite connection between critical points whose values lie on different rays. For the classical trajectory and graph statements we follow Kozen and Stefánsson [7, Lemmas 2.1–2.2 and §2].

Sutherland uses the term Newton flow for $\dot z = -f(z)/f'(z)$ and notes that its solution curves map under f to radial lines [6, p. 42]. The value equation [7, Lemma 2.1] determines their orientation and time parametrisation. The global Newtonian graph [7, §2], described with its endpoint conventions below, is a separate input to the inverse-sheet discussion in Section 13.

A Newton trajectory is a differentiable curve satisfying

$$z'(t) = -\frac{f(z(t))}{f'(z(t))}$$

where $f'(z(t)) \neq 0$. Let $I \subseteq \mathbb{R}$ be an interval on which these assumptions hold; put $w(t) = f(z(t))$. Kozen and Stefánsson record the following identity as a lemma of Shub, Tischler and Williams [7, Lemma 2.1].

Theorem 11.1 (value equation)**.** *For a polynomial f and a differentiable curve $z : I \to \mathbb{C}$ on an interval I, assume $f'(z(t)) \neq 0$ and $z'(t) = -f(z(t))/f'(z(t))$ throughout I. Then $w = f \circ z$ satisfies $w' = -w$, and*

$$f(z(t)) = e^{-(t-t_0)} f(z(t_0)) \qquad (t, t_0 \in I).$$

The computation is one line: $w' = f'(z)\,z' = f'(z) \cdot (-f(z)/f'(z)) = -f(z) = -w$. Equivalently,

$$\frac{d}{dt}\big(e^t f(z(t))\big) = 0,$$

and integration on the real interval gives

$$f(z(t)) = e^{-(t-t_0)} f(z(t_0)), \qquad t_0 \le t, \quad t_0, t \in I.$$

Lean proves the theorem together with this integrated identity. The separate local complex-parameter calculations are the value derivative and the scaled derivative. They require complex differentiability and do not themselves integrate a trajectory on a real interval. For an existing trajectory with nonzero initial value, its value moves inward on one positive ray; a zero value has no argument and remains zero. Thus $|f| < 1$ is preserved for later times in that trajectory's existing interval. No global existence or description of a whole ray preimage follows from this scalar equation.

Corollary 11.2 (ray separation)**.** *Let $a < b$ and let the value trajectory $t \mapsto f(z(t))$ be continuous on $[a, b]$. Assume the Newton equation and $f'(z(t)) \neq 0$ on (a, b) only. Then*

$$f(z(b)) = e^{a-b} f(z(a)).$$

If these endpoint values are nonzero, they lie on one positive ray. Therefore critical points with values on distinct positive rays cannot be endpoints of such a finite connection. The trajectory in the z plane need not be radial.

The interior identity passes to the endpoints by continuity, not by evaluating $-f/f'$ at a critical endpoint. The corollary constructs no trajectories and supplies no global monodromy theorem. The Lean ray exclusion assumes the exponential endpoint identity; it does not construct or integrate a connecting trajectory. Kozen and Stefánsson draw the same ray consequence for the Newtonian graph: under f, every edge maps onto a segment of a ray through the origin whose endpoints are 0 or critical values [7, §2].

12 Separating the rays of critical values

A constant perturbation translates the critical values and leaves the critical points fixed. We can therefore separate their positive rays by describing the exceptional translation parameters: two nonzero values collide in argument exactly when their ratio is positive real.

Along a Newton trajectory the argument of f stays constant while its modulus decreases. By Corollary 11.2, arranging pairwise distinct arguments of the nonzero critical values is therefore a sufficient way to exclude finite connections between critical points. It is not asserted to be necessary. This condition imposes no lower bound on the distances between critical values and no upper bound on their moduli.

The weaker conditions really can coexist with a connection, even when all roots lie in the open unit disc. Take

$$f(z) = z^3 - \frac{3}{16}z + \frac{3}{64}.$$

On $|z| = 1$, the lower terms have total modulus at most $15/64 < 1$, so Rouché's theorem places all three roots inside. The critical points are $-1/4$ and $1/4$, with distinct positive values $5/64$ and $1/64$. On the intervening real interval f is positive and strictly decreasing. The equation

$$f(x(t)) = \frac{5}{64}e^{-t}, \qquad 0 < t < \log 5,$$

therefore defines a differentiable path with $x'(t) = -f(x(t))/f'(x(t))$. Its endpoint limits are $-1/4$ and $1/4$. It is an actual critical-to-critical Newton trajectory, not merely a pair of values on one ray. The time convention is that of the value equation [7, Lemma 2.1]; the singular field is not evaluated at the endpoints.

A common translation of distinct values can separate their arguments. The next calculation describes the translations to avoid.

Theorem 12.1 (ray-collision locus)**.** *Let $a \neq b$ be complex. Every common translation β for which $a + \beta$ and $b + \beta$ lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \qquad r \in \mathbb{R}_{>0},\ r \neq 1.$$

Indeed, write $b + \beta = r(a + \beta)$ with $r > 0$. Since $a \neq b$, one has $r \neq 1$, and solving for β gives the displayed formula. Equivalently, $\beta = -a + (a - b)/(1 - r)$, so the forbidden translations lie on the real affine line through $-a$ and $-b$.
A finite union of these loci has empty interior. Adding a constant to f translates all critical values by that constant and leaves the critical points unchanged. Section 15 records the formal avoidance and root-retention estimates. A constant cannot separate equal critical values; coefficient perturbation and control of the remaining geometric margins are separate requirements.

Local polynomial expansions

For $f(z) = \prod_j (z - a_j)$ with $a_j \neq 0$, put $r = \min_j |a_j|$ and $p_m = \sum_j a_j^{-m}$. On $|z| < r$, the exact expansion

$$\log|f(z)| = \log|f(0)| - \Re \sum_{m \geq 1} \frac{p_m}{m} z^m$$

expresses the local level geometry through the reciprocal power sums. If $q = |z|/r < 1$, the tail after N has modulus at most $nq^{N+1}/((N + 1)(1 - q))$; differentiated geometric series give the corresponding first and second derivative bounds. A research note gives recurrences for these reciprocal power sums and tests based on the first two derivatives for a locally contained piecewise path. The assumption $|z| < r$ is essential: this disc contains no zero of f, so a logarithm exists there. The expansion does not continue through a root and does not select a pair of roots to join. A proposed local path must still satisfy the containment inequalities.

13 Why the proposed spanning-tree estimate fails

The spanning-tree estimate in [10, Proposition 12] fails already for a quadratic Cassini lemniscate. That manuscript, posted by `shtuka` on 24 March 2026, claims the unrestricted statement [10, Theorem 1, p. 1]. The defect was located publicly in the problem's discussion thread: on 25 March 2026 Tao observed that the invocation of Lemma 8 there is unjustified and that the flow lines need not organise into connected trees, and on 26 March 2026 the manuscript's author agreed that the statement of Proposition 12 itself is incorrect. We give the explicit numerical contradiction first. We then examine the

saddle model and give an unformalised decomposition into inverse sheets, including the arcs that join them.

Recent work on polynomial lemniscates separates component counts from metric path questions. Ghosh and Ramachandran record that the open set $\{|f| < 1\}$ has $1 + \#\{j : |f(c_j)| \geq 1\}$ components, where $c_1, \dots, c_{n-1}$ are the critical points listed with multiplicity [5, Lemma 2.5]; the underlying component-wise Riemann–Hurwitz count appears in the proof of [8, Proposition 2.1]. For the binomial family $z^n - a$, the condition $|a| < 1$ therefore puts the filled unit lemniscate in the connected regime. Connectedness alone gives no path-length bound. Bishop, Eremenko and Lazebnik describe the possible shapes: every rational lemniscate is a lemniscate graph whose vertices are the critical points on the level set, each of even degree at least four [20, Definition 1.2 and Proposition 2.4], and a lemniscate graph is realised by a polynomial lemniscate up to a homeomorphism of the plane exactly when it is the boundary of its unbounded face [20, Corollary 1.6]. This topological description also gives no path-length bound.

The March manuscript's Proposition 12 claims the following estimate. For $u = -\log|f|$, a connected component V of $\{u > c\}$ carrying $m \geq 2$ simple zeros, and the regularity and Morse hypotheses in that proposition, it constructs, for every $\varepsilon > 0$, an embedded spanning tree $G_\varepsilon \subset V$ with

$$\operatorname{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t)\, dt + \varepsilon. \tag{5.1}$$

Here $P_V(t) = \mathcal{H}^1(V \cap \{u = t\})$ is the length of the level set in V, and $\alpha > 0$ is the truncation parameter in the proposed estimate. The final theorem depends on this bound.

The estimate itself is false. Take $f_a(z) = z^2 - a^2$ with $a = 9/10$ and choose $2\alpha = -\log a$. For $0 < c < \alpha/2$, the component V of $\{-\log|f_a| > c\}$ contains $\{|f_a| \leq a\}$ and both roots. The only critical point is a nondegenerate saddle at 0; distinctness of critical values is vacuous.

Here the level-length integral can be bounded directly. Put $r = e^{-t}$. On a regular level the inverse parametrisations $z = \pm\sqrt{a^2 + re^{i\theta}}$ give

$$P_V(t) = r \int_0^{2\pi} |a^2 + re^{i\theta}|^{-1/2}\, d\theta.$$

The elementary identity

$$|a^2 + re^{i\theta}|^2 = (a^2 + r)^2 \cos^2(\theta/2) + (a^2 - r)^2 \sin^2(\theta/2)$$

implies

$$|a^2 + re^{i\theta}|^{-1/2} \leq (a^2 + r)^{-1/2} |\cos(\theta/2)|^{-1/2}.$$

The angular factor has mean at most 2. Indeed, symmetry reduces its integral to $4\int_0^{\pi/2} (\cos s)^{-1/2}\, ds$, and $\cos s \geq 1 - 2s/\pi$ bounds the latter integral by 4π. Changing variables from t to r therefore gives

$$\frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t)\, dt \leq 2 \int_0^{a} \frac{dr}{\sqrt{a^2 + r}} = 4(\sqrt{a^2 + a} - a) < \frac{41}{25}.$$

The isolated critical level is handled by the corresponding improper integral. Every connected set containing both roots has length at least $2a = 9/5$. Since $9/5 - 41/25 = 4/25$, any $0 < \varepsilon < 4/25$ contradicts (5.1). Thus neither a different local saddle neighbourhood nor a perfect topological decomposition can recover the printed coefficient $1/(2\pi)$. The exact numerical inequality and the resulting contradiction for a tree-length bound also have a Lean proof, in the repository's research corpus, a library the checked build does not compile. The level-length majorant is an explicit analytic input.

At an interior index-one critical point p, the proof invokes a Morse chart

$$u = u(p) + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because $p \in V$ and V is open, a sufficiently small closed disc around p lies entirely in V. In that disc the full Morse chart has four sectors: two components of $u > u(p)$ and two components of $u < u(p)$. The level set $\{u = u(p)\}$ is a lemniscate graph with a vertex of degree four at p [20, Definition 1.2 and Proposition 2.4]. A global component argument cannot delete one local sector from a disc already contained in V.

The four-sector calculation identifies an error in the proof, while the Cassini example above disproves the printed metric inequality itself. Possible alternative constructions include cutting an adjoining regular annulus along a separatrix or regular flow arc, retaining a four-pronged saddle neighbourhood, or using the inverse-sheet decomposition below. None can restore (5.1). A revised estimate would have to include the length needed to join the pieces, seek only one short pair instead of a tree spanning all roots, or use a different global metric inequality. The shorter descriptions of the same three-ended block do not repair the four-sector topology.

The following ordinary topological argument assumes that the critical values have pairwise distinct arguments and moduli and that the critical points are simple. Its ingredients are classical: the Newtonian graph of Shub, Tischler and Williams [7, §2], the component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1], and the division of the Riemann surface of the inverse function into sheets by outward critical-value slits. Dubinin describes the sheet adjacency tree directly in [21, §1, pp. 1169–1170] and uses a related network for capacity in [9, §2, pp. 34–35]. The latter paper assumes bounded critical values and controls capacity; it does not supply a Euclidean tree-length bound. The remark below specifies the sheets, critical transpositions and descent arcs for the stated component.

The assumptions exclude a component containing a multiple critical point or two critical values on the same ray. For $z^n - b$ with $n > 2$ and $0 < |b| < 1$, the component containing the origin fails the first condition. They hold, for example, for $f(z) = z^3 - (3/25)z + 1/500$: its critical points are $\pm 1/5$ and its critical values are $-7/500$ and $9/500$. On the unit circle the two lower terms have total modulus at most $61/500 < 1$, so all three roots lie in the open unit disc by Rouché's theorem. In the statement, "excellent" means that the critical levels of the Morse function are pairwise distinct. This regularity permits a decomposition into sheets; it does not bound the total length of the resulting tree.

Remark 13.1 (unformalised decomposition into inverse sheets). Let f be monic, and let U be a component of $\{|f| < 1\}$ containing $k \geq 2$ roots, counted with multiplicity. Suppose every critical point of f in U is simple, its critical value is nonzero, and these critical values have pairwise distinct arguments and pairwise distinct moduli. All preimages and sheets below are taken inside U, and only critical points in U determine the cuts. Then:

1. $-\log|f| : U \setminus f^{-1}(0) \to (0,\infty)$ is a proper excellent Morse function, with exactly $k-1$ nondegenerate saddles;
2. cutting $\mathbb{D} \setminus \{0\}$ along the critical-value rays decomposes its preimage in U into conformal strips;
3. cutting each ray only from its critical value to the unit circle gives k conformal sheets, one per root, whose critical transpositions form a tree;
4. for each critical point $c \in U$, the two inverse lifts of $[0, f(c)]$ join two roots through c inside $U \cap \{|f| \leq |f(c)|\}$, and the union of these arcs is an embedded geometric realisation of that tree.

Small neighbourhoods of the saddles can be chosen with diameter $O(\sqrt{\delta})$ at value radius δ.

No supplied Lean theorem proves the whole decomposition. The sources check finite ray, level-separation and saddle-scale facts. The linked component-local declarations below also prove covering and continuous inverse-branch prerequisites. The Morse, conformal strip, sheet-count, monodromy and embedded-tree assertions remain ordinary arguments.

Inverse-branch prerequisites. For a nonconstant complex polynomial, each component of $\{|f| < 1\}$ contains a root. The restriction to that component is onto the unit disc and has finite fibres. Over values avoiding the critical values attained in that component, it is a covering map. Choose finitely many nonzero slit starts that include every critical value attained in the component, and remove the outward rays beginning at those starts from the value disc. On the remaining domain, each root in the component determines a unique continuous inverse branch, taking 0 to that root. The archived Lean source states these prerequisites and supplies proof scripts; a kernel-check receipt for this module is not supplied here. The full conformal sheet decomposition, sheet count, monodromy and embedded-tree assertions remain ordinary arguments here; the prerequisites give no length estimate.

Proof of Remark 13.1. The map $f : U \to \mathbb{D}$ is proper of degree k, and U is simply connected. To apply the count from the proof of [8, Proposition 2.1] only at a regular level, join all roots and critical points in U by finitely many compact paths in U. Choose a regular $t < 1$ above the maximum of $|f|$ on their union. One component of $\{|f| < t\}$ then contains exactly the roots and critical points of U; Riemann–Hurwitz there gives $k-1$ simple critical points. No regularity of the level-one boundary is needed. The holomorphic Morse lemma makes these points nondegenerate saddles of $-\log|f|$, and their distinct moduli separate the critical levels. A compact range in $(0,\infty)$ stays away from both the level-one boundary and the deleted roots, proving properness of the Morse function.

Remove 0 and the rays determined by these $k-1$ values. Each remaining sector is simply connected and contains no branch value of $f : U \to \mathbb{D}$, even if it contains critical values coming from outside U. Each component of its preimage in U maps biholomorphically to the sector. If the sector has angular range (θ_1, θ_2), the holomorphic coordinate $-\log f$ maps this component onto the semi-infinite strip

$$\{x + iy : x > 0,\ -\theta_2 < y < -\theta_1\}.$$

Thus its real coordinate is $-\log|f|$, while its imaginary coordinate is $-\arg f$. The minus sign in both coordinates is needed for a holomorphic map; keeping $+\arg f$ would reverse orientation.

For the sheet tree, remove only the outward slits $\{tf(c) : 1 \le t < 1/|f(c)|\}$, for critical points $c \in U$. The remaining value domain is star-shaped about 0, hence simply connected, and its preimage in U splits into k sheets labelled by the roots. Local monodromy at each slit endpoint is a transposition of two of these sheets. Deleting the finitely many fibres over the branch values from U leaves a connected domain, so the monodromy action is transitive. The $k-1$ transpositions therefore give a connected graph with k vertices, which is a tree.

For $c \in U$, the inward segment from $f(c)$ to 0 contains no other branch value of $f : U \to \mathbb{D}$, by the distinct-argument hypothesis. Distinct moduli were needed only to separate the Morse levels. Its two inverse lifts converge to c by the holomorphic Morse coordinate and start at the two root labels exchanged by its transposition. These labels are distinct, as are the roots, because a root in U cannot also be a critical point under the nonzero-critical-value hypothesis. Lifts belonging to different critical points cannot cross away from roots, since an intersection would give one nonzero value on two critical rays. Their union is therefore an embedded tree. Each edge is rectifiable: at a simple critical endpoint, parametrising the value segment by its distance s from the critical value gives speed $O(s^{-1/2})$, which is integrable. The root endpoint is regular, and the derivative is bounded on compact subsegments between the endpoints. This proves finiteness of each length, not a uniform bound for it. The same local Morse coordinate identifies a value disc of radius δ with a spatial neighbourhood of diameter $O(\sqrt{\delta})$. The constant and the permitted range of δ depend on f and the chosen saddle. Indeed, in the coordinate $f(z) - f(c) = u^2$, the inverse map has derivative of modulus $\sqrt{2/|f''(c)|}$ at $u = 0$. No bound uniform over degenerating critical points is asserted. For a fixed polynomial there are finitely many saddles, so their individual constants have a finite maximum. □

The complete argument is also written out in the proof of the decomposition into inverse sheets. This remark supplies a topological construction with no useful bound on the sum of lengths: the canonical inverse-ray tree can already have the Cassini deficit discussed above.

The component restriction cannot be dropped. For example, take

$$f(z) = (z^2 - 4)^2 - \tfrac{1}{4}.$$

Its critical points are $0, \pm 2$, with values $63/4, -1/4, -1/4$. The positive roots $\sqrt{7/2}$ and $\sqrt{9/2}$ lie in one component U: on the intervening real interval, $-1/4 \le f \le 0$. But

$f(iy) = (y^2+4)^2 - 1/4 > 1$ for real y, so this component cannot contain the negative roots or -2. Its only critical point is the simple point 2. Thus the theorem applies in U, although the other component has the same critical value. This example concerns component topology; no root-unit-disc hypothesis is imposed here.

Corollary 11.2 also excludes finite saddle-to-saddle Newton connections inside U. The theorem does not treat coincident arguments, simultaneous critical levels or multiple saddles in that component. Nor does it identify a slit domain with the quotient by all trajectories: the quadratic example following Problem 15.5 satisfies its hypotheses but has a non-Hausdorff orbit quotient. Boundary tangencies and quantitative estimates for joining the strips remain separate questions. The embedded tree supplies no length bound for the historical question.

14 Numerical path searches

The earlier searches discretised root configurations and tested candidate connections. Their finite scope is recorded here so that a negative search is not mistaken for a general path theorem.

A search was run over random monic polynomials with roots in the unit disc. For each sample the region $\{|f| < 1\}$ was rasterised and shortest grid paths were computed between every pair of roots. The reported grid distances were

degree 5,	500 trials :	1.1052648928
degree 6,	1500 trials :	0.8450414343
degree 8,	1500 trials :	0.6203916714
degree 10,	1500 trials :	0.4303640486.

No counterexample candidate was found in these trials. The reported grid distances are below 2, but they do not certify lengths of curves in the lemniscate. A polygon whose vertices satisfy $|f| < 1$ can cross the boundary along an edge. Joining an approximate root to its grid vertex introduces another edge requiring the same check. Turning a candidate into a proof needs a continuous certificate on every edge, for instance strict positivity of the real polynomial $1 - |f(z(t))|^2$ on $[0,1]$ for each straight edge $z(t) = u + t(v-u)$, established by exact coefficients or outward-rounded interval arithmetic with subdivision, together with exact root enclosures. No such certificate is attached to the numbers above. They are grid distances for the sampled configurations, and the apparent decrease with degree is a property of the sample, from which we draw no conjecture. Even a finite collection of proved instances would not establish the universal statement. These searches do not test `ani`'s degree-seven counterexample. Numerical tests of a particular path construction should also examine near-degenerate saddles, thin necks, boundary-critical configurations and almost-connected separatrices. Any paths found on a grid still require verification along their entire edges.

15 Further questions about topology, length and perturbation

The degree-seven example refutes the unrestricted formulation. The questions below concern the topology and quantitative estimates needed on restricted classes. They pre-

serve the distinction between a valid inverse-sheet decomposition, bounds for its pieces, and a bound for a contained curve assembled from those pieces.

1. Local saddle geometry and the refuted tree estimate

Problem 15.1 (corrected local saddle assembly)**.** The printed $1/(2\pi)$ spanning-tree estimate of Proposition 12 is false: the Cassini polynomial $z^2 - a^2$ at $a = 9/10$ makes the proposed upper bound strictly smaller than the distance between its roots. Any later argument must include the positive length needed to join the pieces, select only one short pair instead of spanning every root, or use a different global metric inequality. A four-pronged or cut-annulus local model may still be useful for another estimate, but it cannot recover (5.1).

The full-disc model $x^2 - y^2$, an annulus in which lower branches rejoin, two saddles joined by a separatrix and simultaneous saddle levels are mandatory tests. Repeating the one-lower/two-upper assertion does not answer the problem.

2. Cutting a compact region into flow strips

Let p be a polynomial with simple roots and simple critical points, and set $u = -\log|p|$ away from the roots. Simplicity of the roots makes every critical value nonzero; critical points themselves may lie at 0. For regular values $c < T$, let V be a component of $\{u > c\}$ and assume that $\overline{V \cap \{c < u < T\}}$ is a compact genus-zero surface, its lower boundary is one smooth Jordan curve, its upper boundary consists of m smooth level curves each enclosing one root, all interior critical points are nondegenerate index-one saddles, the normalised gradient field, defined away from the saddles,

$$X = \frac{\nabla u}{|\nabla u|^2} = -\frac{p}{p'},$$

is transverse to the level boundaries, and no maximal X-trajectory has two saddle endpoints. The equality uses the identification of a planar vector with a complex number: $\nabla u = -\overline{p'/p}$. In particular, $du(X) = 1$ wherever X is defined.

Problem 15.2 (finite strip decomposition after ray cuts)**.** For every $\eta > 0$, construct disjoint Morse neighbourhoods N_j with $\sum_j \operatorname{diam} N_j < \eta$ and a finite set of complete separatrix or regular-flow cuts so that the closure of each remaining component in the cut-open surface is flow-diffeomorphic to a rectangle

$$[a_S, b_S] \times [0, 1], \qquad u = t,$$

with connected level sections and no uncut annular component. Give an explicit finite bound for the number of strips in terms of the number s of saddles and the number m of upper boundary curves, or exhibit the minimal missing hypothesis or a counterexample.

No particular formula such as $2s + 1$ is presumed. Boundary tangencies, simultaneous levels and branch reunion through an annulus require explicit cuts. The closure is taken in the cut-open surface, with its separate boundary copies, not after regluing them in the plane. The quadratic example following Problem 15.5 explains why replacing this construction by the ordinary orbit quotient would fail.

3. Length estimates for joining flow strips

For a strip S, write

$$\Gamma_t^S = S \cap \{u = t\}, \qquad P_S(t) = \mathcal{H}^1(\Gamma_t^S),$$

and write

$$\Phi_S(t) = \int_{\Gamma_t^S} |\nabla u|\, ds$$

for its transverse flux. On a regular sub-band, u is harmonic and the side boundaries are flow lines. The divergence theorem therefore makes $\Phi_S(t)$ independent of t. Normalise the transverse measure by

$$d\mu_{t_0}(x) = \frac{|\nabla u(x)|}{\Phi_S(t_0)}\, ds(x).$$

The choice of measure is determined by the length calculation. Since $u = t$ along the normalised flow, its Euclidean speed is $1/|\nabla u|$. Transporting the flux measure to level t therefore gives

$$\begin{aligned}\int_{\Gamma_{t_0}^S} \operatorname{len}(\gamma_x)\, d\mu_{t_0}(x) &= \int_{a_S}^{b_S} \int_{\Gamma_t^S} \frac{1}{|\nabla u|} \frac{|\nabla u|}{\Phi_S(t_0)}\, ds\, dt \\ &= \frac{1}{\Phi_S(t_0)} \int_{a_S}^{b_S} P_S(t)\, dt.\end{aligned}$$

Here $0 < \Phi_S(t_0) < \infty$ for a nonempty regular strip. Tonelli's theorem applies to the nonnegative integrand. At a critical endpoint level, exhaustion through regular sub-bands gives the same identity, allowing the common value to be infinite. A finite right side gives a trajectory of length at most that mean; an identity with both sides infinite gives no length estimate. A strip flux need not be an integer multiple of 2π. For $f(z) = z$ and $u = -\log|z|$, take an annular sector of angular width θ. On a circular level arc, $|\nabla u| = 1/r$ and $ds = r\, d\phi$, so its flux is $\Phi_S = \theta$. For $f(z) = z^n$, the same calculation gives $\Phi_S = n\theta$. A closed regular level curve enclosing k roots has flux $2\pi k$ by the argument principle. Restricting it to a strip replaces that full winding by the strip's share of the flux. The averaging measure therefore depends on the width of the strip. Summing separate strip estimates cannot replace those widths by the winding number of a closed level curve. They do not reopen (5.1). Cassini already excludes the printed coefficient $1/(2\pi)$ for any embedded tree that contains all m roots: Problem 15.1 records that no later gluing argument can retain that estimate.

Problem 15.3 (strip gluing after the printed coefficient)**.** Assuming Problem 15.2, do one of the following, for every $\eta > 0$.

1. Produce an embedded tree G containing all m roots whose length obeys

$$\operatorname{len}(G) \le C \int_{2\alpha}^{\infty} P_V(t)\, dt + \eta \tag{6.1}$$

   for an explicit constant $C > 1/(2\pi)$, uniform over a stated class of polynomials and an explicitly restricted range of truncations α. The restrictions must account for attachment cost and for a valid allocation of the strip fluxes Φ_S.

2. Connect only one pair of distinct roots, rather than spanning every root.

3. Give a different global metric inequality that is not (5.1).

The printed coefficient $1/(2\pi)$ is not an open target. Better additive control of saddle, annular-cut and root-cap cost cannot recover (5.1).

The restriction on α is essential in the first alternative. For a fixed $f(z) = z^2 - a^2$, with $0 < a < 1$, choose $0 < c < -2\log a$ and let V be the component of $\{-\log|f| > c\}$ containing both roots. At sufficiently deep levels, the Cassini parametrisation used above gives

$$P_V(t) \le \frac{2\pi e^{-t}}{\sqrt{a^2 - e^{-t}}}.$$

Consequently $\int_{2\alpha}^{\infty} P_V(t)\,dt \to 0$ as $\alpha \to \infty$, whereas every spanning tree has length at least $2a$. No fixed C can cover arbitrary truncations. This is a limitation of the unrestricted formulation of (6.1), separate from the earlier counterexample to the particular coefficient $1/(2\pi)$.

The error term assigned to the collar in the printed argument

$$q = \frac{1}{2\pi}\int_{\alpha}^{2\alpha} P_V(t)\,dt > 0$$

uses the same excluded coefficient. It cannot be treated as an unused error allowance in an otherwise complete spanning-tree estimate. Shortest trajectories chosen independently in different strips may have endpoints that do not meet; the lengths needed to join them must also be bounded. Even such a bound would not restore the printed $1/(2\pi)$ tree estimate.

4. Coefficient perturbation and stability

Once a finite critical-value family is injective, Lean proves that an arbitrarily small translation makes every value nonzero and pairwise positive-ray separated. Lean also proves the explicit root-retention estimate: for a monic polynomial f of degree $n \ge 1$, a constant shift of modulus below $\varepsilon > 0$ keeps all roots in the unit disc provided

$$((n+1)\varepsilon)^{1/n} + \max_{f(b)=0}|b| < 1.$$

Adding a constant leaves the difference of two critical values unchanged, so it cannot separate values that were equal to begin with.

For a constant shift alone, factorisation gives the sharper elementary bound

$$(f+\beta)(z) = 0 \quad\Longrightarrow\quad \prod_{f(b)=0}|z-b| = |\beta| \quad\Longrightarrow\quad \min_{f(b)=0}|z-b| \le |\beta|^{1/n},$$

where the product counts roots with multiplicity. Hence $\max_{f(b)=0}|b|+|\beta|^{1/n} < 1$ already ensures retention in the open unit disc. The factor $n+1$ in the Lean estimate is not needed for this ordinary constant-shift argument. The coefficient one in the displacement bound is optimal, as $f(z) = z^n$ shows. The estimate does not match labelled root lists or bound the cost of transporting an existing curve.

Qualitatively, a small linear perturbation also supplies distinct critical values as well as simple critical points. For $\deg f \geq 2$, a multiple critical point of $f + \lambda z$ must satisfy $f''(c) = 0$ and $\lambda = -f'(c)$; only finitely many parameters are excluded. Near any other parameter, label the distinct critical points by holomorphic functions $c_j(\lambda)$. Their values

$$v_j(\lambda) = f(c_j(\lambda)) + \lambda c_j(\lambda) \quad \text{satisfy} \quad v_j'(\lambda) = c_j(\lambda).$$

Thus $v_i - v_j$ cannot vanish identically for $i \neq j$, and its zeros are isolated. Avoiding the finitely many pairs in a sufficiently small parameter neighbourhood gives simple critical points with distinct values. The neighbourhood can be chosen inside any prescribed disc about $\lambda = 0$, proving the required arbitrarily small choice.

If distinct moduli are also needed for the excellent Morse function, the translation β can avoid them at the same time as the ray collisions. For each distinct pair v_i, v_j, the equality $|v_i + \beta| = |v_j + \beta|$ is the real affine line

$$2\Re((v_i - v_j)\overline{\beta}) = |v_j|^2 - |v_i|^2.$$

For ray separation it is enough to avoid, for each pair, the real line through $-v_i$ and $-v_j$: the ray-collision formula puts every forbidden translation on that line. Avoid also the finitely many points $-v_j$ to keep the translated values nonzero. Thus the full excluded set is contained in two real affine lines per pair of critical values, together with finitely many points. Its complement meets every open disc, so imposing distinct moduli costs no lower bound on the size of the perturbation. Root retention is also qualitative: choose a circle $|z| = R < 1$ enclosing the original open-disc roots. If $|\lambda|R + |\beta| < \min_{|z|=R} |f(z)|$, Rouché's theorem keeps all roots of $f + \lambda z + \beta$ inside that circle. These ordinary arguments establish genericity and root retention; they do not estimate the collar or the cost of transferring a selected curve. The first two requirements below therefore have qualitative proofs, while their simultaneous quantitative use is the remaining task.

For the next problem, fix a specified valid replacement for the refuted metric inequality. Its slack is the difference between the permitted length and the bound supplied by that replacement. The symbols q and q_g refer to this same specified quantity for f and g, respectively; they are not the historical collar integral with coefficient $1/(2\pi)$. Without a choice of replacement inequality, the requirement $q_g > q/2$ has no defined geometric content.

Problem 15.4 (two-stage generic perturbation with slack)**.** For fixed root discs, compact collar K, regular levels $\alpha/2, \alpha, 2\alpha, 5\alpha/2$, and a specified valid replacement metric inequality with slack $q > 0$ at f, prove that an arbitrarily small

$$g_{\lambda,\beta}(z) = f(z) + \lambda z + \beta$$

can be chosen so that:

1. $f + \lambda z$ has simple relevant critical points and injective complex critical values;
2. the checked constant translation β makes those values nonzero and pairwise ray-separated;

3. no critical point enters the protected collar or truncation boundary;
4. the relevant component remains in K, contains exactly the corresponding perturbed roots and has slack $q_g > q/2$; and
5. root displacement and straight-line transfer back to the original roots consume less than a prescribed fraction of q/m.

If the one-coefficient perturbation λz cannot ensure all five properties, give the smallest additional lower-coefficient direction that can, or an explicit obstruction.

The cited declarations supply finite planar avoidance and the constant translation. The ordinary argument above additionally supplies a linear perturbation with simple critical points and distinct critical values. What remains for the stated problem is simultaneous control of the protected collar, the component, the quantitative slack and the root-to-root transfer cost. No formalisation of the new genericity argument or solution of those stability requirements is claimed.

5. Newton flow on a compact region between regular levels

Classical Newton-flow theory describes maximal trajectories and the graph they form. Away from the roots and critical points, every orbit satisfies

$$p(z(t)) = e^{-(t-t_0)}p(z(t_0)), \qquad u(z(t)) = u(z(t_0)) + t - t_0, \qquad u = -\log|p|,$$

by the lemma of Shub, Tischler and Williams [7, Lemma 2.1]. For an initial point that is neither a root nor a critical point, the maximal Newton trajectory has one of four endpoint patterns: each finite-time end is a critical point, an infinite forward end is a root, and an infinite backward end is the point at infinity [7, Lemma 2.2]. Stationary trajectories at simple roots are not part of this classification. The maximal nonconstant trajectories with endpoints among the roots and critical points form a connected Newtonian graph with finitely many edges [7, Definition 2.1 and §2]. Critical-to-critical edges are included; pairwise distinct arguments exclude them by the value equation, not by the definition of the graph.

The smooth field needed for Morse theory is not $-p/p'$, which is undefined at a critical point with nonzero value. On the root-free band the Euclidean gradient satisfies

$$\nabla u = -\overline{p'/p} = \frac{|p'|^2}{|p|^2}\left(-\frac{p}{p'}\right) \qquad (p' \neq 0).$$

It has the same oriented nonstationary trajectories as the Newton field, with a different time parameter. At a simple critical point c, its derivative is the real Hessian of u, whose eigenvalues are $\pm|p''(c)/p(c)|$. Thus it extends as a hyperbolic saddle there. A finite critical endpoint in Newton time is not a finite arrival at an equilibrium of the smooth field. These two parametrisations must be distinguished in a statement about a Morse–Smale flow.

Problem 15.5 (relative global Newton-flow theorem)**.** Let p be a nonconstant polynomial, let $a < b$ be finite regular values of $u = -\log|p|$, and suppose every critical point in the compact band

$$\{z : a \le u(z) \le b\}$$

is simple, with the critical values in the band on pairwise distinct positive rays. Describe the maximal trajectories of the smooth gradient field ∇u relative to the lower and upper level boundaries, each of which may have several components. Construct a cut-open model by regular flow strips, specifying the separate copies of cut boundaries and the equivalence relation used for its trajectory parameter space. Prove that this parameter space is Hausdorff, or identify the obstruction to the proposed identifications. The ordinary orbit quotient of the uncut band is not required to be Hausdorff. State the boundary convention and the comparison with Newton trajectories through the positive change of time above, without evaluating $-p/p'$ at its singular points.

The endpoint classification on this band follows from the cited Newton classification and the time change. Each nonconstant trajectory runs from the lower boundary to the upper boundary, from the lower boundary to a saddle, or from a saddle to the upper boundary. Indeed, Newton time equals the change in u, so the part of a Newton trajectory in the band has bounded Newton-time interval. At either end it must meet a regular boundary or tend to a critical point. Ray separation excludes two different critical endpoints, and strict increase of u excludes two ends at the same critical point. In smooth-gradient time, regular boundary crossings occur in finite time, whereas convergence to a saddle takes infinite time by uniqueness for the smooth differential equation. The remaining trajectories are the stationary saddles. This is an ordinary consequence of the classical classification, not an additional checked Lean theorem.

The planar transversality condition also follows from the hypotheses. Along a nonstationary gradient trajectory, the value of p has constant argument, so a trajectory between two different saddles would contradict ray separation. A homoclinic trajectory is excluded by the strict increase of u. The stable-manifold theorem gives local stable and unstable curves tangent to the Hessian eigenspaces. Their global branches for different saddles therefore do not meet; at one saddle they meet transversely, since their tangent lines are the two distinct Hessian eigenspaces. The regular lower boundary is an entrance and the regular upper boundary an exit. With this boundary convention, the gradient satisfies the Morse–Smale transversality condition. This does not make the quotient by entire trajectories Hausdorff.

For an explicit obstruction, take $p(z) = z^2 - 1/4$ and the compact band

$$\log 2 \le -\log|p(z)| \le \log 8.$$

Its only critical point is the simple saddle at 0, whose value is $-1/4$; both boundary levels are regular and ray separation is vacuous. The positive real separatrix, including its upper-boundary endpoint, is

$$(0, 1/(2\sqrt{2})], \qquad \frac{dx}{d\tau} = \frac{2x}{1/4 - x^2}.$$

Integration gives $\tau - \tau_0 = \frac{1}{8}\log(x/x_0) - (x^2 - x_0^2)/4$, so this trajectory tends to 0 as smooth-gradient time $\tau \to -\infty$. Thus this orbit is not closed in the band: its closure also contains the stationary orbit $\{0\}$. In the quotient topology its image is a nonclosed singleton, so the orbit quotient is not even T_1, hence not Hausdorff. Reparametrising the nonstationary trajectories leaves this quotient unchanged.

On an actual cut-open flow rectangle, by contrast, collapsing the trajectories gives the transverse interval. Identifying boundary copies is a further quotient and requires a separate separation argument. The Reeb graph uses yet another equivalence relation: it identifies connected components of level sets, not complete trajectories. Neither that graph nor the embedded Newtonian graph can be substituted for the orbit quotient without specifying a new equivalence relation.

Global solution theory and the finite graph are the classical results cited above. The relative endpoint classification and transversality are proved above. They do not construct the required finite cut-open space, justify its boundary identifications or control the length needed to join the pieces. The quadratic example settles the uncut-orbit-space question negatively; it does not obstruct a construction that retains the required separate boundary copies.

The unrestricted assertion in Erdős #1041 is false by Theorem 1.2. The positive results do not give a classification of the polynomials that admit a short connection. Their hypotheses and containment levels differ: low critical value, separated simple critical value, and the individually proved polynomial families treat separate sufficient conditions. In particular, the scaled $(5/2)\mu^{1/n}$ bound is at level $(25/13)\mu$, which is smaller than 2μ; it therefore improves both the constant and the level of the $71/10$ construction. The latter is retained for its independent proof and component-sensitive estimates. The critical-value moment does not choose a contained pair. The ordinary generic slit-sheet topology has classical antecedents; the earlier proposed sufficient estimate concerned the lengths of particular inverse-ray curves. Their minimum need not equal the infimum over all contained curves. The degree-seven counterexample above rules out the corresponding universal estimate for these particular curves too, without invalidating conditional estimates on smaller classes.

Adjacent inverse-map methods. Crane [22, Lemma 2.1, Theorem 2.2 and §§3–4] uses inverse-branch hyperbolic geometry, Dubinin's radial-slit input and capacity for a derivative-normalised Smale ratio. Dubinin's four-point result [23, Theorem 1 and Corollary 4] assumes bounded critical values and estimates distortion. These are relevant methods, not direct sources of the present positive moment or connector constant, and their normalisations are not interchangeable with a root-disc hypothesis.

A Numerical estimates for the packing inequality

The circle-slice inequality can be relaxed to finitely many radii. The numerical values below compare that relaxation with specific configurations. They are historical optimisation measurements; only the directed rational bounds used in Section 4 enter its certificate.

At $a = 1$ the constants of the failure inequalities have the rounded values $\delta \approx 0.4586751$, $D \approx 5.3770730$, and $d_{\mathrm{low}} \approx 2.1700770$; the identity $\cosh(D/2) = e^2$ is exact. The quantity being estimated is $\sum_j \lambda(d_j)$ for k roots at this fixed area, not a Euclidean path length. The four columns compare the earlier bound

$$\max\left\{\frac{\delta}{2} + (k-1)\lambda(D - d_{\mathrm{low}}), k\,\operatorname{artanh}(e^{-2})\right\},$$

where $\lambda(d) = -\log\tanh(d/2)$ as in Section 4; the hyperbolic packing bound used for the earlier threshold $2/5$; the linear-programming value M_{circ} of the relaxation of Lemma 4.3; and the largest reported objective value at a numerically constructed configuration. The last column is a search result, not a certified feasible value or a proved optimum.

k	earlier bound	packing	M_{circ}	search
2	0.3103	0.5899	0.3102	0.3103
3	0.4085	0.6878	0.3893	0.3784
4	0.5447	0.7475	0.4450	0.4306
5	0.6809	0.7906	0.4846	0.4694
6	0.8170	0.8244	0.5155	0.4986
7	0.9532	0.8521	0.5408	0.5208
8	1.0894	0.8757	0.5622	0.5413
10	1.3617	0.9142	0.5972	0.5759
14	1.9064	0.9710	0.6487	0.6228
20	2.7234	1.0298	0.7022	0.6717
30	4.0851	1.0955	0.7624	0.7283
100	13.617	1.2871	0.9428	n/a

Both right-hand columns come from floating-point searches. In addition, M_{circ} uses a 2600-point d-grid and a 1400-point r-grid. Restricting the allowed d values and checking only finitely many circle constraints have opposite effects on the feasible set. Without a separate continuous feasibility or dual certificate, the computed value is not a proved lower or upper bound for the continuous relaxation. At $k = 2$, the grid value 0.31018 differs from the analytic value, approximately 0.310338. Their proximity supplies no rigorous bracket.

The certified lower bounds for k, rounded up to integers, are as follows at $a = 1$.

x	0.40	0.45	0.50	0.55	0.60	0.63	0.65	0.70
earlier bound	3	4	4	5	5	5	5	6
packing bound	2	2	2	2	2	3	3	4
circle-slice bound	3	4	6	8	10	12	14	18

In one explicit-Euler floating replica with step 10^{-3} and a geometric grid of 40 initial areas from 10^{-6} to 1, the largest computed time to reach the area bound falls from 0.89703 with the earlier packing lower bound for k to 0.65503 with the certified circle-slice bounds, and to 0.63003 if the relaxation's linear-programming value replaces those bounds. Shrinking the relaxation by the worst measured slack factor 1.045 moves it

only to 0.60703. These computations suggest that this particular relaxation may stop near $e^{-0.607} = 0.545$; they do not prove that it does, and they exclude neither a stronger lower bound on the root count nor a larger certified range.

The search uses only ten or fourteen radii in its two settings and a coarse a-grid with upward rounding. The reported loss from the first restriction at $a = 1$, $k = 8$ is about one per cent; the reported rounding loss is comparable. These are empirical estimates, not certified error bounds relative to the unrestricted continuous optimum.

The continuous optimisation remains unevaluated: maximise $\sum_j \lambda(d_j)$ subject to $d_j \geq d_{\mathrm{low}}$ and $\sum_j w(d_j, r) \leq \pi$ for every $r > 0$, allowing the relaxed radial distributions used in the computation. The numerical optimiser spreads mass over several radii and suggests logarithmic growth in k. Neither that growth law nor its coefficient is proved. Evaluating the relaxation would clarify this packing argument; its effect on the final path threshold would still need to be established.

B Guide to the results

The main constructions have different hypotheses. The table locates their conclusions and distinguishes the formal results from the additional analytic remarks. The detailed evidence associations are in the evidence record.

Family or mechanism	Conclusion and qualification
Degree seven	The fixed polynomial has no connected two-root subset of Hausdorff measure at most 2 in its strict lemniscate.
Trinomials	Every root-to-origin segment is contained; any two roots are joined with length below 2.
Low critical values	Unformalised area-growth criterion with cutoff 13/25 and scaled level $(25/13)\mu$. Its recorded rational comparison also covers $\mu \leq 529/1000$; the analytic reduction remains required.
Separated simple value	Unformalised square-root, Bergman and capacity construction. The chosen critical point must be simple and its value isolated as stated.
Collinear and sparse families	Formal segment constructions for the specified root geometry or coefficient patterns. The stronger $2\sqrt{1-\lvert h\rvert^2}$ cyclic-fibre estimate remains an ordinary argument.
Critical-value means	Formal exponent $2/(n-1)$. The sharp $4/(n-1)$ refinement and higher exponents under complex power-sum cancellations remain unformalised. No path selection follows.
Perimeter and prescribed paths	The Cassini example refutes the stated spanning-tree coefficient. The degree-eight example refutes the proposed Gamma-function constant, while a uniform perimeter bound is still open. The implication from $\beta\sigma^{1/n}$ to a path of length at most $\beta\rho$ in K_μ needs the unformalised first-critical-level split. Several prescribed straight-segment rules fail.

The inverse-ray estimate in the earlier note is retained only as a historical sufficient condition. It does not follow from these value, segment or perimeter calculations.

C Formal statements and computational checks

The formal sources supplied with these manuscripts have not been rebuilt for this revision. Their recorded statements include the fixed counterexample for preconnected sets, the negation of the Formal Conjectures statement, its `answer(False)` form, and the total-variation formulation. The independent ordinary reconstruction in Section 2 uses Rouché's theorem, the component degree and Hausdorff measure. Its Cayley signs, critical-disc estimates and all four Bernstein tests were recomputed with rational arithmetic in an earlier revision; the present edit retains those finite calculations. Independent human review of the construction and its correspondence with the 1958 wording is not recorded.

The 13/25 calculation has the recorded full replay described in Section 4. Its analytic reduction is an ordinary proof and remains unformalised. The same distinction applies to the fourth-power mean estimate and the longer analytic family arguments. The numbered statements without a complete formal proof have therefore been retained as remarks, with their original mathematical content and local input notes. The following list records exactly what the existing formal deductions assume.

4.1. No Lean proof covers the whole assertion. Lean checks the degree-two case, and the closing inequality $(13/25)e^X < 1$ at the stopping time X certified in the proof. The computation that certifies X, done by an exact-arithmetic program, and the analytic argument in degrees three and higher are not checked in Lean.

4.2. The Lean proof assumes Remark 4.1.

4.4. The Lean proof takes the separation bound (7) and the individual and summed bounds (8) as hypotheses; their derivation from the failure assumption, given in that section, is not checked in Lean.

4.5. The Lean proof assumes Remark 4.1 in degrees above two; the degree-two case is checked without it.

5.1. The Lean proof assumes the two-parameter bound (CF) in degrees $n \geq 3$, which the proof obtains by averaging over levels and value directions. The degenerate case, degree two and the numerical constants are checked without that input.

5.2. The Lean proof assumes the bound (CF) from the proof of Remark 5.1, for a component containing at least k_0 roots; the rational inequalities are checked.

5.3. The Lean proof assumes the averaging construction in the proof of Remark 5.1, rerun with the area–capacity inequality for the area of C; the rational cutoffs are checked.

5.6. The Lean proof assumes the two-component split at the first critical level constructed in the ordinary argument: two one-root components whose closures meet at a critical point, each with boundary length at most $\beta\rho$ and each joining its root to that point within half its perimeter.

7.1. The Lean proof assumes the ordinary analytic construction: the connector obtained through the square-root map, its Bergman length bound (7), and the capacity bound for the area of the two-sheeted component.

7.2. The Lean proof assumes the same construction as Remark 7.1 for the second assertion. Inequality (6) and every numerical step are checked without it.

13.1. No Lean proof covers the whole assertion. Lean checks the ray-disjointness, level-separation and saddle-scale steps of the proof; the Morse, monodromy and strip statements are not checked.

8.2. The formal capacity conclusion takes as input the classical transfinite-diameter formula for a filled monic lemniscate. The root count at the first merger does not require that input.

No supplied proof establishes the reported Runge construction of unbounded one-root perimeters or the stronger assertion that every critical joining point fails in the near-pentagon example. Neither is used in the positive theorems. The fixed value of the degree-seven parameter is not a proof of the reported parameter family.

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator pending	Lemma 2.1	Lean · Comparator pending
Theorem 3.1	Lean · Comparator	Passage, p. 10	Verification record
Passage, p. 10	Lean†	Lemma 4.3	Lean · Comparator
Passage, p. 13	Lean†	Passage, p. 17	Lean†
Passage, p. 18	Lean†	Passage, p. 20	Lean†
Passage, p. 21	Lean†	Proposition 5.5	Lean · Comparator
Passage, p. 24	Lean†	Theorem 6.1	Lean · Comparator
Passage, p. 27	Lean†	Passage, p. 30	Lean†
Proposition 8.1	Lean · Comparator	Proposition 8.2	Lean†
Theorem 9.1	Lean · Comparator	Theorem 9.2	Lean · Comparator

Statement	Formal proof	Statement	Formal proof
Corollary 9.3	Lean · Comparator	Theorem 9.4	Lean · Comparator
Theorem 9.5	Lean · Comparator	Theorem 9.6	Lean · Comparator
Theorem 9.7	Lean · Comparator	Theorem 10.1	Lean · Comparator
Theorem 10.2	Lean · Comparator	Lemma 10.3	Lean · Comparator
Theorem 10.4	Lean · Comparator	Theorem 10.5	Lean · Comparator
Corollary 10.6	Lean · Comparator	Proposition 10.7	Lean · Comparator
Theorem 10.8	Lean · Comparator	Theorem 11.1	Lean · Comparator
Corollary 11.2	Lean · Comparator	Theorem 12.1	Lean · Comparator
Passage, p. 70	Verification record		